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(54) **PRECURSOR GLASSES AND
GLASS-CERAMICS COMPRISING A
CRYSTALLINE PHASE HAVING A
JEFFBENITE CRYSTALLINE STRUCTURE**

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(2013.01); **C03C 21/002** (2013.01); **C03C**
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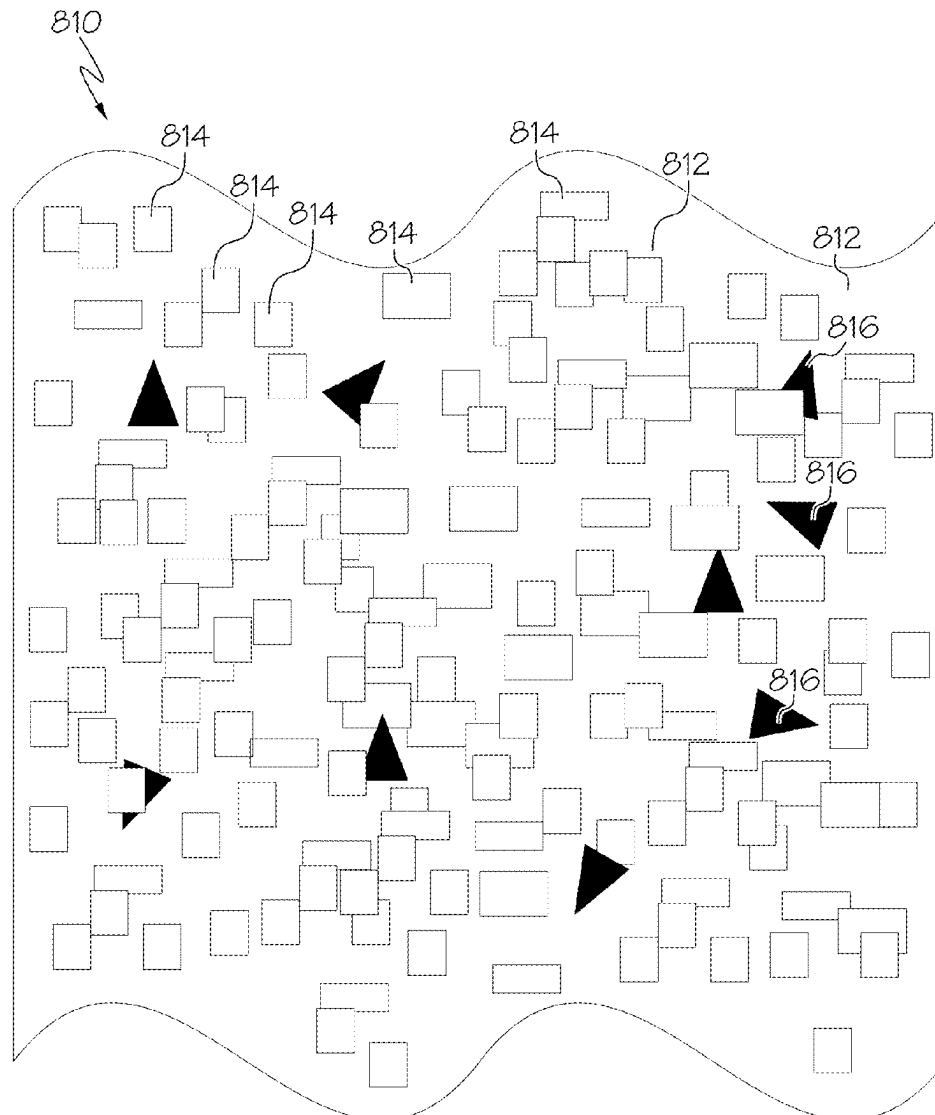
(22) Filed: **May 22, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/794,547, filed on
Aug. 5, 2024.

(57) **ABSTRACT**

Disclosed herein is glass-ceramic and glass-ceramic articles including a crystalline phase comprising a jeffbenite crystalline structure, glass that may be cerammed to become such, and methods of making and using the same.



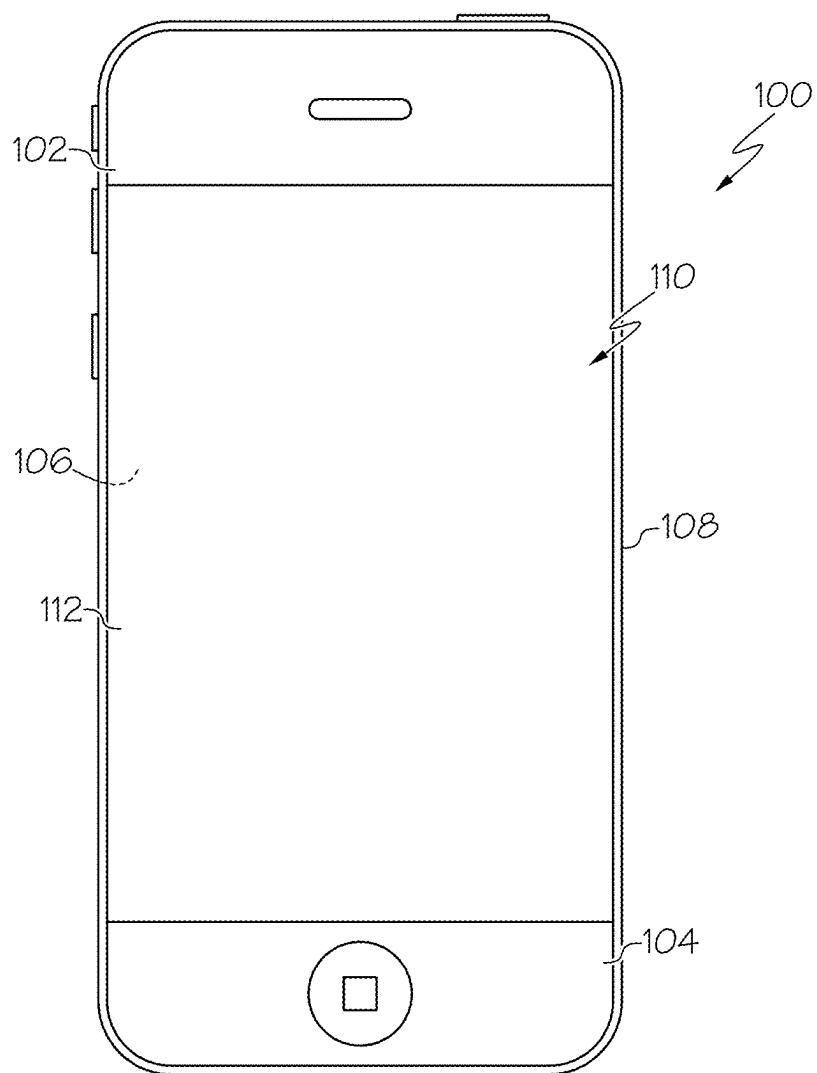


FIG. 1

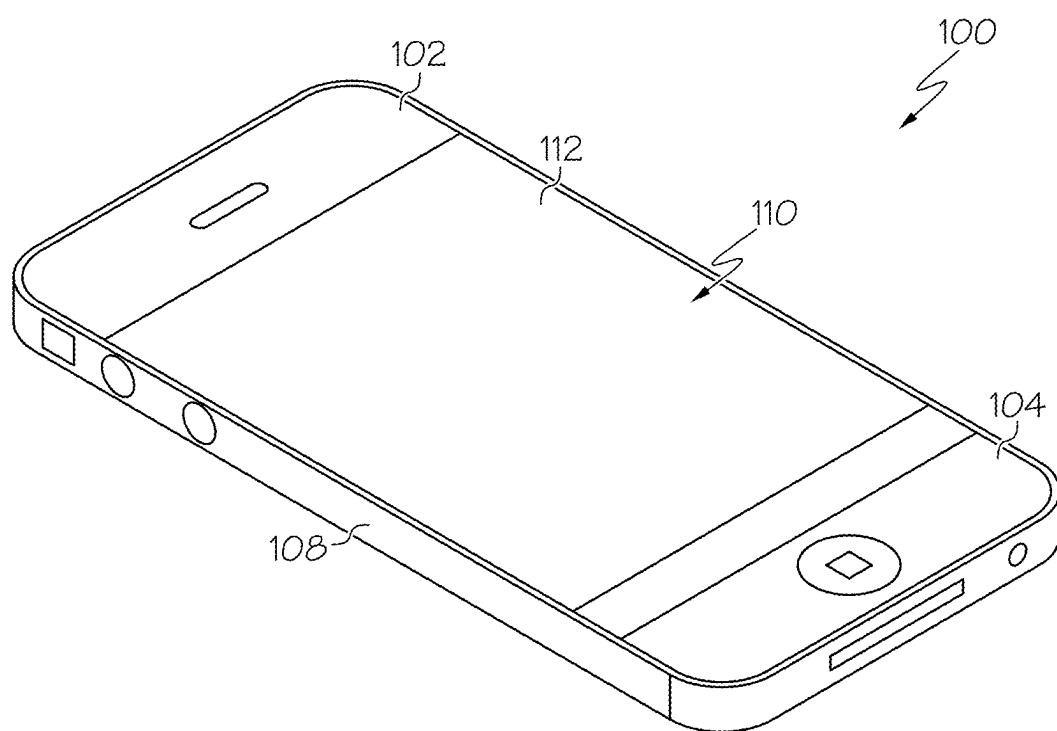


FIG. 2

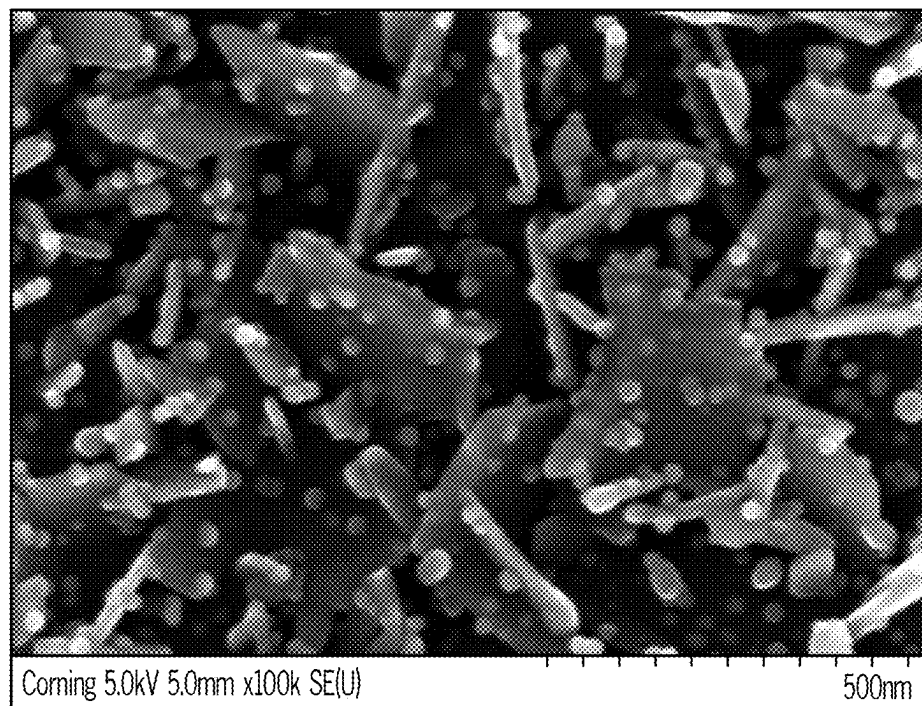


FIG. 3

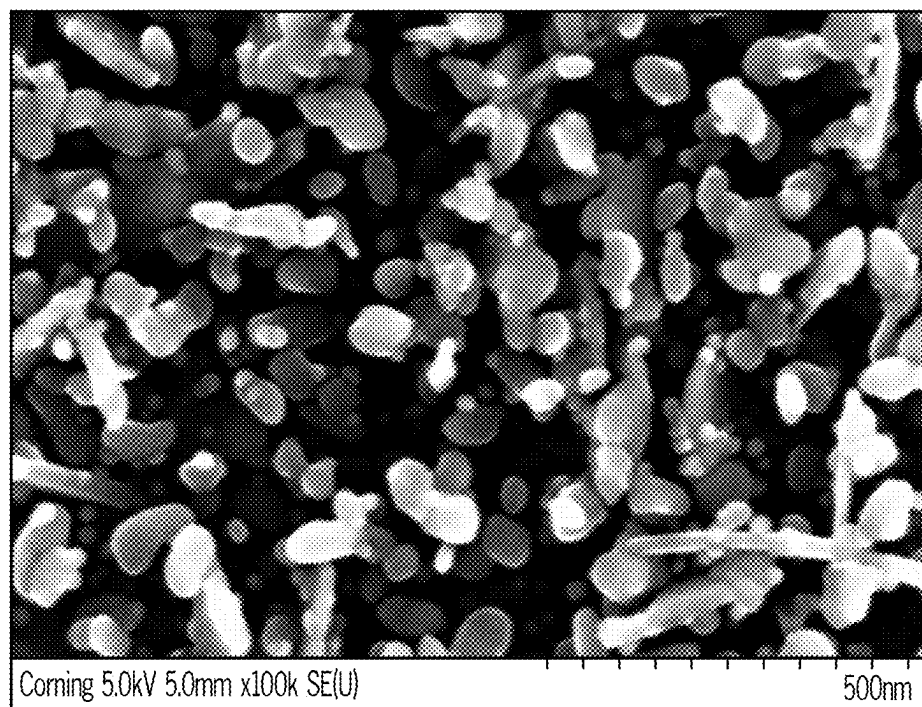


FIG. 4

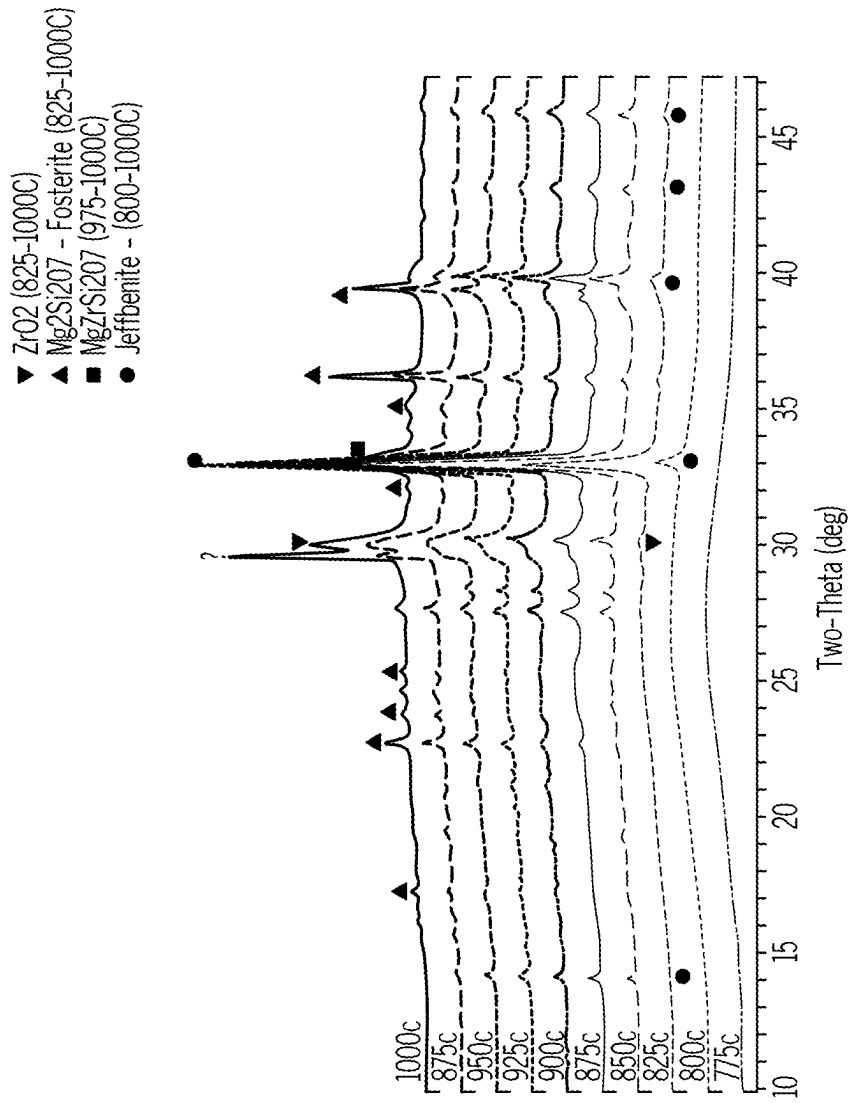


FIG. 5

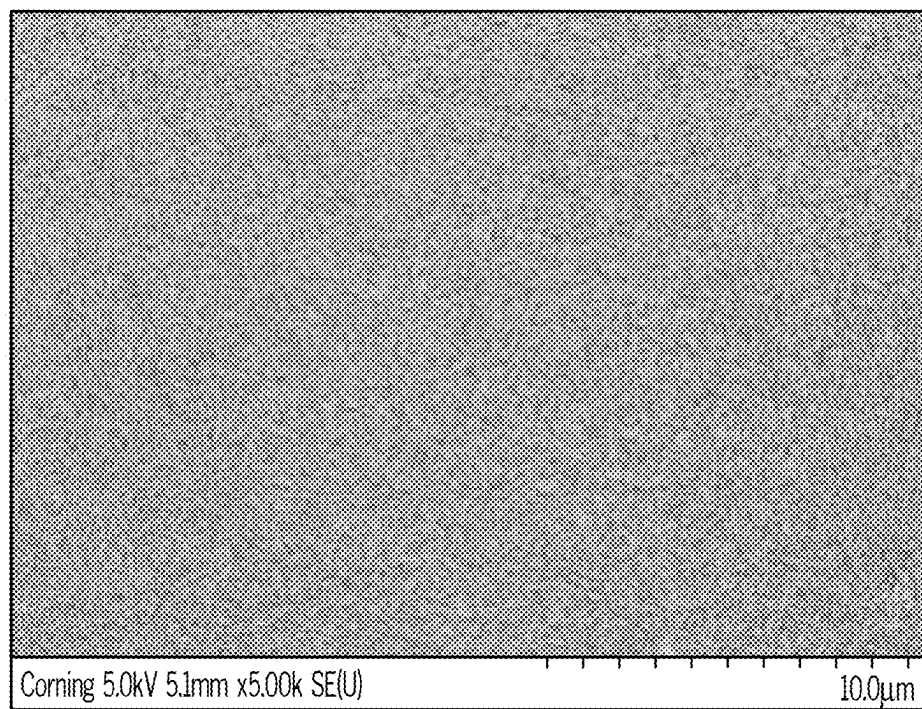


FIG. 6A

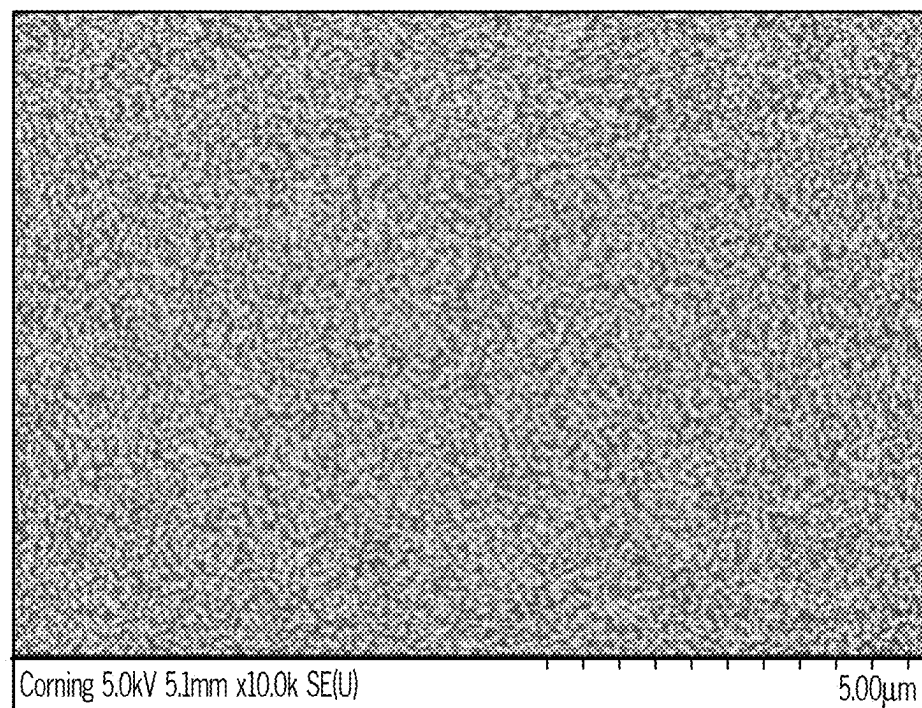


FIG. 6B

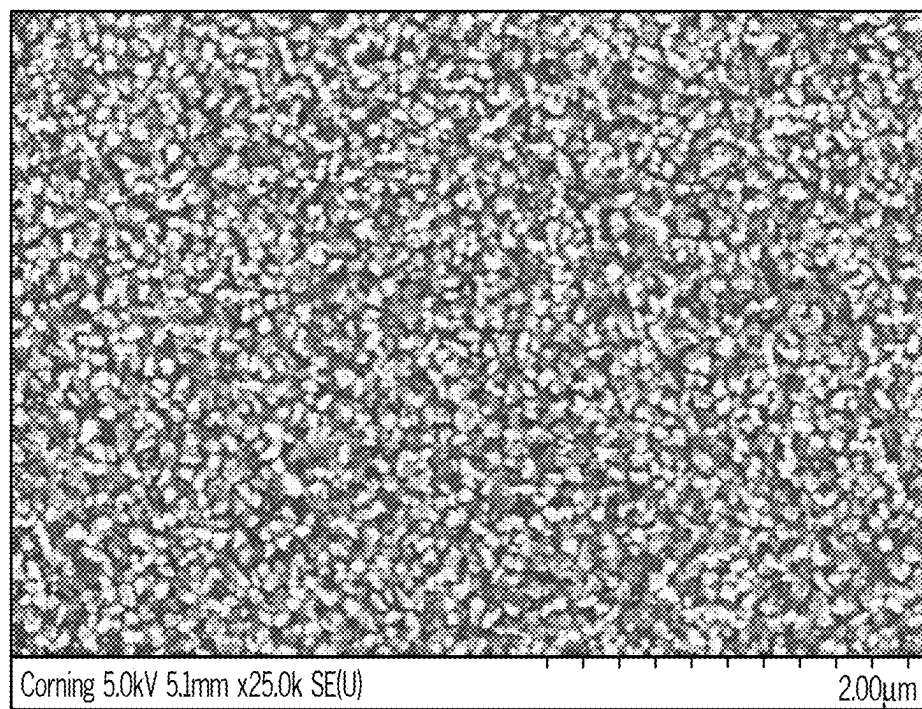


FIG. 6C

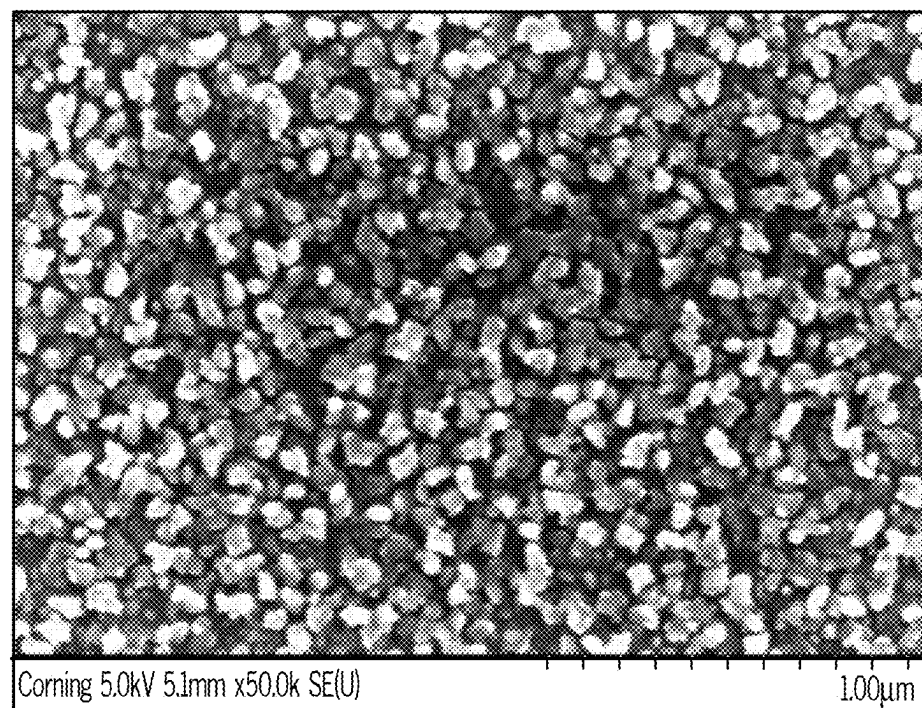


FIG. 6D

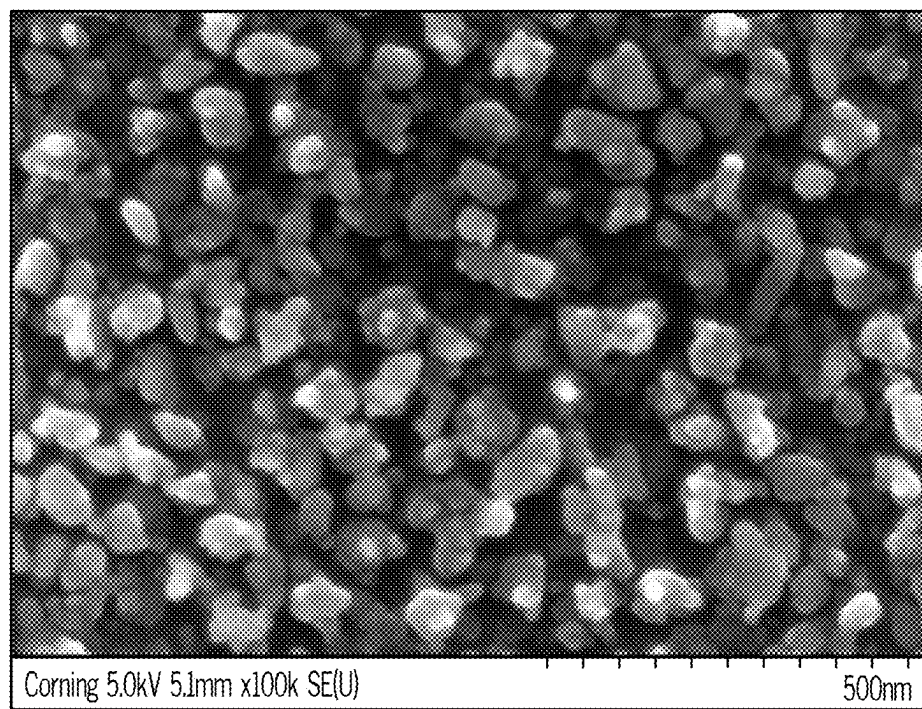


FIG. 6E

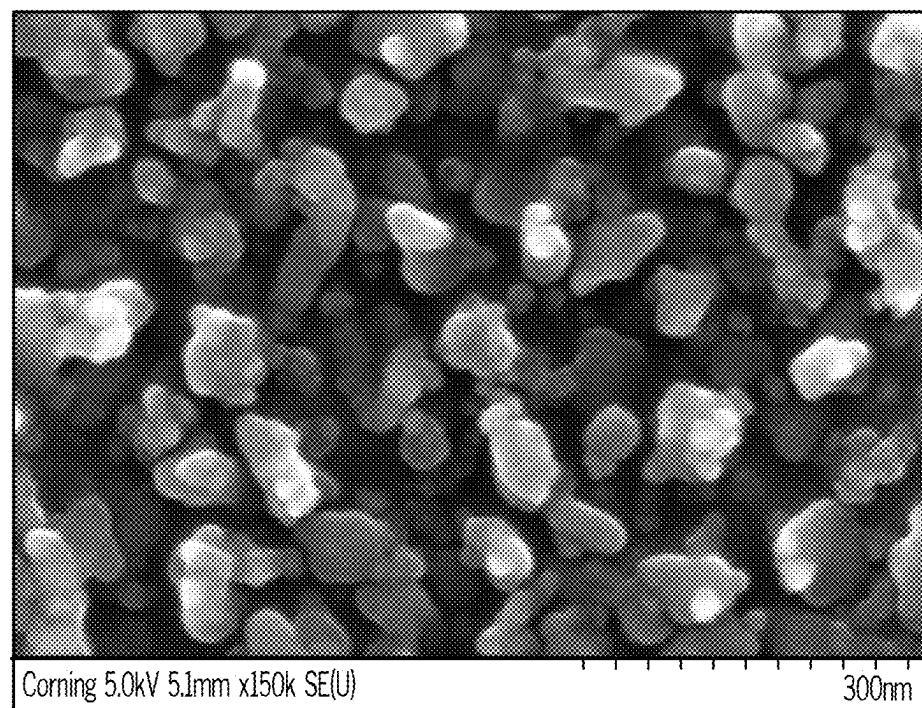


FIG. 6F

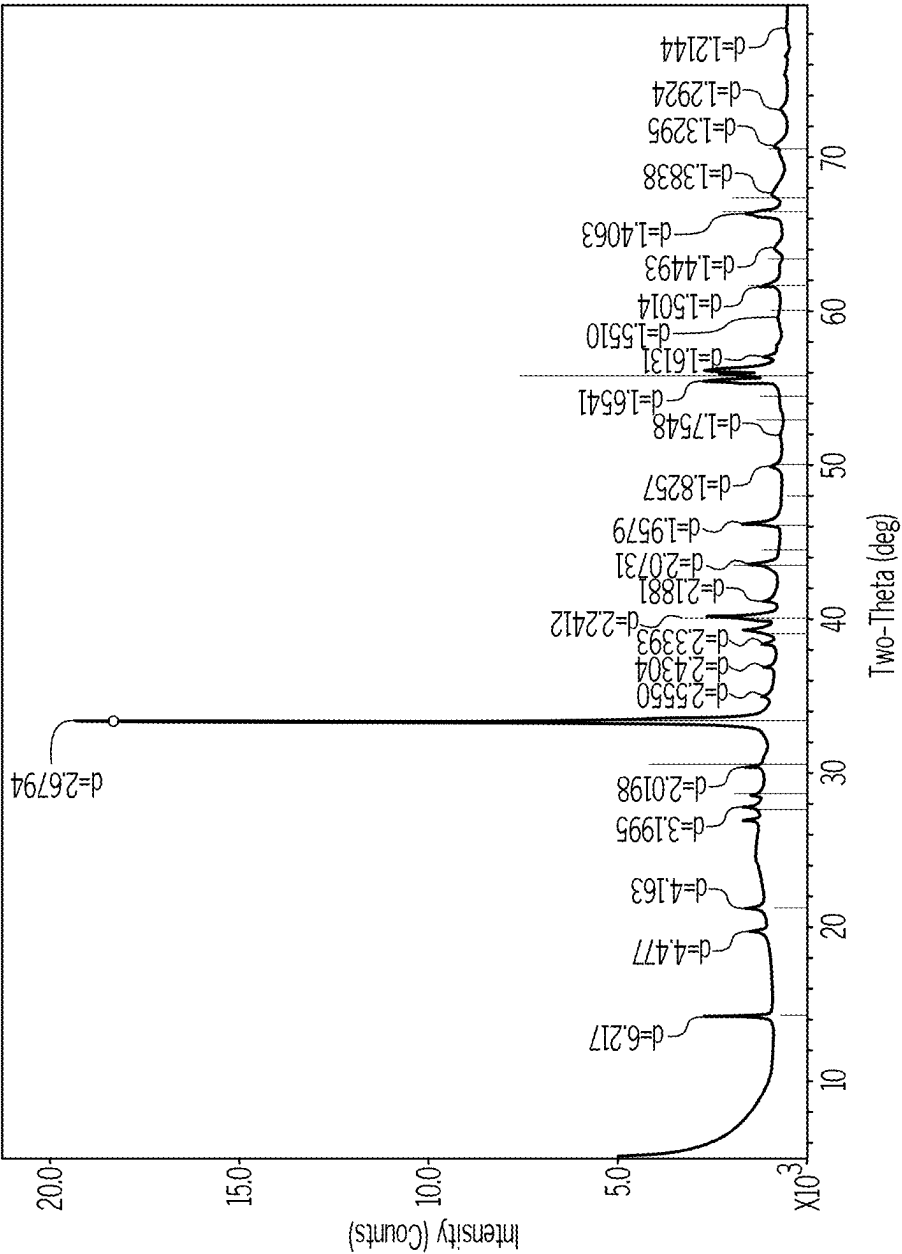


FIG. 7

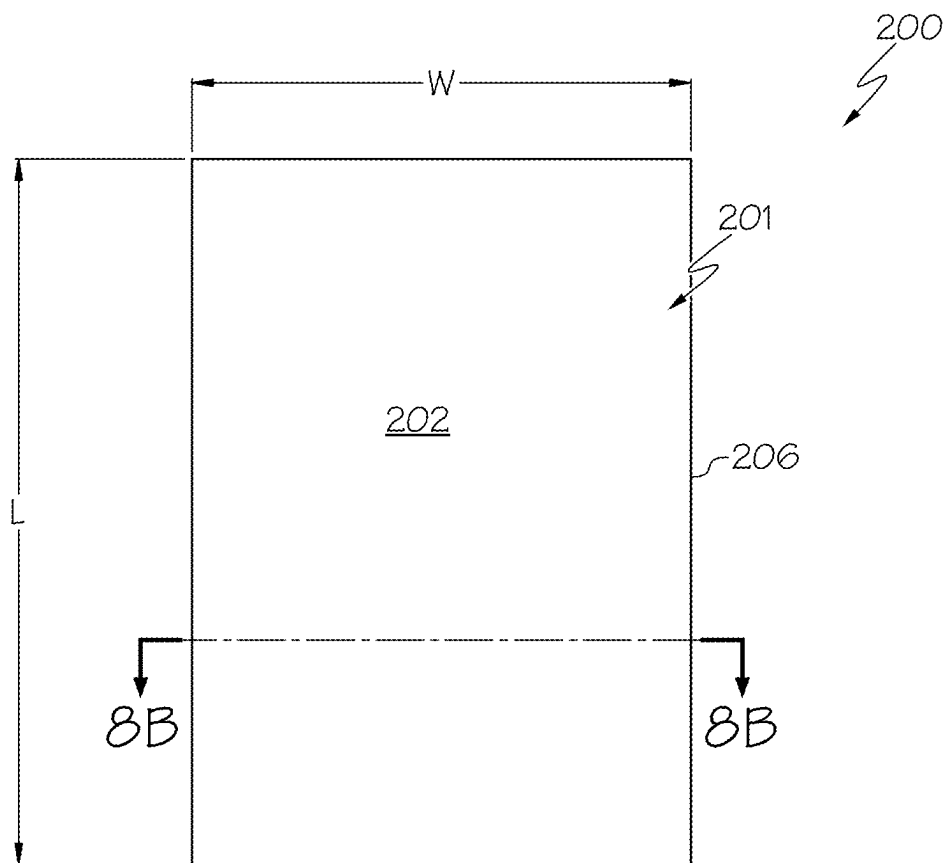


FIG. 8A

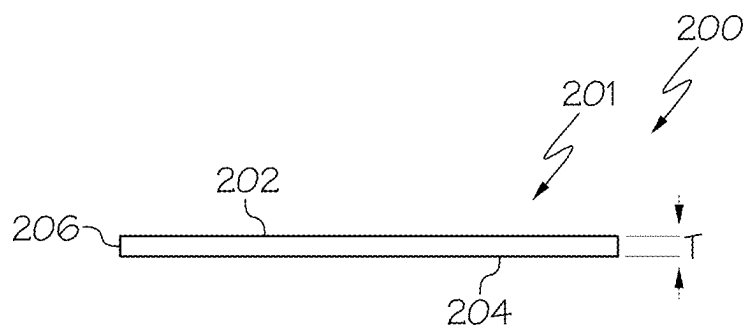


FIG. 8B

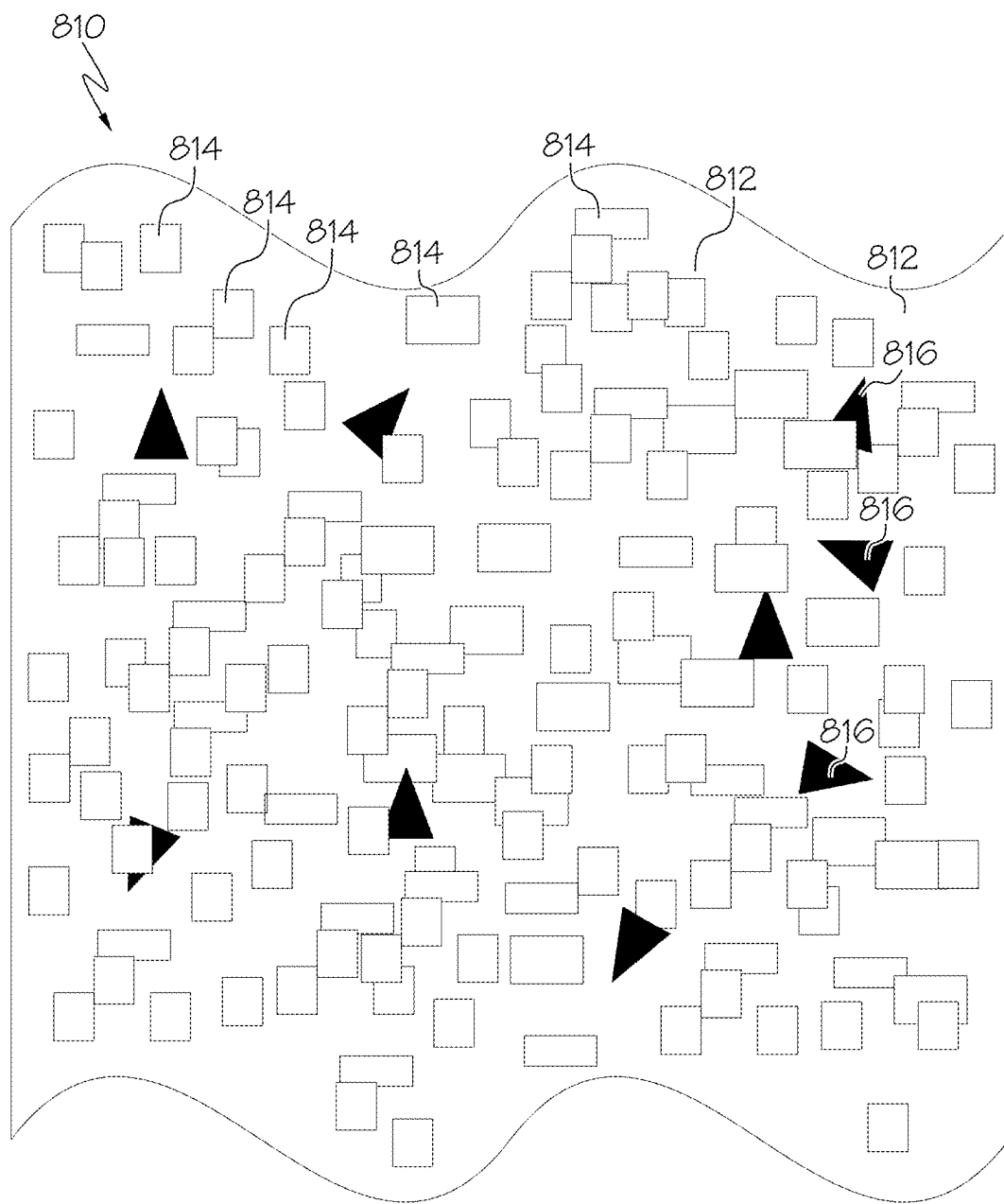


FIG. 9

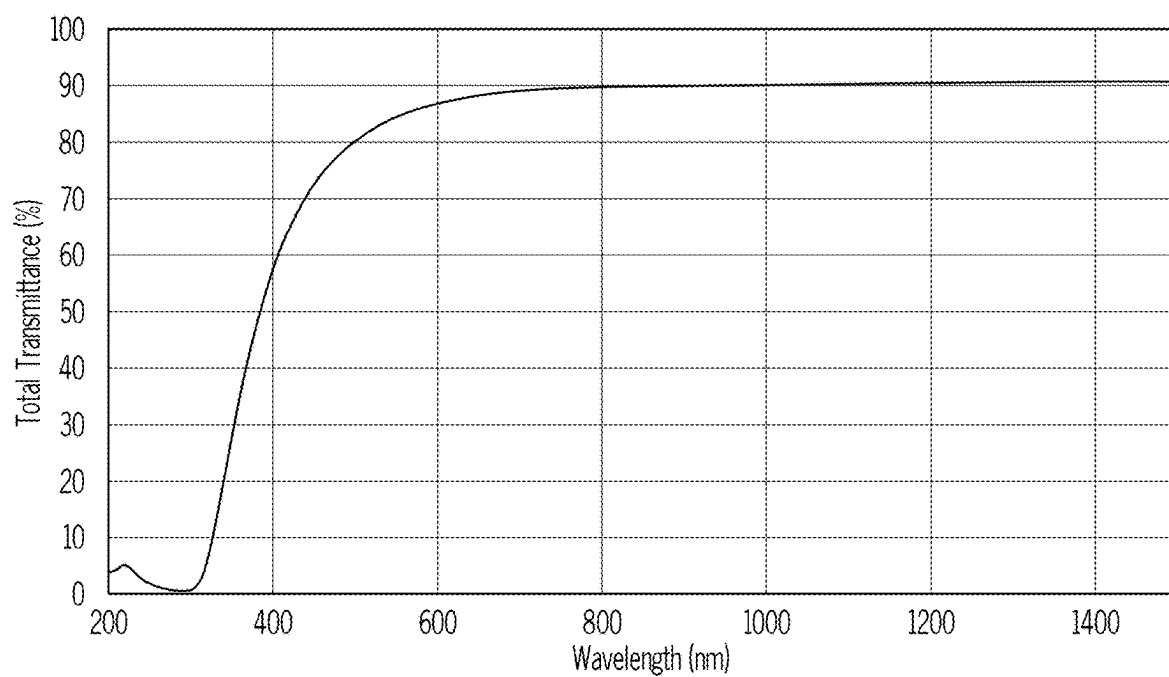


FIG. 10

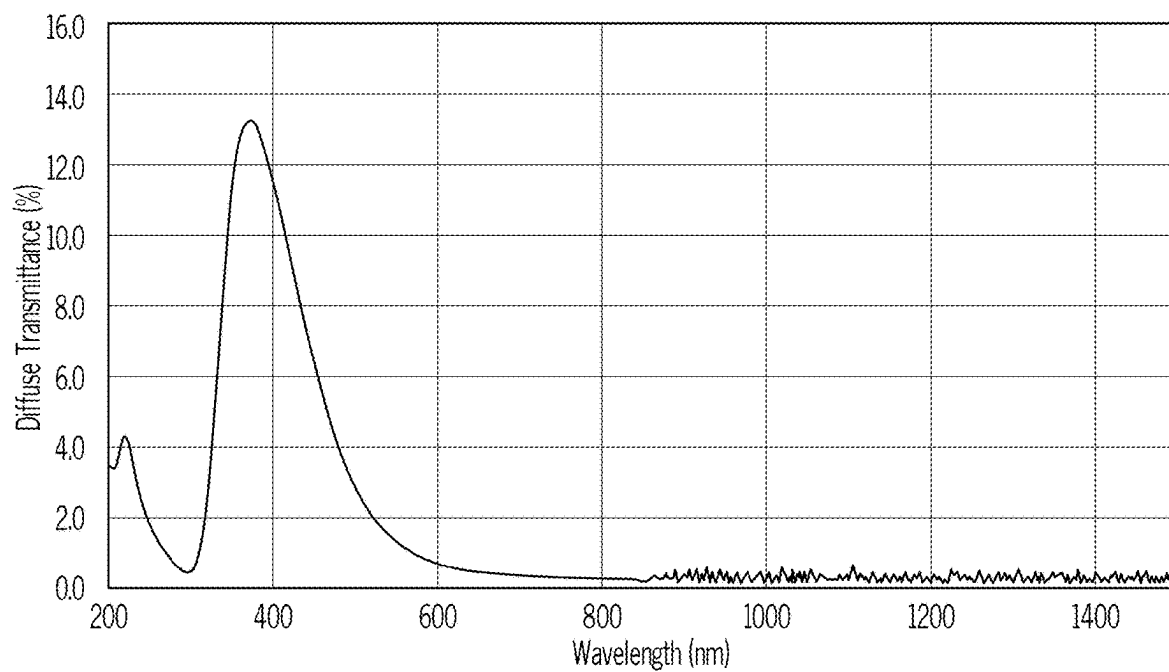


FIG. 11

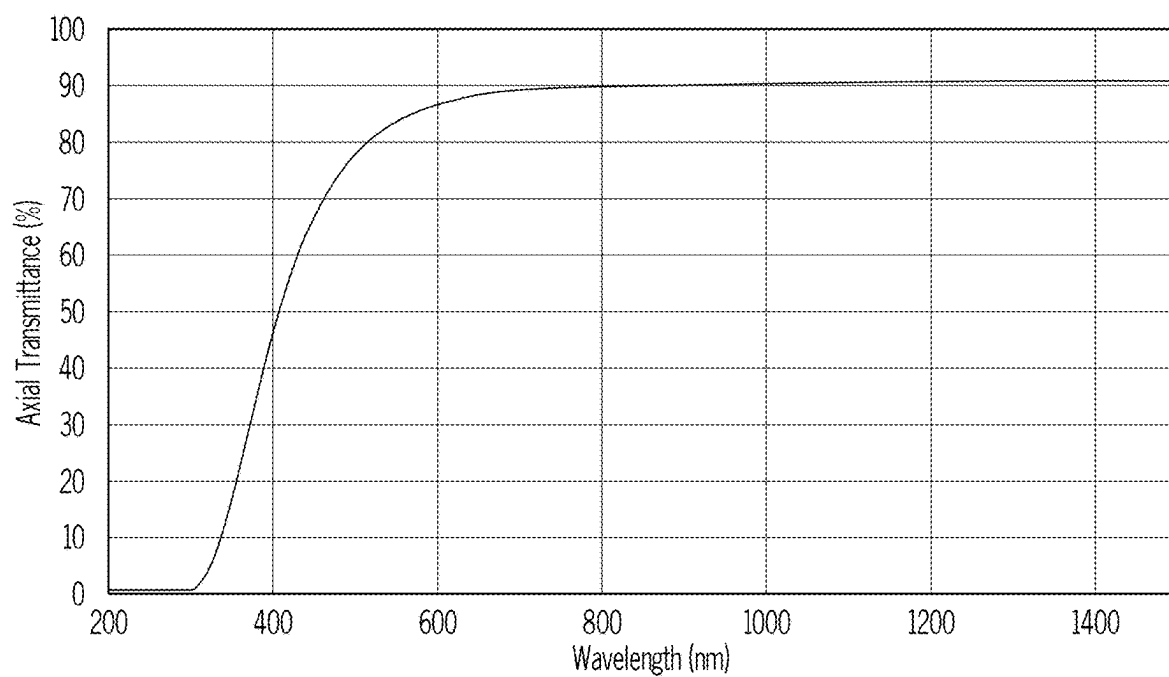


FIG. 12

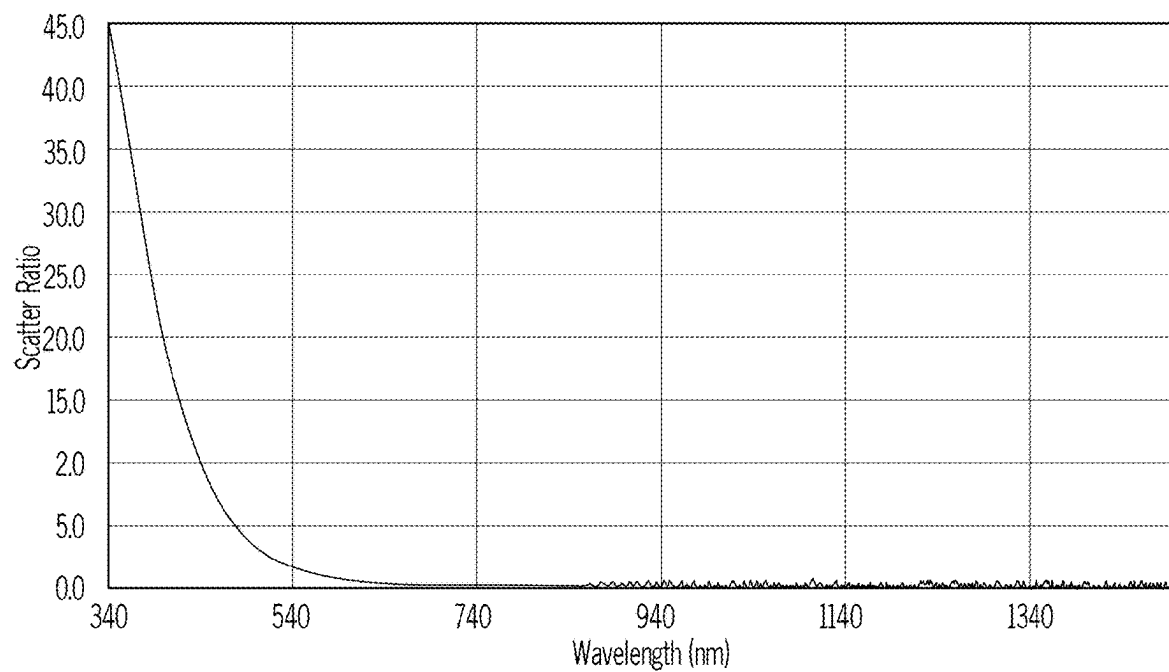


FIG. 13

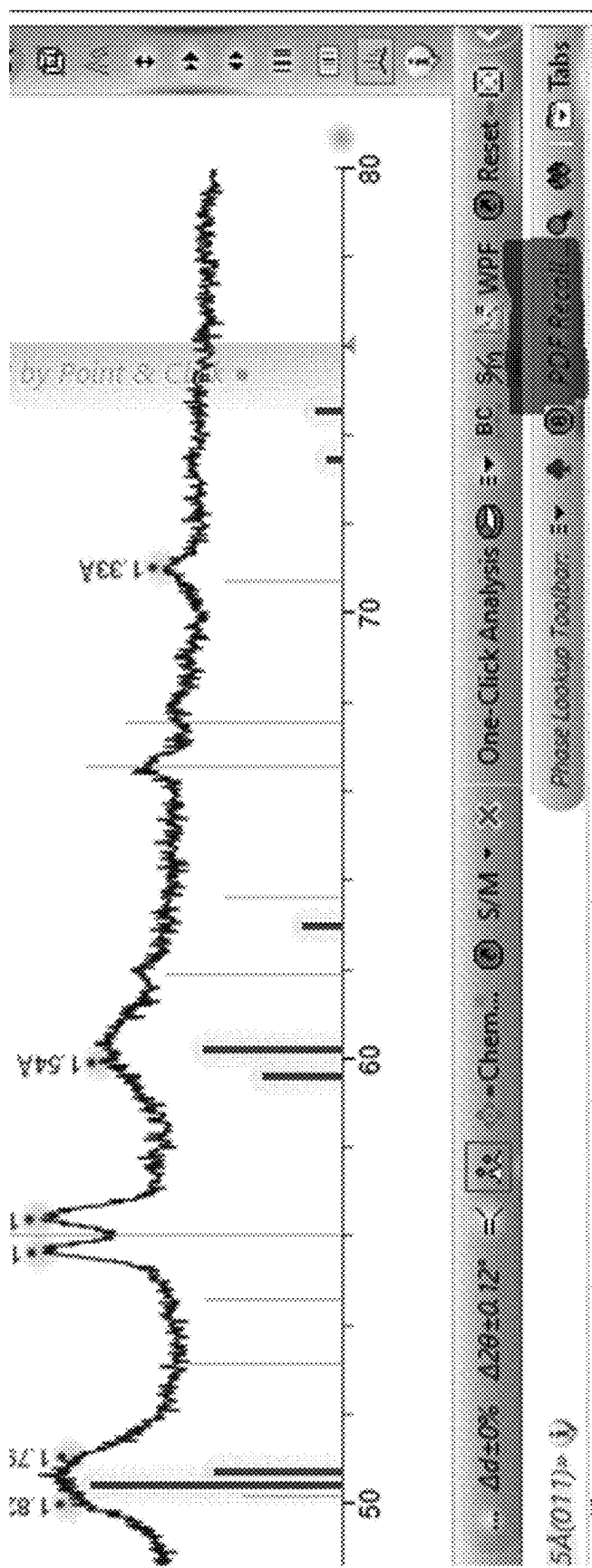


FIG. 14

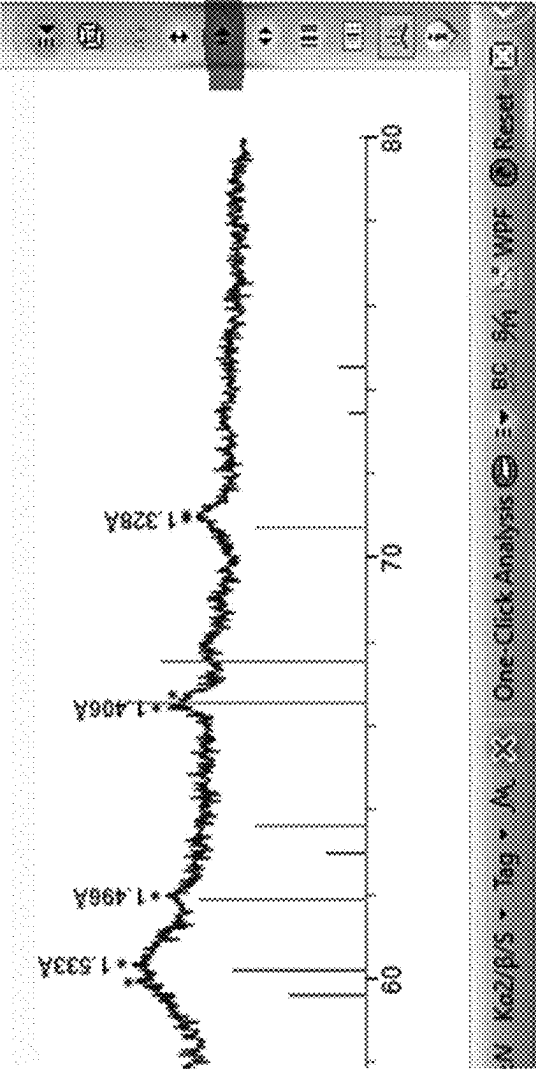


FIG. 15

● Scans (1) Recent Files			🌀 Phases (2) Hits (0)			📊 Peaks (24) Lines (18)			📈 Profiles (0) Points (0)					
#	Angle	d(Å)	BG*	Height	Height%	Area	Area%	FWHM	P/N	(h k l)	Phase ID	d(Å)	%θ	Δ2θ(°)
1	24.392	3.6452	1499.8	127.2	3.5%	16.3	0.9%	0.109	3.2					
2	27.911	3.1940	1504.6	187.4	5.1%	58.5	3.4%	0.265	4.6	(1 0 5)	Jeffberite • Mg...	3.2081	9.0	-0.126
3	30.390	2.9389	1463.3	1205.7	32.9%	1527.8	87.9%	1.077	23.3	(0 1 1)	ZrO2 • Zirconiu...	2.9502	100.0	-0.119
4	30.686	2.9130	1490.1	1262.9	34.5%	1395.4	80.2%	0.939	24.1	(2 1 1)	Jeffberite • Mg...	2.9092	24.0	0.041
5	33.503	2.6726	1228.3	3665.7	100.0%	1739.0	100.0%	0.403	52.4	(2 0 4)	Jeffberite • Mg...	2.6729	100.0	-0.005
6	40.320	2.2245	967.7	643.3	17.6%	399.7	23.0%	0.528	16.6	(2 0 6)	Jeffberite • Mg...	2.2217	19.0	-0.136

FIG. 16

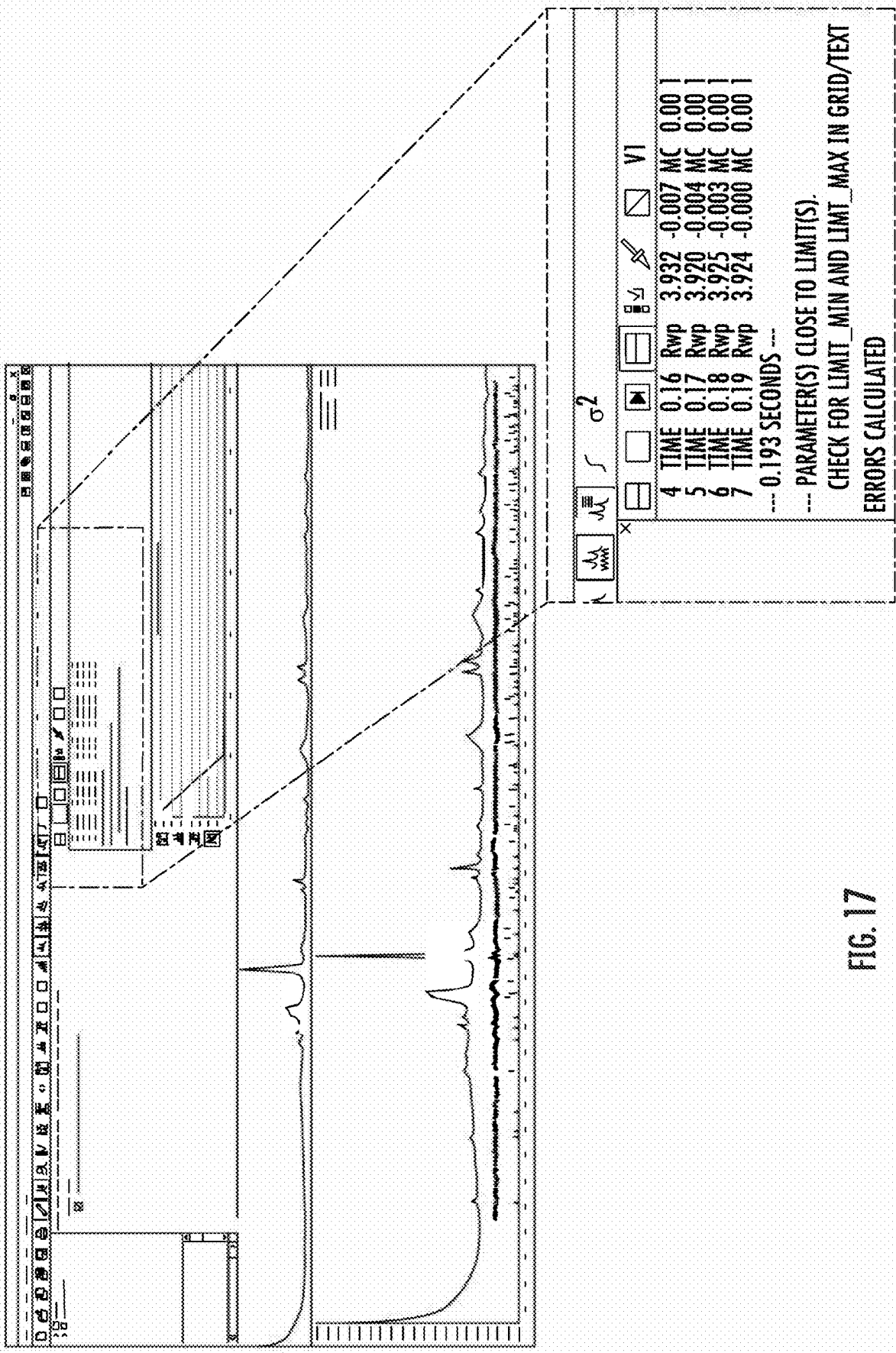


FIG. 17

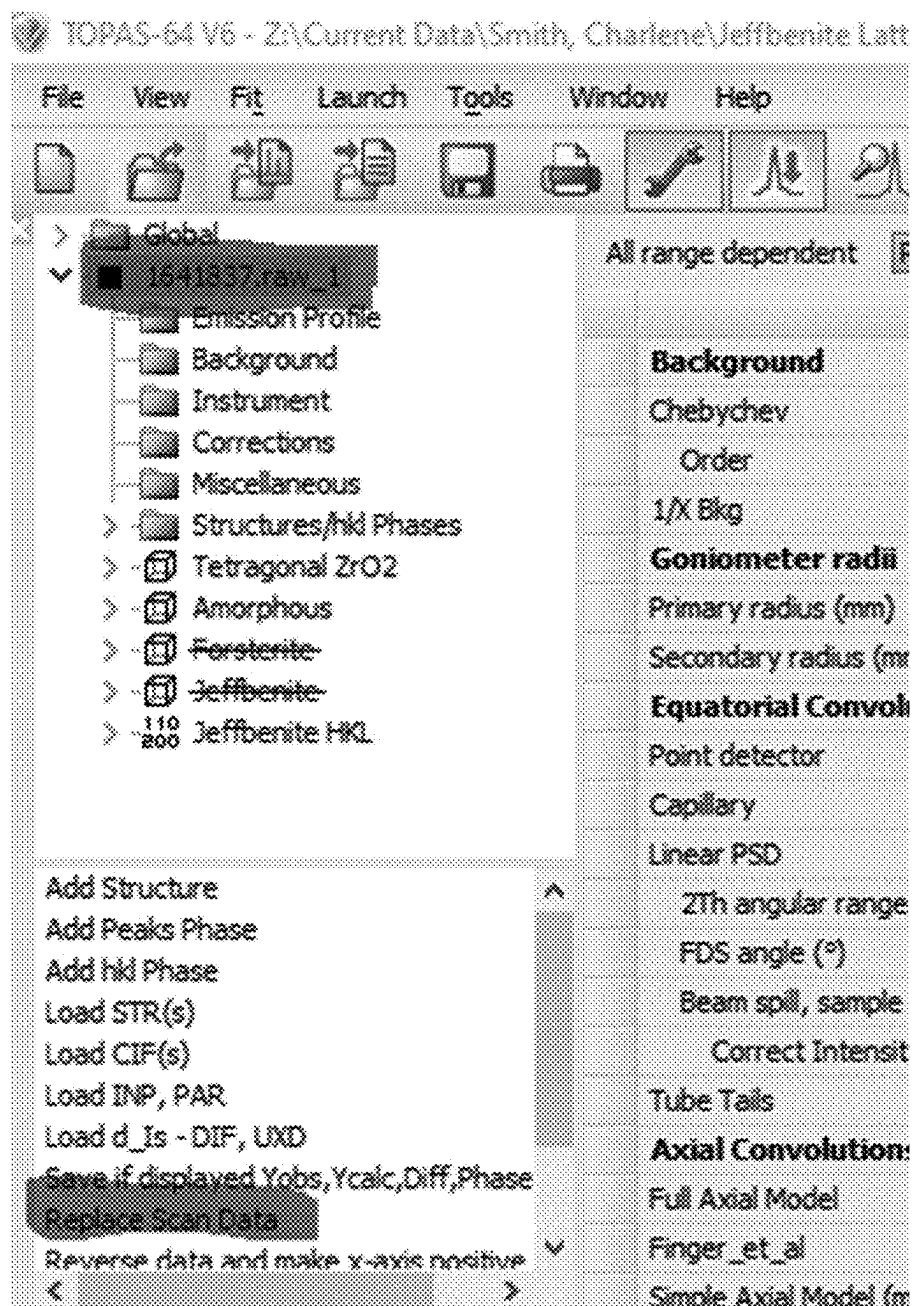


FIG. 18

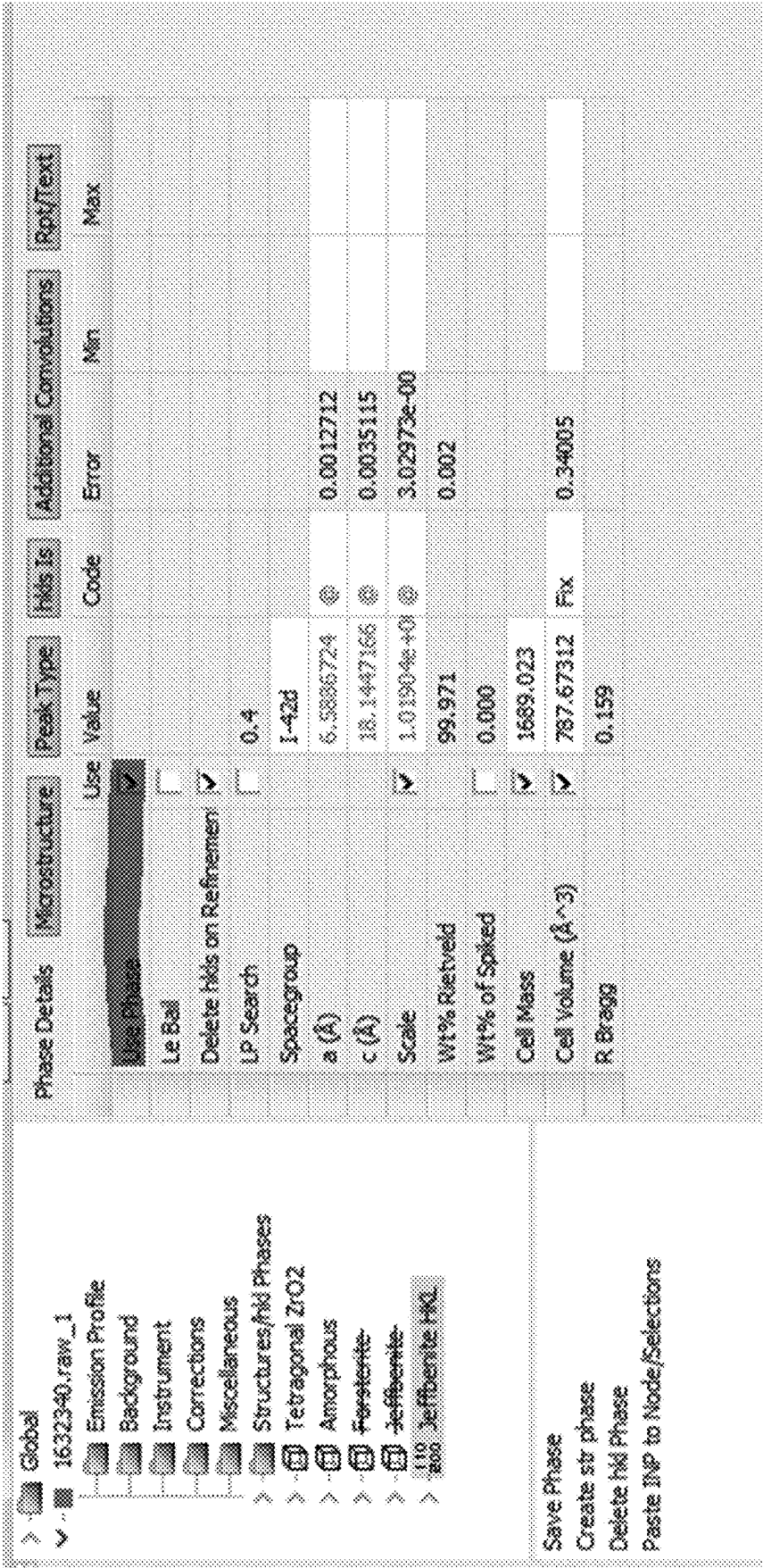


FIG. 19

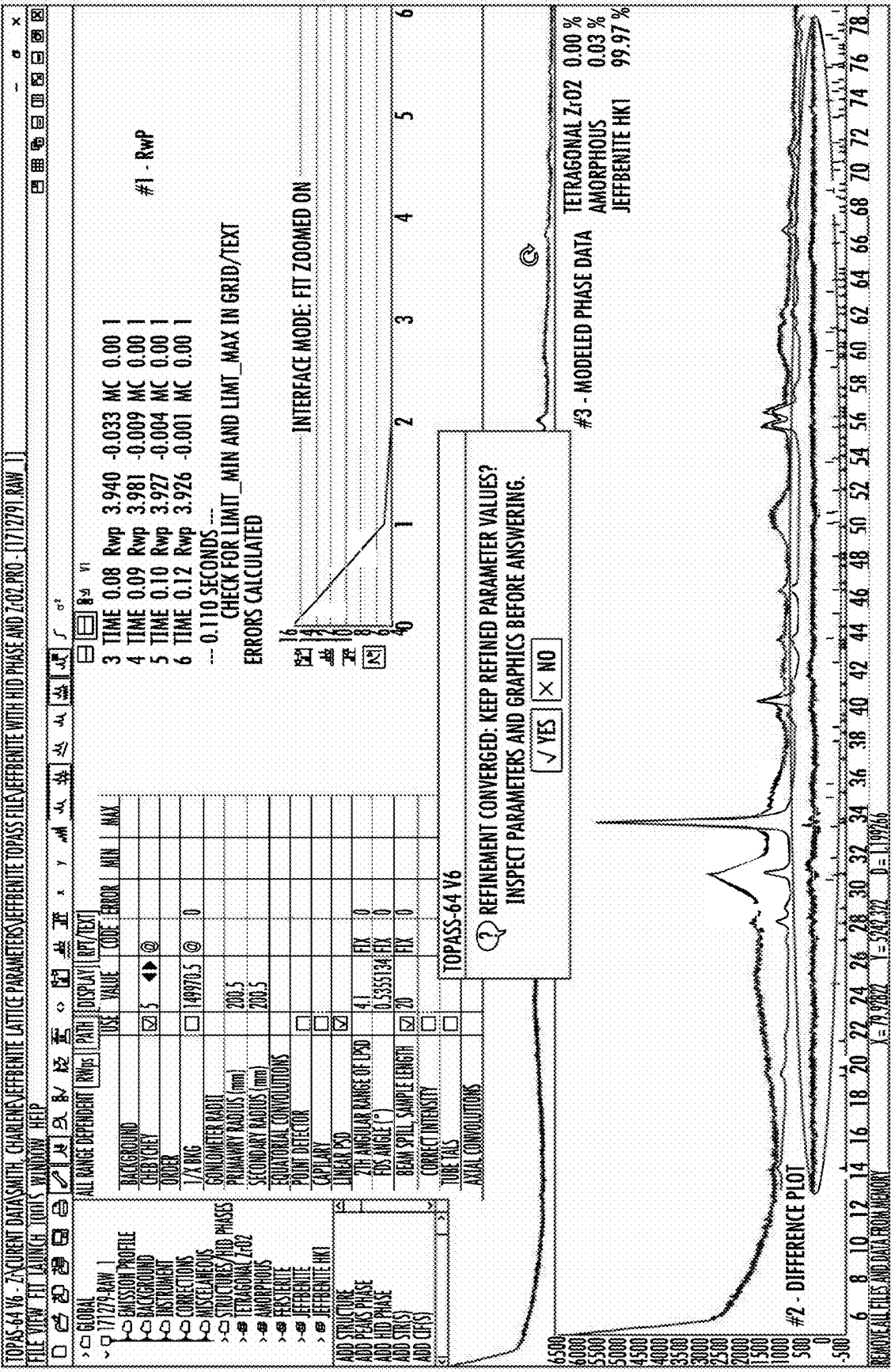


FIG. 20

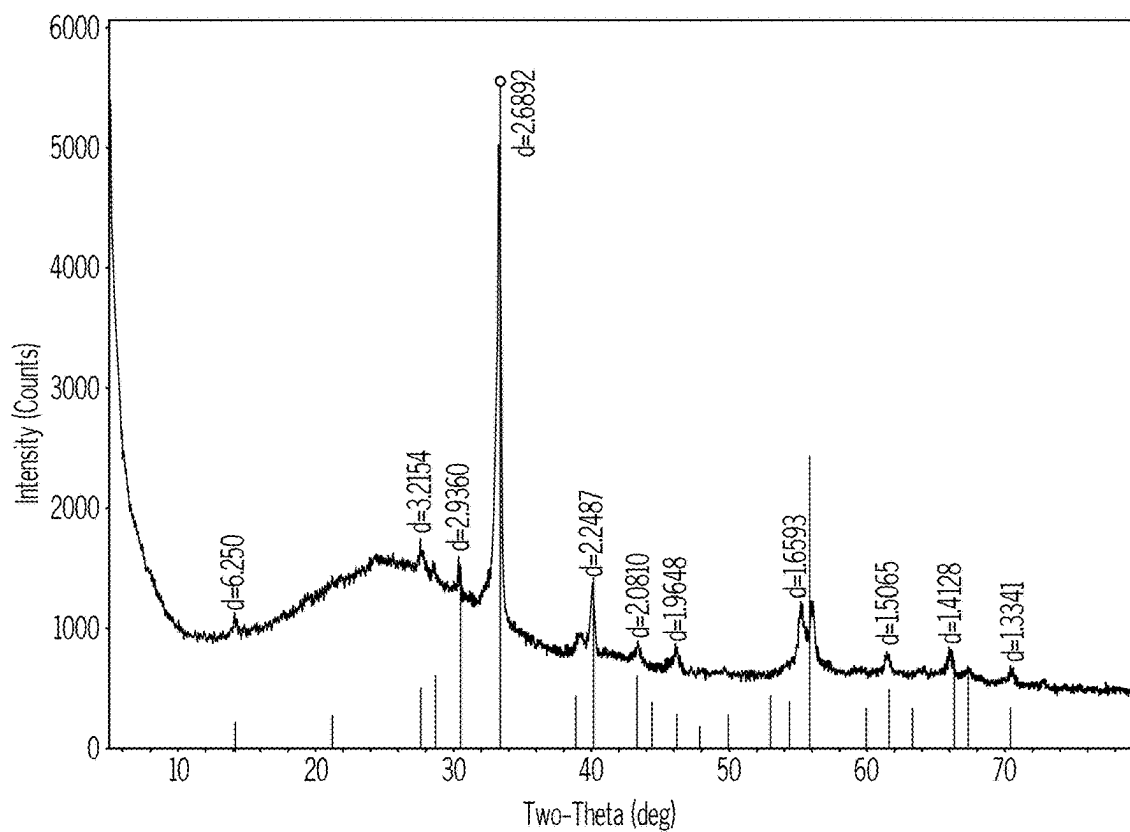


FIG. 21

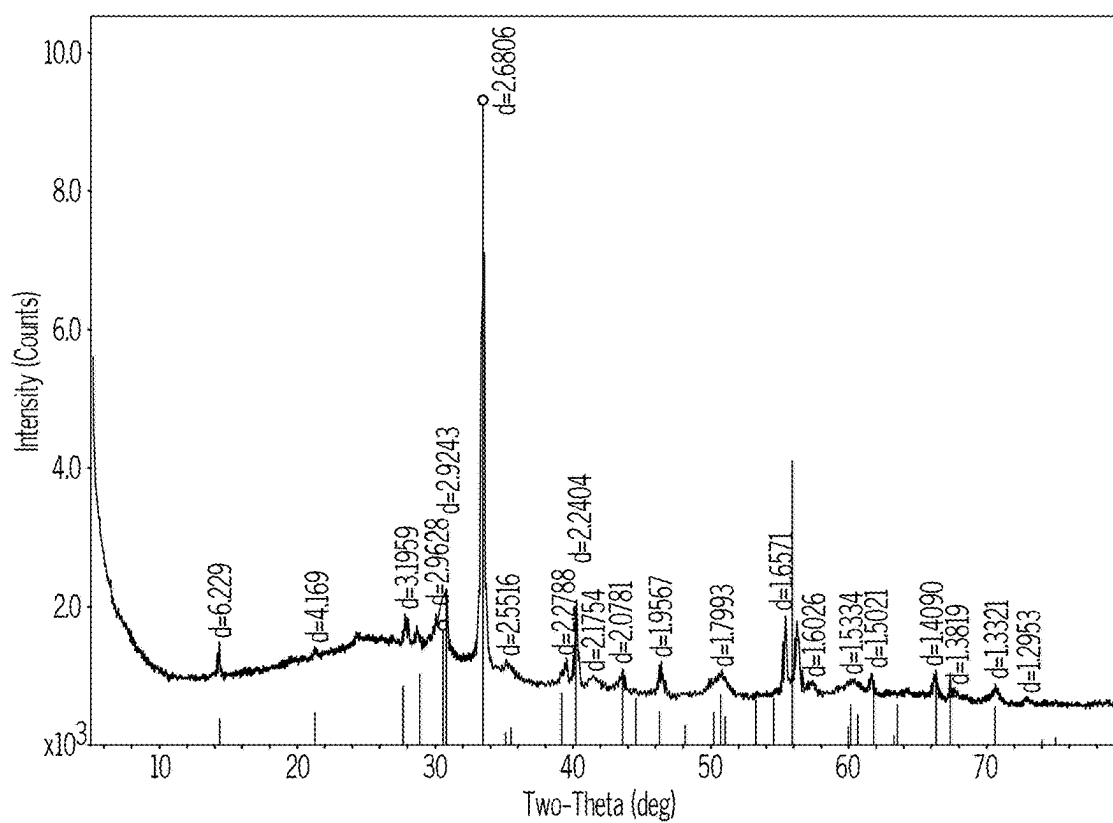


FIG. 22

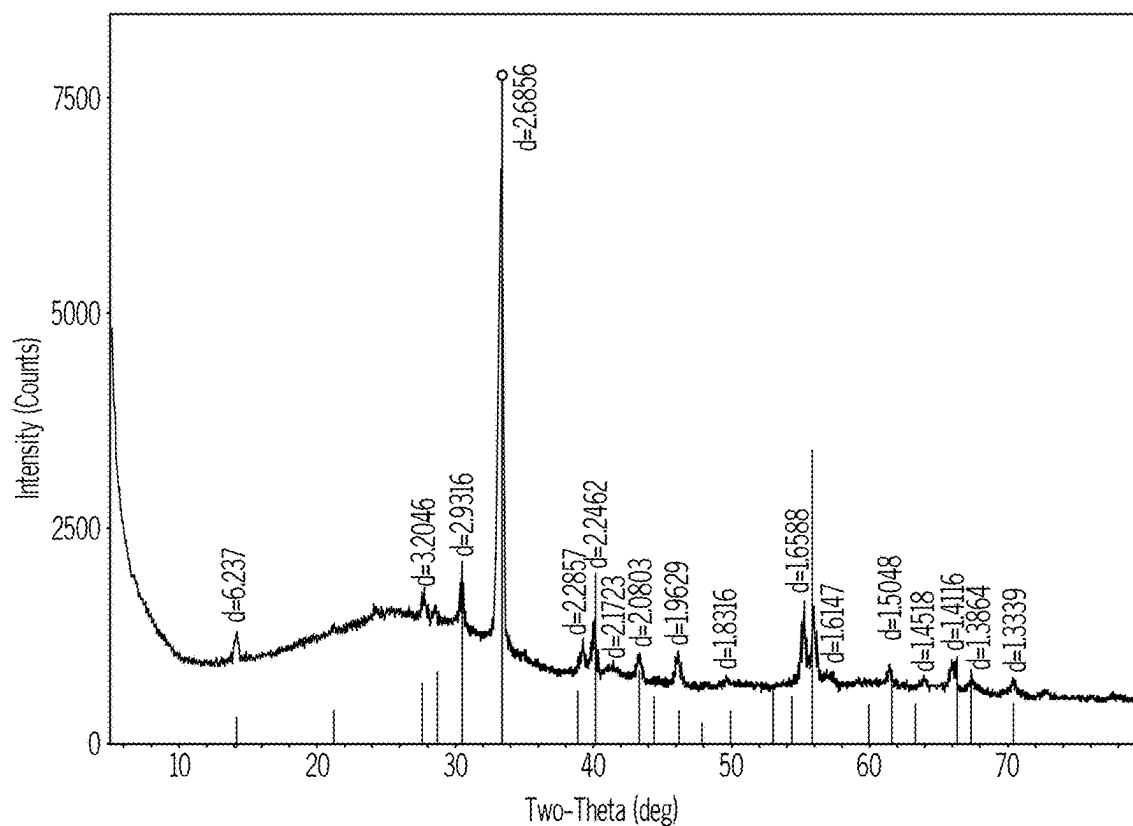


FIG. 23

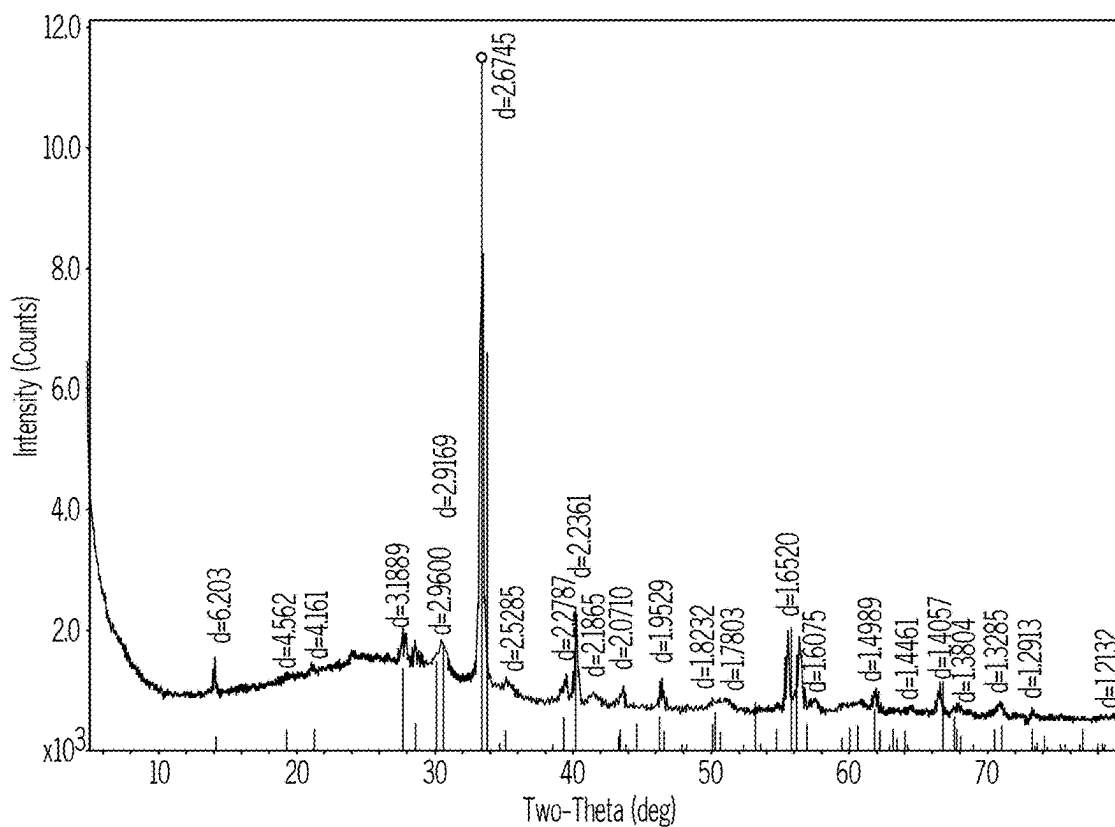


FIG. 24

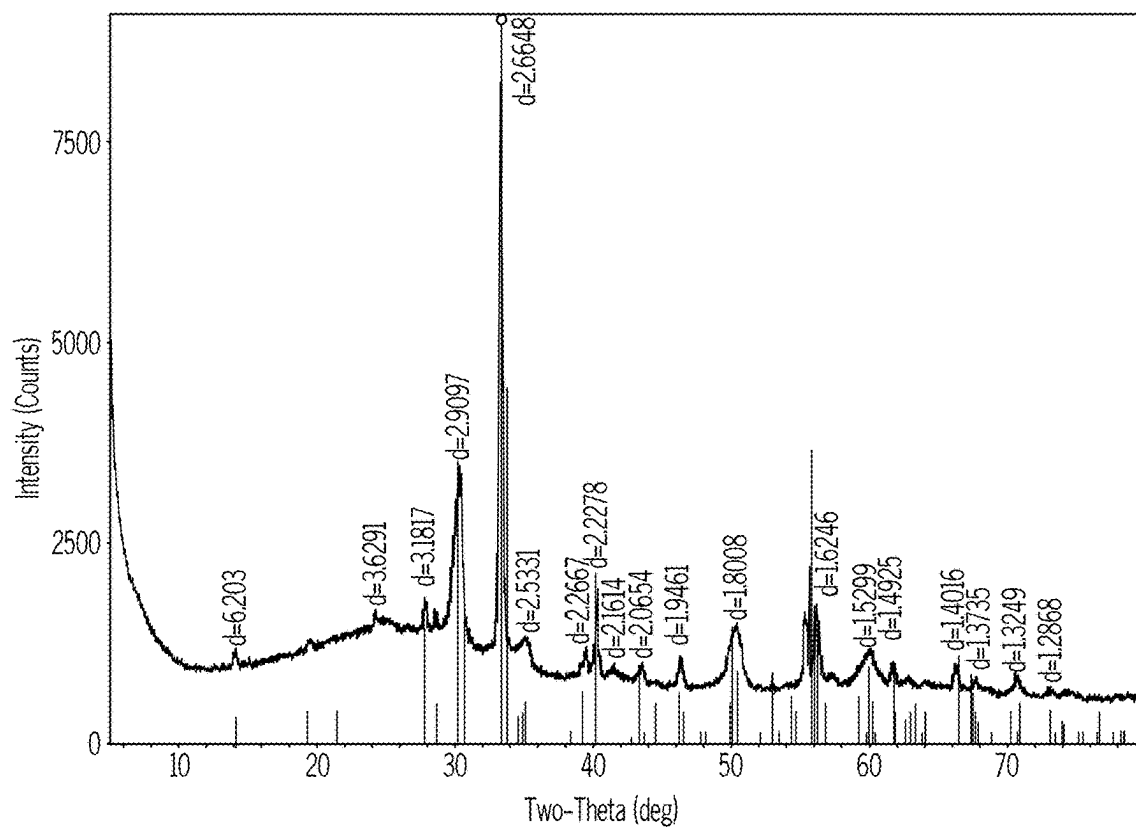


FIG. 25

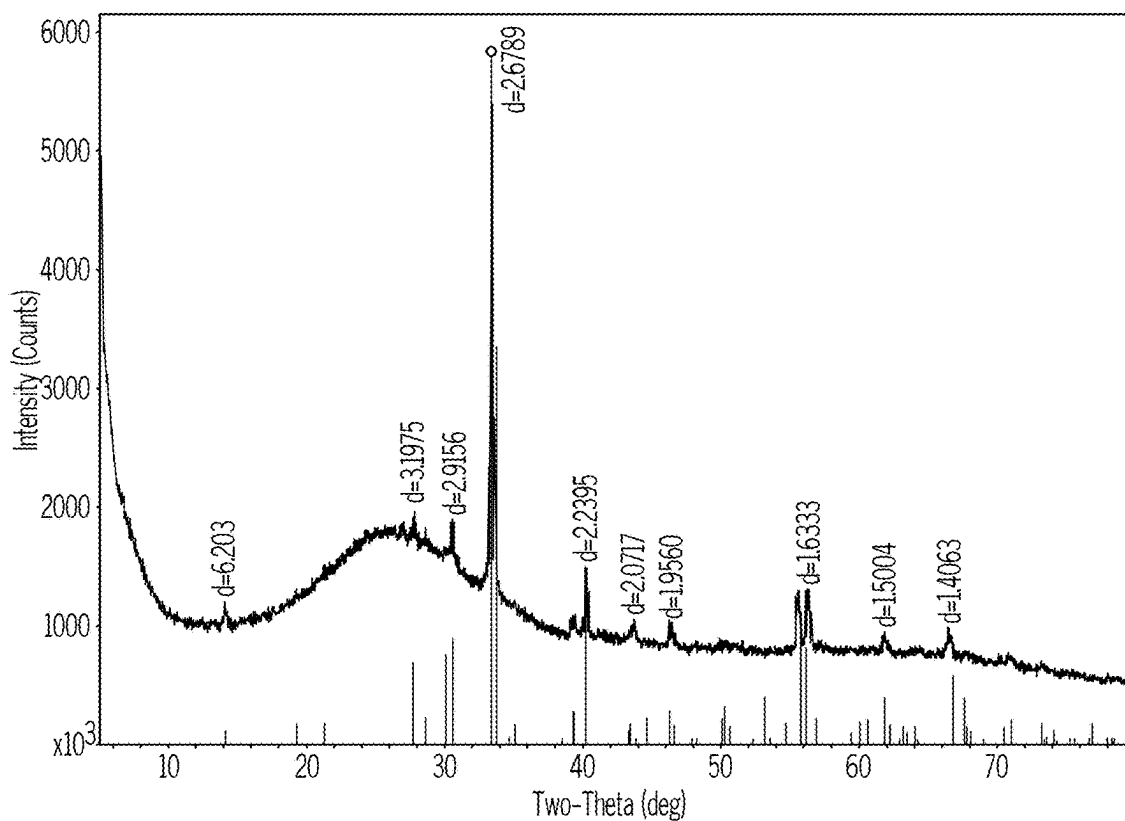


FIG. 26

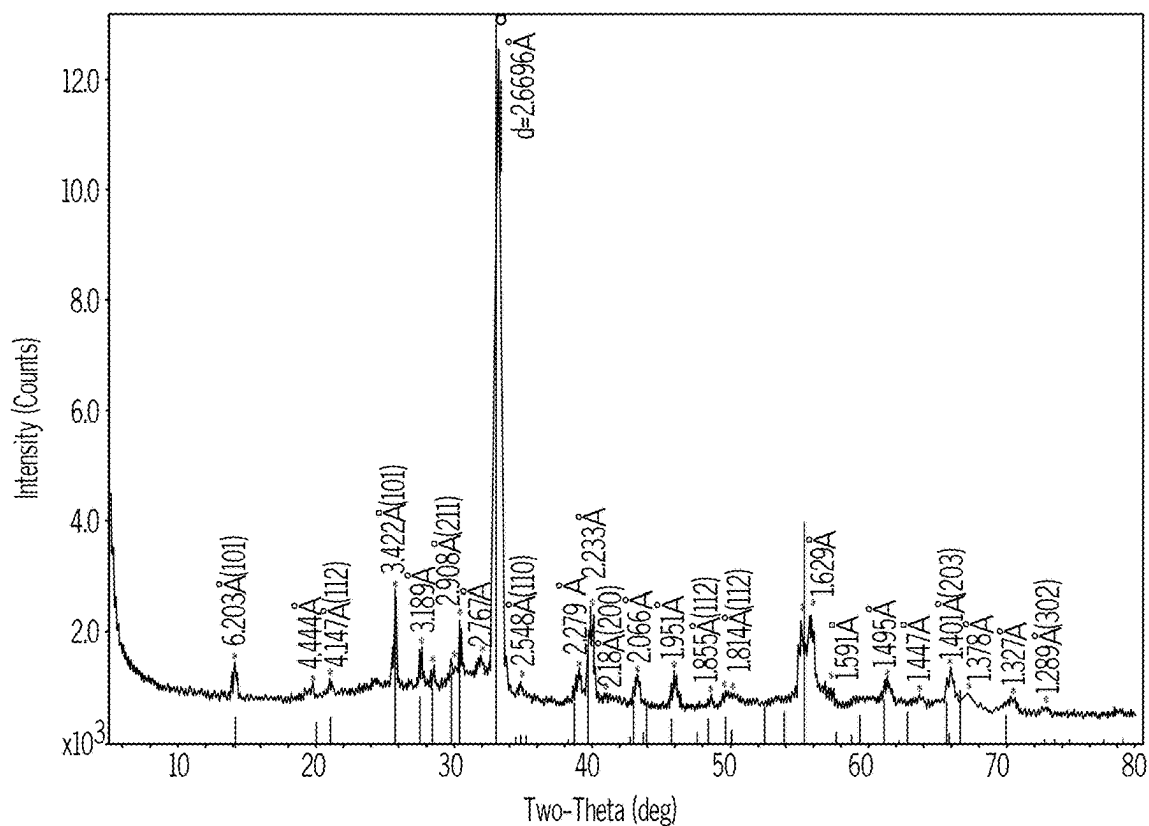


FIG. 27

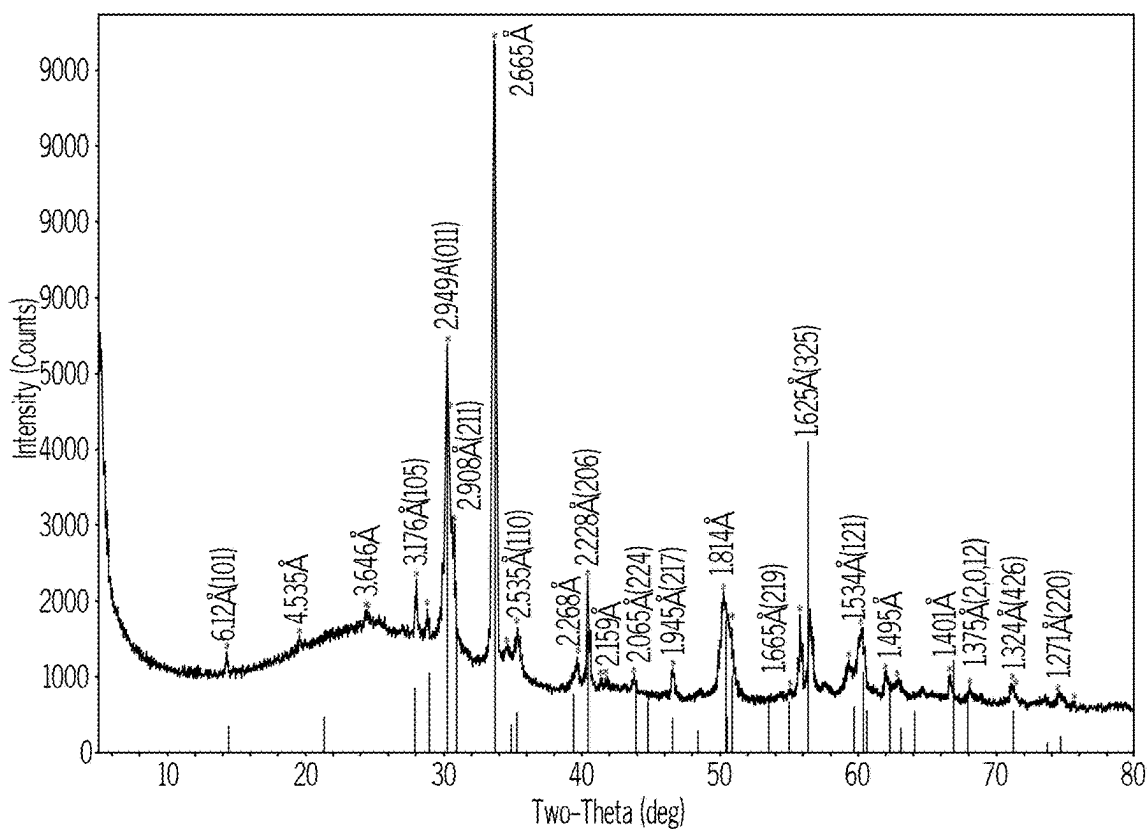


FIG. 28

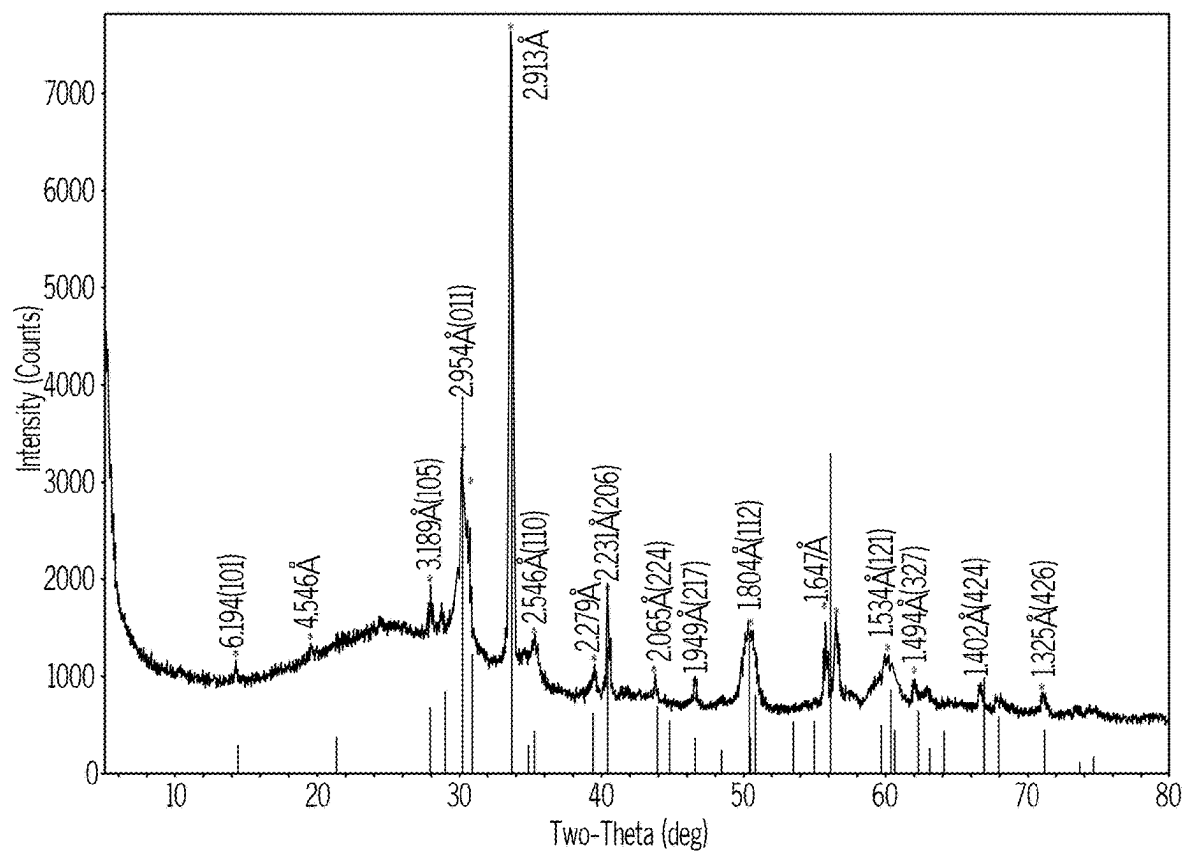


FIG. 29

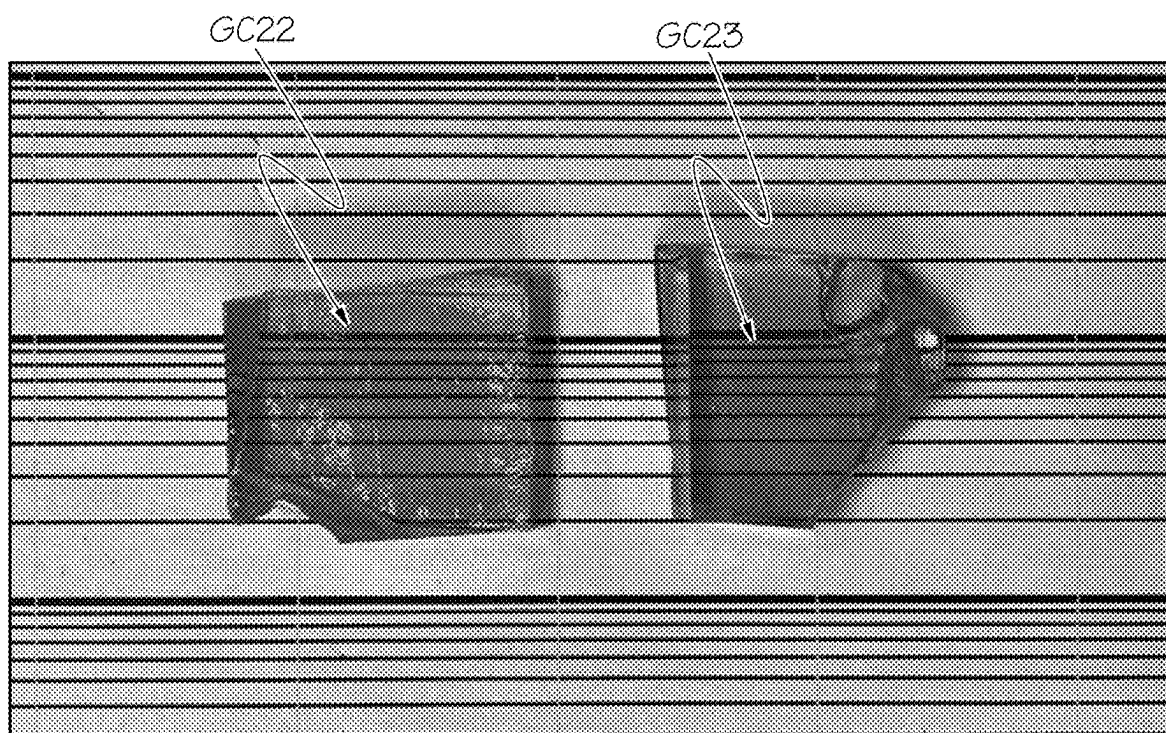


FIG. 30

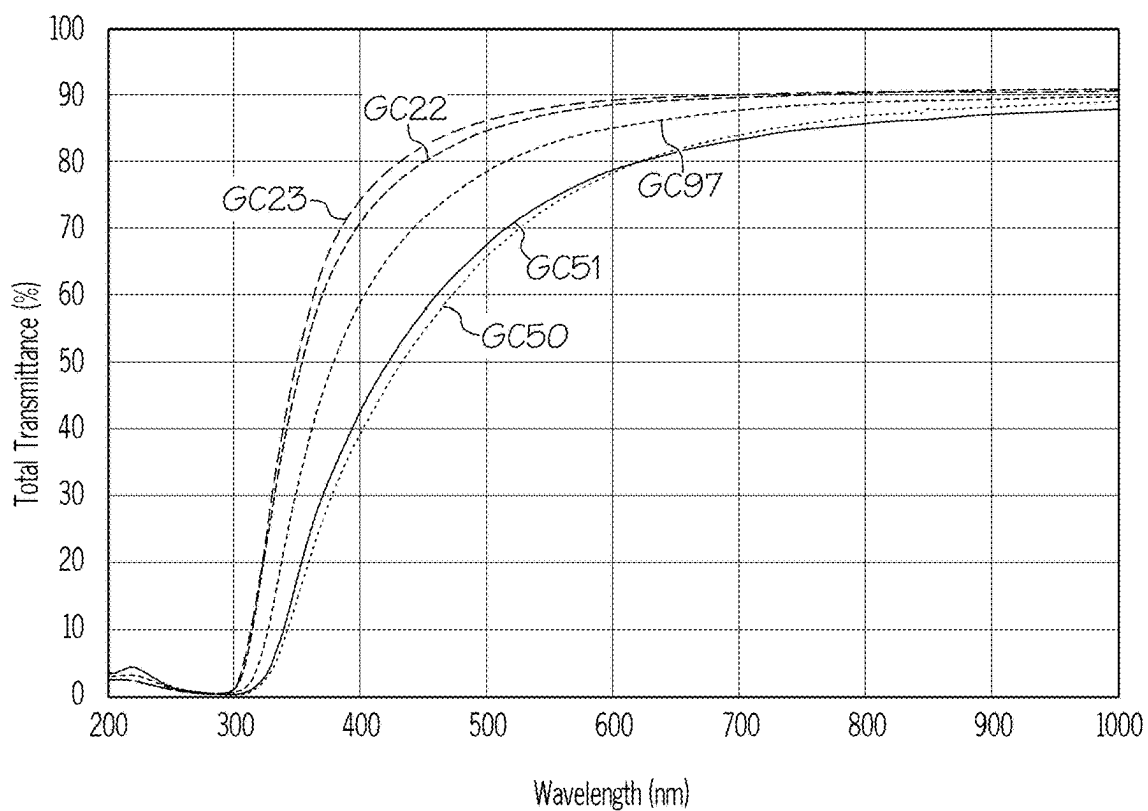


FIG. 31

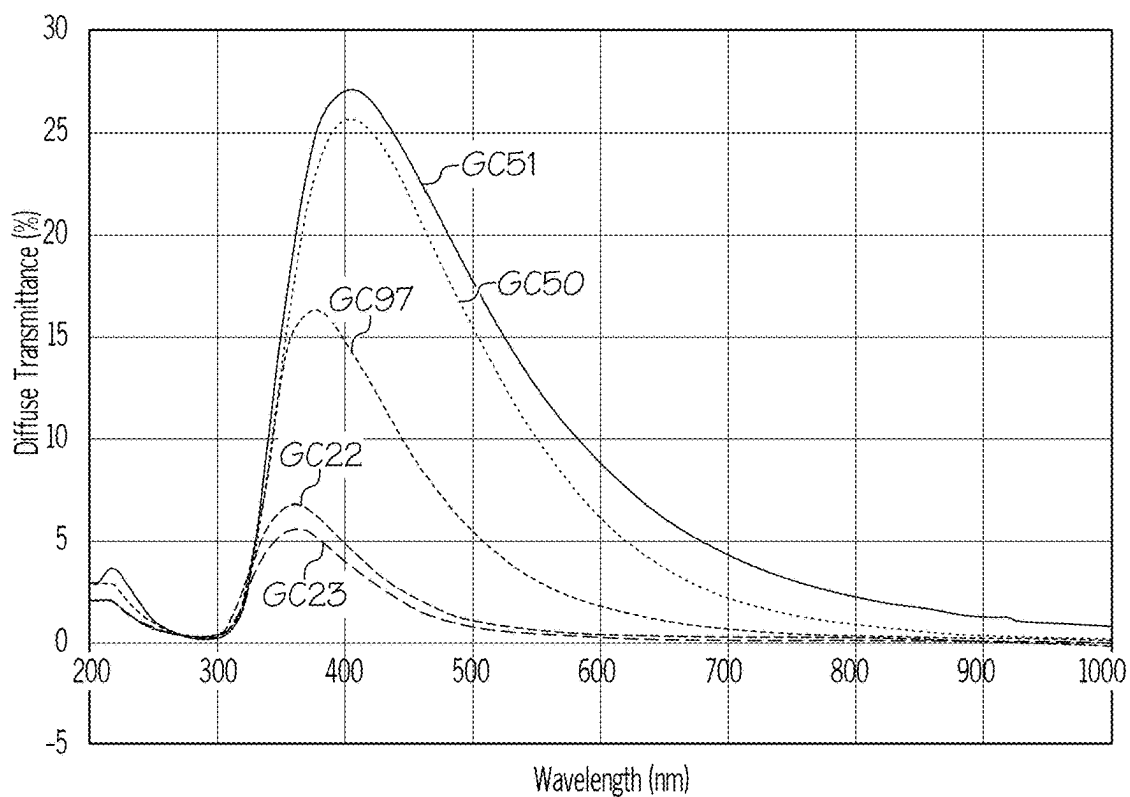


FIG. 32

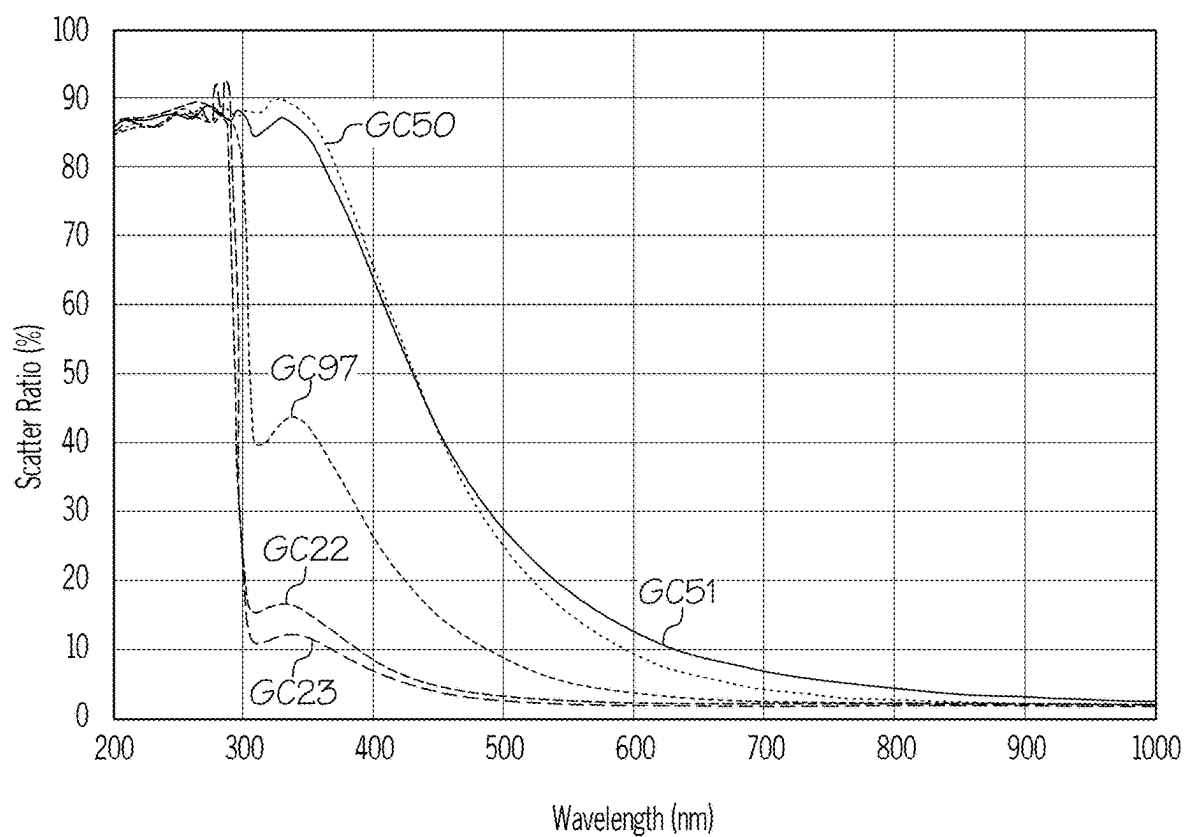


FIG. 33

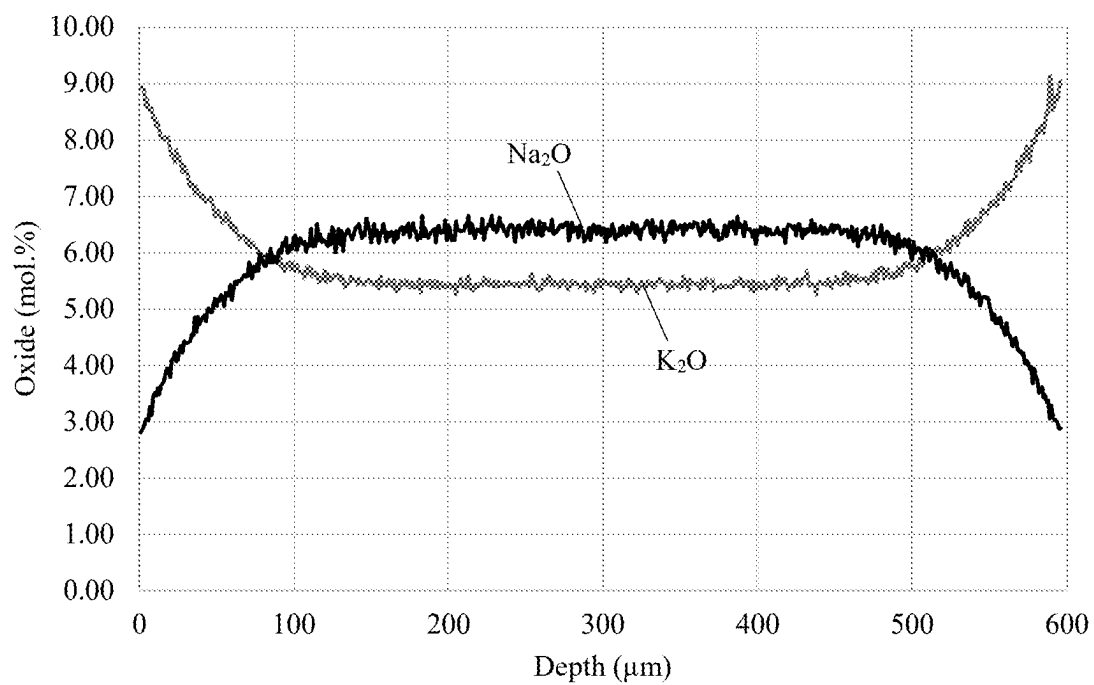


FIG. 34

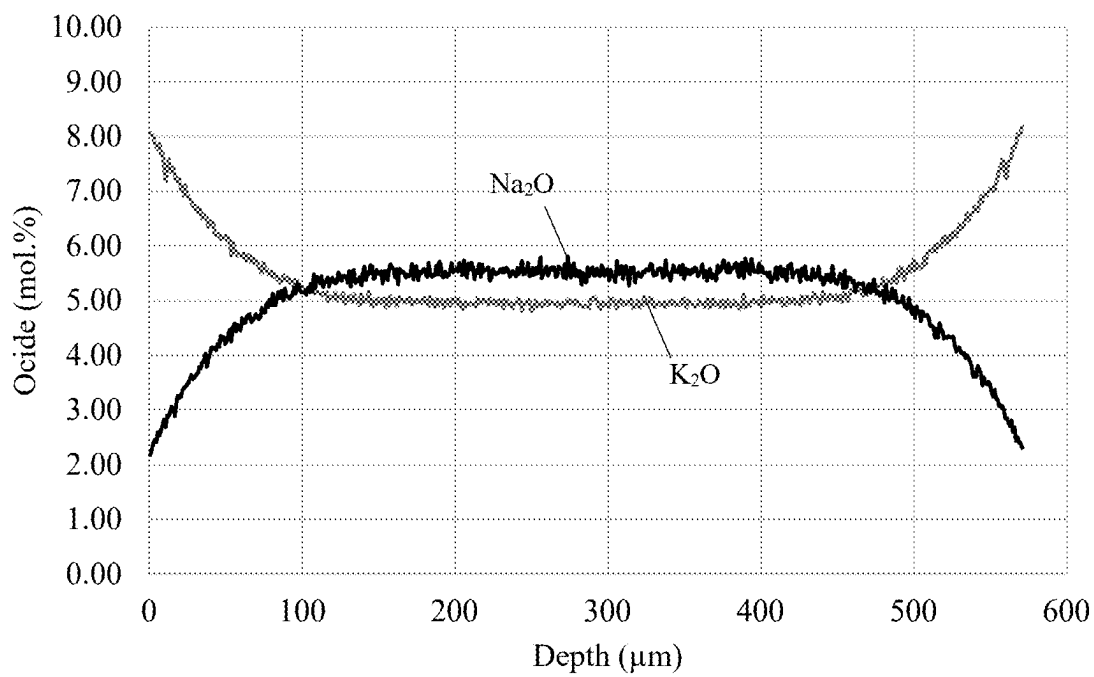


FIG. 35

**PRECURSOR GLASSES AND
GLASS-CERAMICS COMPRISING A
CRYSTALLINE PHASE HAVING A
JEFFBENITE CRYSTALLINE STRUCTURE**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This Application is a continuation of U.S. application Ser. No. 18/794,547 filed Aug. 5, 2024, which claims the benefit of priority of U.S. Application No. 63/532,167 filed Aug. 11, 2023, the content of which is relied upon and incorporated herein by reference in its entirety.

FIELD

[0002] The present specification relates to glasses and glass-ceramic articles made therefrom.

BACKGROUND

[0003] Glass articles, such as cover glasses, glass backplanes, housings, and the like, are employed in both consumer and commercial electronic devices, such as smart phones, tablets, portable media players, personal computers, and cameras. The mobile nature of these portable devices makes the devices and the glass articles included therein particularly vulnerable to accidental drops on hard surfaces, such as the ground. Moreover, glass articles, such as cover glasses, may include “touch” functionality, which necessitates that the glass article be contacted by various objects including a user’s fingers and/or stylus devices. Accordingly, the glass articles must be sufficiently robust to endure accidental dropping and regular contact without damage.

[0004] Accordingly, a need exists for alternative materials that have improved mechanical properties relative to glass.

SUMMARY

[0005] Features and advantages of the precursor glasses and glass-ceramics described herein will be set forth in the detailed description that follows, and in part will be readily apparent to those skilled in the art from that description or recognized by practicing the embodiments described herein, including the detailed description that follows, the claims, as well as the appended drawings.

[0006] The present Applicant has discovered that the jeffbenite crystalline structure associated with inclusions in “super-deep” diamonds may provide many useful advantages and properties if formed in glass as a crystalline phase of glass-ceramic, as explained herein. The Applicant believes that a crystalline phase having a jeffbenite crystalline structure has never before been grown in or otherwise formed in glass-ceramic. The Applicant further believes that a crystalline phase having a jeffbenite crystalline structure has never before been formed into or otherwise incorporated into glass-ceramic articles, such as sheets of glass-ceramic, glass-ceramic containers, windows, panels, housings, plates, counters, kitchenware, rods, fibers, or other such articles. Further, in contrast to jeffbenite in diamonds or in isolation, Applicant believes that crystalline phases having a jeffbenite crystalline structure have never been included in articles with isotropic material properties (e.g., properties such as the tensile strength, elasticity, and fracture toughness that remain the same when tested in different directions), as may be effectively achieved by inclusion of relatively small crystal grains, as disclosed herein, randomly-oriented and

homogenously distributed within the residual glass, to form glass-ceramic, or within another isotropic solid media (e.g., polymer). Applicant believes that a glass-ceramic with crystalline phase having a jeffbenite crystalline structure has never before been manufactured, even in nature; grown from precursor glasses; or formed at temperatures (e.g., <1600K, <1400K) and pressures (e.g., <12 GPa, such as <10 GPa, <8 GPa, <1 GPa, such as even 1 atm), as disclosed herein. Enabled by the present discoveries, crystals having the jeffbenite crystalline structure may be now produced in a single batch, in part because extreme temperatures and pressures, associated with formation super-deep within the Earth’s mantle, are unnecessary.

[0007] Furthermore, the glass-ceramics described herein may achieve excellent hardness and stiffness values, and therefore enable thin and light mobile phone and tablet display faces. The same features allow opaque or colored glass-ceramics for phone and tablet housings. Moreover, these glass-ceramics may be free of lithium and still amenable to strengthening by ion exchange, reducing need for this scarce resource.

[0008] Aspect 121 includes a glass-ceramic comprising a glass phase and one or more crystalline phases, wherein at least one of the one or more crystalline phases comprises jeffbenite.

[0009] Aspect 122 includes an article, comprising: a surface; and a body interior to the surface, wherein the body comprises the glass-ceramic of Aspect 121.

[0010] Aspect 123 includes the article of Aspect 122, wherein the surface is a first surface, the article further comprising a second surface facing away from the first surface with the body positioned between the first and second surfaces such that the article is a sheet.

[0011] Aspect 124 includes a glass-ceramic article comprising: a first surface; a second surface opposite the first surface; a perimeter defining a shape of the glass-ceramic article; and a phase assemblage comprising one or more crystalline phases and a glass phase, the one or more crystalline phases comprising a crystalline phase comprising a jeffbenite crystalline structure.

[0012] Aspect 125 includes the article of Aspect 124, wherein the crystalline phase comprising the jeffbenite crystalline structure is a primary crystalline phase.

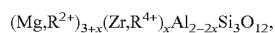
[0013] Aspect 126 includes the article of Aspect 124 or 125, wherein the phase assemblage comprises greater than or equal to 10 wt. % to less than or equal to 85 wt. % of the one or more crystalline phases and greater than or equal to 15 wt. % to less than or equal to 90 wt. % of the glass phase, by weight of the glass-ceramic article.

[0014] Aspect 127 includes the article of Aspects 124-126, wherein the amount of jeffbenite crystalline structure in the one or more crystalline phases, by total weight of the one or more crystalline phases, is greater than or equal to 90 wt. % to less than or equal to 100 wt. %.

[0015] Aspect 128 includes the article of Aspects 124-127, wherein at least some grains of the at least one crystalline phase comprising the jeffbenite crystalline structure have a largest dimension greater than or equal to 20 nm to less than or equal to 500 nm.

[0016] Aspect 129 includes the article of Aspects 124-128, wherein the glass-ceramic article is substantially free of lithium.

[0017] Aspect 130 includes the article of Aspects 124-129, wherein the crystalline phase comprising the jeffbenite crystalline structure has a composition according to the formula:



where R^{2+} represents one or more divalent metal cations, R^{4+} represents one or more tetravalent metal cations, and x is greater than or equal to 0 to less than or equal to 1.

[0018] Aspect 131 includes the article of Aspect 130, wherein R^{2+} is one or more divalent metal cations selected from Ca^{2+} , Mn^{2+} , Fe^{2+} , Zn^{2+} , and wherein R^{4+} is one or more tetravalent metal cations selected from Ti^{4+} , Sn^{4+} , Hf^{4+} .

[0019] Aspect 132 includes the article of Aspects 124-131, wherein the glass-ceramic article has an average transmittance of at least 75% for a light in a wavelength range from 400 nm to 800 nm at an article thickness of 0.6 mm.

[0020] Aspect 133 includes the article of Aspects 124-131, wherein the glass-ceramic article has an average transmittance in a range from 20% to less than 75% for light in a wavelength from 400 nm to 800 nm at an article thickness of 0.6 mm.

[0021] Aspect 134 includes the article of Aspects 124-131, wherein the glass-ceramic article has an average transmittance in a range of less than 20% for light in a wavelength range from 400 nm to 800 nm at an article thickness of 0.6 mm.

[0022] Aspect 135 includes a glass-ceramic article, comprising in mole percent (mol. %) on an oxide basis: greater than or equal to 35 mol. % to less than or equal to 65 mol. % SiO_2 ; greater than or equal to 2.5 mol. % to less than or equal to 20 mol. % Al_2O_3 ; greater than or equal to 7 mol. % to less than or equal to 65 mol. % MgO ; greater than or equal to 0 mol. % and less than or equal to 10 mol. % CaO ; greater than or equal to 0 mol. % and less than or equal to 5 mol. % B_2O_3 ; greater than or equal to 0 mol. % to less than or equal to 7 mol. % ZrO_2 ; greater than or equal to 0 mol. % to less than or equal to 15 mol. % Na_2O ; greater than or equal to 0 mol. % to less than or equal to 15 mol. % K_2O ; greater than or equal to 0 mol. % to less than or equal to 9 mol. % FeO ; greater than or equal to 0 mol. % to less than or equal to 10 mol. % MnO_2 ; and greater than or equal to 0 mol. % to less than or equal to 15 mol. % ZnO ; wherein the glass-ceramic comprises a phase assemblage comprising one or more crystalline phases and a glass phase, the one or more crystalline phases comprising a crystalline phase comprising a jeffbenite crystalline structure.

[0023] Aspect 136 includes the article of Aspect 135, further comprising less than or equal to 3 mol. % Li_2O .

[0024] Aspect 137 includes the article of Aspect 135, wherein the glass-ceramic article is substantially free of Li_2O .

[0025] Aspect 138 includes the article of Aspects 135-137, wherein the glass-ceramic article comprise greater than or equal to 1.5 mol. % to less than or equal to 3 mol. % ZrO_2 .

[0026] Aspect 139 includes the article of Aspects 135-138, wherein the glass-ceramic article comprises

greater than 0 mol. % to less than or equal to 12 mol. % HfO_2 , wherein a sum of ZrO_2 and HfO_2 is greater than 1 mol. %.

[0027] Aspect 140 includes the article of Aspects 135-139, wherein the glass-ceramic article comprises greater than 0 mol. % to less than or equal to 4 mol. % P_2O_5 .

[0028] Aspect 141 includes the article of Aspects 135-140, wherein the glass-ceramic article comprises from greater than 0 mol. % to less than or equal to 7 mol. % La_2O_3 .

[0029] Aspect 142 includes the article of Aspects 135-141, wherein Na_2O (mol. %)+ K_2O (mol. %) is greater than or equal to 2 mol. % to less than or equal to 15 mol. %.

[0030] Aspect 143 includes the article of Aspects 135-142, wherein Na_2O (mol. %)/(Na_2O (mol. %)+ K_2O (mol. %)) is greater than or equal to 0.2.

[0031] Aspect 144 includes the article of Aspects 135-143, further comprising greater than or equal to 0.3 mol. % to less than or equal to 7 mol. % TiO_2 .

[0032] Aspect 145 includes the article of Aspects 135-144, wherein ZrO_2 (mol. %)+ TiO_2 (mol. %) is greater than or equal to 2 mol. %.

[0033] Aspect 146 includes the article of Aspects 135-145, wherein ZrO_2 (mol. %)/(ZrO_2 (mol. %)+ TiO_2 (mol. %)) is greater than or equal to 0.3.

[0034] Aspect 147 includes the article of Aspects 135-146, comprising greater than or equal to 1 mol. % to less than or equal to 12 mol. % ZnO .

[0035] Aspect 148 includes the article of Aspects 135-147, further comprising greater than or equal to 1 mol. % to less than or equal to 8 mol. % BaO .

[0036] Aspect 149 includes the article of Aspects 135-148, further comprising greater than or equal to 0.2 mol. % to less than or equal to 1.7 mol. % of at least one of CaO and SrO .

[0037] Aspect 150 includes the article of Aspects 135-149, further comprising greater than or equal to 0 mol. % and less than or equal to 0.2 mol. % SnO_2 .

[0038] Aspect 151 includes the article of Aspects 135-150, further comprising greater than or equal to 1×10^{-6} mol. % to less than or equal to 10 mol. % of a colorant, the colorant comprising at least one of Au , Ag , Cr_2O_3 , CuO , NiO , CoO , Co_3O_4 , TiO_2 , Cr_2O_3 , V_2O_5 , MnO , and CeO_2 .

[0039] Aspect 152 includes the article of Aspect 151, wherein the glass-ceramic article comprises: a L^* value greater than or equal to 0 and less than or equal to 100; an a^* value greater than or equal to -128 and less than or equal to 127; and a b^* value greater than or equal to -128 and less than or equal to 127.

[0040] Aspect 153 includes the article of Aspects 135-152, wherein the glass-ceramic article has an elastic modulus greater than or equal to 50 GPa and less than or equal to 200 GPa.

[0041] Aspect 154 includes the article of Aspects 135-153, wherein the glass-ceramic article has a Vickers hardness greater than or equal to 600 kgf/mm^2 to less than or equal to 1400 kgf/mm^2 .

[0042] Aspect 155 includes the article of Aspects 135-154, wherein the glass-ceramic article as an ion exchanged glass-ceramic article.

[0043] Aspect 156 includes the article of Aspect 155, wherein the glass-ceramic article has a diffusivity during a Na^+ for K^+ ion exchange process in a 450°C . salt bath greater than or equal to $200\ \mu\text{m}^2/\text{hour}$.

[0044] Aspect 157 includes the article of Aspect 155 or Aspect 156, wherein the glass-ceramic composition may have a diffusivity during a Na^+ for K^+ ion exchange process in a 550°C . salt bath greater than or equal to $1500\ \mu\text{m}^2/\text{hour}$.

[0045] Aspect 158 includes the article of Aspects 155-157, wherein the ion exchanged glass-ceramic article has a depth of compression of about $30\ \mu\text{m}$ or greater.

[0046] Aspect 159 includes the article of Aspects 155-158, wherein the ion exchanged glass-ceramic article has a depth of compression greater than or equal to 3% of the thickness of the ion exchanged glass-ceramic article.

[0047] Aspect 160 includes the article of Aspects 155-159, wherein the ion exchanged glass-ceramic article has a central tension greater than or equal to 10 MPa and less than or equal to 200 MPa.

[0048] Aspect 161 includes the article of Aspects 155-160, wherein the ion exchanged glass-ceramic article has a surface compressive stress greater than or equal to 100 MPa and less than or equal to 1 GPa.

[0049] Aspect 162 includes the article of any of the preceding aspects, further comprising a limitation, attribute, or feature disclosed herein but not expressly or inherently disclosed in International Application No. PCT/US2023/012905 filed Feb. 13, 2023, which is incorporated by reference herein in its entirety.

[0050] Aspect 163 includes the article of any of the preceding aspects, wherein the article is a sheet having a total thickness variation greater than $0.1\ \mu\text{m}$ and less than 1 mm.

[0051] Aspect 164 includes the article of any of the preceding aspects, wherein the article has a dulled corner or edge.

[0052] Aspect 165 includes the article of any of the preceding aspects, wherein the article is coated with a coating other than conductive carbon.

[0053] Aspect 166 includes the article of any of the preceding aspects, wherein the article is coated and the coating has a thickness greater than 20 nm and less than $50\ \mu\text{m}$.

[0054] Aspect 167 includes a device including the article of any of the preceding aspects adhered thereto.

[0055] Aspect 168 includes a device including the article of any of the preceding aspects in bending.

[0056] Aspect 169 includes a device including the article of any of the preceding aspects in strain.

[0057] Aspect 170 includes a device including the article of any of the preceding aspects within 10 cm of electrical components of the device.

[0058] Aspect 171 includes a device including the article of any of the preceding aspects and an antenna.

[0059] It is to be understood that both the foregoing general description and the following detailed description describe various embodiments and are intended to provide an overview or framework for understanding the nature and character of the claimed subject matter. The accompanying drawings are included to provide a further understanding of the various embodiments, and are incorporated into and constitute a part of this specification. The drawings illustrate

the various embodiments described herein, and together with the description serve to explain the principles and operations of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0060] FIG. 1 is a top view of an electronic device comprising a glass-ceramic article;

[0061] FIG. 2 is a perspective view of an electronic device comprising a glass-ceramic article;

[0062] FIG. 3 is an SEM micrograph of the crystalline structure of a glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure;

[0063] FIG. 4 is an SEM micrograph of the crystalline structure of a glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure;

[0064] FIG. 5 is a high temperature X-ray diffraction two-theta plot indicating the ranges of stability of the various crystalline phases in glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure and a tetragonal zirconia crystalline phase as a function of temperature;

[0065] FIG. 6A is an SEM micrograph of the crystalline structure of a glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure;

[0066] FIG. 6B is an SEM micrograph of the crystalline structure of a glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure;

[0067] FIG. 6C is an SEM micrograph of the crystalline structure of a glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure;

[0068] FIG. 6D is an SEM micrograph of the crystalline structure of a glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure;

[0069] FIG. 6E is an SEM micrograph of the crystalline structure of a glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure;

[0070] FIG. 6F is an SEM micrograph of the crystalline structure of a glass-ceramic comprising a crystalline phase having a jeffbenite crystalline structure;

[0071] FIG. 7 is an X-ray diffraction two-theta plot indicating a crystalline phase having a jeffbenite crystalline structure in an example glass-ceramic article;

[0072] FIG. 8A schematically depicts a glass-ceramic article;

[0073] FIG. 8B schematically depicts a cross section of the glass-ceramic article;

[0074] FIG. 9 schematically depicts a glass-ceramic article;

[0075] FIG. 10 depicts the total transmittance of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure as a function of wavelength;

[0076] FIG. 11 depicts the diffuse transmittance of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure as a function of wavelength;

[0077] FIG. 12 depicts the axial transmittance of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure as a function of wavelength;

[0078] FIG. 13 depicts the scatter ratio of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure as a function of wavelength;

[0079] FIG. 14 depicts a window from MDI Jade software used to characterize glass-ceramic articles;

[0080] FIG. 15 depicts a window from MDI Jade software used to characterize glass-ceramic articles;

[0081] FIG. 16 depicts a window from MDI Jade software used to characterize glass-ceramic articles;

[0082] FIG. 17 depicts a window from Bruker Topas software used to characterize glass-ceramic articles;

[0083] FIG. 18 depicts a window from Bruker Topas software used to characterize glass-ceramic articles;

[0084] FIG. 19 depicts a window from Bruker Topas software used to characterize glass-ceramic articles;

[0085] FIG. 20 depicts a window from Bruker Topas software used to characterize glass-ceramic articles;

[0086] FIG. 21 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0087] FIG. 22 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0088] FIG. 23 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0089] FIG. 24 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0090] FIG. 25 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0091] FIG. 26 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0092] FIG. 27 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0093] FIG. 28 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0094] FIG. 29 depicts an XRD spectrum of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure;

[0095] FIG. 30 is a photograph of transparent glass-ceramic articles comprising a crystalline phase having a jeffbenite crystalline structure;

[0096] FIG. 31 depicts the total transmittance of glass-ceramic articles comprising crystalline phases having a jeffbenite crystalline structure as a function of wavelength;

[0097] FIG. 32 depicts the diffuse transmittance of glass-ceramic articles comprising crystalline phases having a jeffbenite crystalline structure as a function of wavelength;

[0098] FIG. 33 depicts the scatter ratio of glass-ceramic articles comprising crystalline phases having a jeffbenite crystalline structure as a function of wavelength;

[0099] FIG. 34 depicts the mol. % of the oxide of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure as a function of depth; and

[0100] FIG. 35 depicts the mol. % of the oxide of a glass-ceramic article comprising a crystalline phase having a jeffbenite crystalline structure as a function of depth.

DETAILED DESCRIPTION

[0101] Reference will now be made in detail to various embodiments of precursor glasses and glass-ceramic articles made therefrom. According to embodiments, a glass-ceramic article includes a first surface, a second surface opposite the first surface, and a perimeter defining a shape of the glass-ceramic article. The glass-ceramic article may further include a phase assemblage comprising one or more

crystalline phases and a glass phase, the one or more crystalline phases comprising a crystalline phase comprising a jeffbenite crystalline structure. Various embodiments of precursor glasses, glass-ceramic articles made therefrom, and methods of making glass-ceramic articles will be referred to herein with specific reference to the appended drawings.

[0102] Ranges may be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0103] Directional terms as used herein—for example up, down, right, left, front, back, top, bottom—are made only with reference to the figures as drawn and are not intended to imply absolute orientation.

[0104] Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order, nor that with any apparatus specific orientations be required. Accordingly, where a method claim does not actually recite an order to be followed by its steps, or that any apparatus claim does not actually recite an order or orientation to individual components, or it is not otherwise specifically stated in the claims or description that the steps are to be limited to a specific order, or that a specific order or orientation to components of an apparatus is not recited, it is in no way intended that an order or orientation be inferred, in any respect. This holds for any possible non-express basis for interpretation, including: matters of logic with respect to arrangement of steps, operational flow, order of components, or orientation of components; plain meaning derived from grammatical organization or punctuation, and; the number or type of embodiments described in the specification.

[0105] As used herein, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a” component includes aspects having two or more such components, unless the context clearly indicates otherwise.

[0106] The term “substantially free,” when used to describe the concentration and/or absence of a particular constituent component in a precursor glass or glass-ceramic composition, means that the constituent component is not intentionally added to the precursor glass or glass-ceramic composition. However, the precursor glass or glass-ceramic composition may contain traces of the constituent component as a contaminant or tramp in amounts of less than 0.05 mol. %.

[0107] In the embodiments of the precursor glass or glass-ceramic compositions described herein, the concentrations of constituent components (e.g., SiO₂, Al₂O₃, and the like) are specified in mole percent (mol. %) on an oxide basis, unless otherwise specified.

[0108] Transmittance data (total transmittance) is measured with a Lambda 950 UV/Vis Spectrophotometer manufactured by PerkinElmer Inc. (Waltham, Massachusetts USA). The Lambda 950 apparatus was fitted with a 150 mm integrating sphere. Data was collected using an open beam baseline and a Spectralon® reference reflectance disk. For

total transmittance (Total Tx), the sample is fixed at the integrating sphere entry point. For diffuse transmittance (Diffuse Tx), the Spectralon® reference reflectance disk over the sphere exit port is removed to allow on-axis light to exit the sphere and enter a light trap. A zero offset measurement is made, with no sample, of the diffuse portion to determine efficiency of the light trap. To correct diffuse transmittance measurements, the zero offset contribution is subtracted from the sample measurement using the equation: $\text{Diffuse Tx} = \text{Diffuse}_{\text{Measured}} - (\text{Zero Offset} * (\text{Total Tx} / 100))$. The scatter ratio is measured for all wavelengths as: $(\% \text{ Diffuse Tx} / \% \text{ Total Tx})$.

[0109] The term “transparent,” when used to describe an article herein, refers to an article that has an average transmittance of at least 75% for a light in a wavelength range from 400 nm to 800 nm (inclusive of endpoints) at an article thickness of 0.6 mm.

[0110] The term “translucent,” unless otherwise specified such as in the claims, when used to describe an article herein, refers to an article that has an average transmittance in a range from 20% to less than 75% for light in a wavelength range from 400 nm to 800 nm (inclusive of endpoints) at an article thickness of 0.6 mm.

[0111] The term “opaque,” when used to describe a glass-ceramic article formed of a glass-ceramic composition herein, means that the glass-ceramic composition has an average transmittance less than 20% when measured at normal incidence for light in a wavelength range from 400 nm to 800 nm (inclusive of endpoints) at an article thickness of 0.6 mm.

[0112] The term “CIELAB color space,” as used herein, refers to a color space defined by the International Commission on Illumination (CIE) in 1976. It expresses color as three values: L* for the lightness from black (0) to white (100), a* from green (−) to red (+), and b* from blue (−) to yellow (+). Unless otherwise specified, the L*, a*, and b* values are indicated for article thicknesses of 0.4 mm to 5 mm (inclusive of endpoints) in the thickness direction of the sample under F2 illumination and a 10° standard observer angle. Unless otherwise specified, this means that each thickness within the range of thicknesses has L*, a*, and b* coordinates falling within the specified range(s) for L*, a*, and b* coordinates. For example, a colored glass article having an L* value within the range from 55 to 96.5 means that each thickness within the range of 0.4 mm to 5 mm (e.g., 0.6 mm, 0.9 mm, 2 mm, etc.) has an L* in the range of 55 to 96.5.

[0113] The dimensions of the grains of a crystalline phase or phases of the glass-ceramics described herein are measured using scanning electron microscopy.

[0114] The term “melting point,” as used herein, refers to the temperature at which the viscosity of the precursor glass or glass-ceramic composition is 200 poise (20 Pa*s).

[0115] The term “softening point,” as used herein, refers to the temperature at which the viscosity of the precursor glass or glass-ceramic composition is $1 \times 10^{7.6}$ poise ($1 \times 10^{6.6}$ Pa*s). The softening point is measured according to the parallel plate viscosity method, which measures the viscosity of inorganic glass from 10^7 to 10^9 poise (10^6 to 10^8 Pa*s) as a function of temperature, similar to ASTM C1351M.

[0116] The term “liquidus viscosity,” as used herein, refers to the viscosity of the glass-ceramic at the onset of devitrification (i.e., at the liquidus temperature as determined with the gradient furnace method according to ASTM C829-81).

[0117] The elastic modulus (also referred to as Young’s modulus) of the glass-based article is provided in units of gigapascals (GPa). The elastic modulus of the glass is determined by resonant ultrasound spectroscopy on bulk samples of each glass-based article in accordance with ASTM C623.

[0118] Vickers hardness may be measured using ASTM C1326 and C1327 (and its progeny, all herein incorporated by reference) “Standard Test Methods for Vickers Indentation Hardness of Advanced Ceramics,” ASTM International, Conshohocken, PA, US. In some embodiments, the glass-ceramics exhibit such Vickers indentation crack initiation load values after being chemically strengthened via ion exchange.

[0119] The fracture toughness may be measured using a chevron notch short beam, according to ASTM C1421-10, “Standard Test Methods for Determination of Fracture Toughness of Advanced Ceramics at Ambient Temperature” prior to ion-exchange strengthening of the glass-ceramic.

[0120] Compressive stress (including surface compressive stress) is measured with a surface stress meter (FSM) such as commercially available instruments such as the FSM-6000, manufactured by Orihara Industrial Co., Ltd. (Japan). Surface stress measurements rely upon the measurement of the stress optical coefficient (SOC), which is related to the birefringence of the glass-ceramic. SOC, in turn, is measured according to Procedure C (Glass Disc Method) described in ASTM standard C770-16, entitled “Standard Test Method for Measurement of Glass Stress-Optical Coefficient,” the contents of which are incorporated herein by reference in their entirety. Depth of compression (DOC) is also measured with the FSM. The maximum central tension (CT) values are measured using a scattered light polariscope (SCALP) technique known in the art.

[0121] The phrase “depth of compression” and “DOC” refer to the position in the glass-ceramic where compressive stress transitions to tensile stress.

[0122] The phrase “glass precursor” or “precursor glass,” as used herein, refers to a glass or glass article containing one or more nucleating agents and/or nucleation sites (e.g., within a body of the material, which may be homogeneously distributed therein and throughout the body), which, upon thermal treatment, at least in part causes (e.g., facilitates) the nucleation of at least one crystalline phase in the glass.

[0123] The phrase “glass-ceramic,” as used herein, refers to a material or article formed from a precursor glass material following nucleation of at least one crystalline phase in the precursor glass such that the glass-ceramic includes a residual glass phase and at least one crystalline phase (see generally FIG. 9 and related disclosure).

[0124] The phrase “primary crystalline phase,” as used herein, refers to a crystalline phase present in the glass-ceramic in an amount (in wt. % of the glass-ceramic) greater than the amount (in wt. % of the glass-ceramic) of any other individual crystalline phases present in the glass-ceramic. For example, if a glass-ceramic comprises crystalline phases A, B, and C and crystalline phase A is the primary crystalline phase, the amount of crystalline phase A in the glass-ceramic is greater than the amount of crystalline phase B in the glass-ceramic and greater than the amount of crystalline phase C in the glass-ceramic.

[0125] Articles formed from glass-ceramics generally have improved fracture toughness relative to articles formed from glass. This improvement may be due to the presence of

crystalline grains in the glass-ceramics, which may impede crack growth. The fracture toughness of glass-ceramics may be improved by decreasing the number of grains per unit volume of the glass-ceramic—that is by increasing the size of the grains of the glass-ceramic. However, the transparency or optical transmission of glass-ceramics may decrease with increasing grain size. In particular, the visible transparency of glass-ceramics may be significantly reduced when the size of the grains is larger than 300 nm. Thus, some glass-ceramics may have relatively good mechanical properties (such as fracture toughness) and relatively poor optical characteristics (such as optical transparency or optical transmittance) or relatively poor mechanical properties and relatively good optical characteristics, but not both relatively good mechanical properties and relatively good optical characteristics.

[0126] In addition, some glass-ceramics may be strengthened by ion exchange processes in which smaller alkali metal ions in the glass-ceramic are exchanged for larger alkali metal ions from, for example, a bath of molten alkali metal salts. As an example, a lithium-containing glass-ceramic may be strengthened by ion exchange by placing the glass-ceramic in a bath of molten alkali metal salts, such as salts of sodium and/or potassium, thereby facilitating the exchange of lithium ions in the glass-ceramic with sodium and/or potassium ions in the bath. However, the recent demand for lithium for use in various applications has increased the cost of lithium raw materials and reduced availability, thereby increasing the overall cost for producing glass-ceramics containing lithium which may be strengthened by ion exchange. The glass-ceramics described herein do not require lithium to facilitate desirable ion exchange performance.

[0127] Disclosed herein are precursor glasses and glass-ceramics formed therefrom which mitigate the aforementioned problems.

[0128] Referring now to FIG. 9 by way of example, a glass-ceramic article **810** is schematically depicted according to one or more embodiments shown and described herein. As noted herein, the phrase “glass-ceramic” refers to a material or article formed from a precursor glass material following nucleation of at least one crystalline phase in the precursor glass such that the glass-ceramic includes a residual glass phase and at least one crystalline phase. Thus, the glass-ceramic article **810** includes both a glass phase **812** (e.g., amorphous glass; single-phase glass or multi-phase glass) and polycrystalline ceramic phase (e.g., grains of the primary crystalline phase **814**, optionally with grains of accessory crystalline phase(s) **816**). The glass phase **812** may be referred to as “residual glass” or a “residual glass phase.” Conceivably, the grains of the crystalline phases **814**, **816** may be unevenly distributed or directionally grown, such as by localizing heat via laser, or by positioning and/or orienting nucleation sites to guide growth of the grains of the crystalline phases **814**, **816**, and contemplated embodiments include such properties; however, generally in the glass-ceramics disclosed herein, the grains of the crystalline phases **814**, **816** may be uniformly or homogeneously distributed and randomly oriented within the glass phase **812**, such as throughout some, most or all of the glass-ceramic article **810**. Further, once nucleated from the precursor glass, the grains of the crystalline phases **814**, **816** may be grown to contact one another, overlap one another, interlock with one another, and/or fill more of the volume of

the glass-ceramic article **810**. Growing the grains of the crystalline phases **814**, **816** to different sizes may influence properties of the glass-ceramic article **810**, and such properties may be isotropic with grains of the crystalline phases **814**, **816** uniformly, homogeneously positioned, and randomly oriented when the grains of the crystalline phases **814**, **816** have grain sizes and the articles have sizes, as disclosed herein. After nucleation and growth of the grains of the crystalline phases **814**, **816**, the glass phase **812** surrounds (e.g., contacts, envelops, encapsulates) individual or clusters of the grains of the crystalline phases **814**, **816**. The glass-ceramic article **810** may have more than one glass phase **812** and/or more than one crystalline phase, as shown with crystalline phases **814**, **816**.

[0129] Embodiments of the glass-ceramics described herein, such as the embodiment depicted in FIG. 9, have a phase assemblage comprising one or more crystalline phases and a glass phase. At least one of the crystalline phases comprises a jeffbenite crystalline structure. A crystalline phase having a jeffbenite crystalline structure refers to a crystalline phase identified as jeffbenite by X-ray diffraction (XRD) analysis. For example, XRD data (such as an XRD spectrum) collected from a sample of glass-ceramic material, along with general compositional information about the composition of the sample (such as the batch composition from which the sample was prepared) may be input into MDI Jade powder XRD Analysis software from Materials Data Inc. The software utilizes the input information, along with the International Center For Diffraction Data's Powder Diffraction File version 4 database (ICDD PDF-4 database), to identify the crystalline phases in the sample based on the compositional information of the sample and the crystalline structure of the phases as determined from the XRD data. In embodiments, the phase assemblage of the glass-ceramic may have at least one crystalline phase comprising a jeffbenite crystalline structure as determined by this methodology.

[0130] Based on the foregoing, unless otherwise specified or further clarified herein, such as in the claims or elsewhere, the phrase “jeffbenite crystalline structure” means a crystalline phase or grains of a crystalline phase identified by XRD analysis as jeffbenite, as described herein, and the further characterizations provided herein that may aid in clarifying various embodiments and forms of crystalline phases having a jeffbenite crystalline structure that may be included and claimed.

[0131] In embodiments, a crystalline phase having a jeffbenite crystalline structure is the primary crystalline phase in the glass-ceramics. The crystalline phase having a jeffbenite crystalline structure may have attributes (e.g., compositional, molecular structural, microstructural) in common with jeffbenite.

[0132] Jeffbenite, named after Jeffrey Harris and Ben Harte, is a mineral recently discovered as inclusions in diamonds from “super-deep” (e.g., >300 km deep) within the mantle of the Earth. Prior to its naming, jeffbenite was called tetragonal-almandine-pyrope-phase (“TAPP”). Jeffbenite may comprise tetragonal $\text{Mg}_3\text{Al}_2\text{Si}_3\text{O}_{12}$. The term “tetragonal” refers to the otherwise cubic lattice being stretched along one of its lattice vectors to become a rectangular prism with a square base (“a by a”) and height (“c,” different from “a”), such as within the space group $142d$. The tetragonal crystal structure of jeffbenite may include cell edge parameter a of about 6.5, such as 6.5231

(1), such as within 0.1 thereof, and parameter c of about 18.2, such as 18.1756(3) angstroms, such as within 0.1 thereof. A crystalline phase having a jeffbenite crystalline structure may have the tetragonal structure of jeffbenite, as described herein.

[0133] The density of jeffbenite (itself) may be about 3.6 g/cm³, such as 3.576 g/cm³, such as within 0.1 g/cm³ thereof. The microhardness of jeffbenite (itself) may be about 7, such as within 1 thereof. Jeffbenite (itself) may be uniaxial (−) with refractive indexes ω of about 1.7, such as 1.733(5), such as within 0.1 thereof, and ε of about 1.7, such as 1.721, such as within 0.1 thereof.

[0134] While Mg₃Al₂Si₃O₁₂ is an ideal form of jeffbenite, jeffbenite can be generally described as a stoichiometric garnet composition, similar to pyrope (Mg₃Al₂(SiO₄)₃)-almandine (Fe₃Al₂(SiO₄)₃), but with a tetragonal crystalline structure, and may include other elements. Put another way, structurally, jeffbenite and crystals having a jeffbenite crystalline structure may be described as (M1)(M2)₂(M3)₂(T1)(T2)₂O₁₂ where M1 comprises magnesium (e.g., is mostly magnesium), M2 comprises aluminum (e.g., is mostly aluminum), M3 comprises magnesium (e.g., is mostly magnesium), and T1 and T2 comprise silicon (are both mostly silicon), and where two tetrahedra of such crystalline structures do not share any oxygen with one another. Jeffbenite may be categorized as an orthosilicate, such as a silicate containing the tetrahedra group SiO₄ where the ratio of silicon to oxygen is 1 to 4.

[0135] In embodiments of the glass-ceramics described herein, the crystalline phase comprising the jeffbenite crystalline structure may at least comprise, mostly consist of (>50 wt. %), consist essentially of, or be tetragonal Mg₃Al₂Si₃O₁₂.

[0136] In embodiments of the glass-ceramics described herein, the crystalline phase comprising the jeffbenite crystalline structure (or a portion thereof) may be modified by the addition of zirconia (ZrO₂). Without intending to be bound by any theory, alumina (i.e., an aluminum contribution) may be at least partially replaced in the jeffbenite crystalline structure by magnesia (i.e., a magnesium contribution) and zirconia (i.e., a zirconium contribution). In such embodiments, the crystalline phase comprising the jeffbenite crystalline structure (or a portion thereof) may have a composition according to the following formula: Mg_{3+x}Zr_xAl_{2-2x}Si₃O₁₂, where x is greater than or equal to 0 to less than or equal to 1. In embodiments, x may be greater than or equal to 0 to less than or equal to 0.6. For example, without limitation, the crystalline phase having the jeffbenite crystalline structure (or a portion thereof) may have the composition(s): Mg₃Al₂Si₃O₁₂, Mg_{3.1}Zr_{0.1}Al_{1.8}Si₃O₁₂, Mg_{3.2}Zr_{0.2}Al_{1.6}Si₃O₁₂, Mg_{3.3}Zr_{0.3}Al_{1.4}Si₃O₁₂, Mg_{3.4}Zr_{0.4}Al_{1.2}Si₃O₁₂, Mg_{3.5}Zr_{0.5}Al₁Si₃O₁₂, Mg_{3.6}Zr_{0.6}Al_{0.8}Si₃O₁₂, Mg_{3.7}Zr_{0.7}Al_{0.6}Si₃O₁₂, Mg_{3.8}Zr_{0.8}Al_{0.4}Si₃O₁₂, or Mg_{3.9}Zr_{0.9}Al_{0.2}Si₃O₁₂.

[0137] In embodiments of the glass-ceramics described herein, the crystalline phase comprising the jeffbenite crystalline structure (or a portion thereof) may be further modified by the addition of titania, tin oxide, iron oxide (FeO), manganese oxide, and/or zinc oxide. For example, titania (i.e., a titanium contribution) and/or tin oxide (i.e., a tin contribution) may be substituted for up to 50% of the zirconium in the jeffbenite crystalline structure. Similarly, iron oxide (i.e., an iron contribution), manganese oxide (i.e., a manganese contribution), and/or zinc oxide (i.e., a zinc

contribution) may be substituted for a portion of the magnesium in the jeffbenite crystalline structure. In such embodiments, the crystalline phase comprising the jeffbenite crystalline structure (or a portion thereof) may have a composition according to the following formula: (Mg,Fe,Mn,Zn)_{3+x}(Zr,Ti,Sn)_xAl_{2-2x}Si₃O₁₂, where x is greater than or equal to 0 to less than or equal to 1. In embodiments, x may be greater than or equal to 0 to less than or equal to 0.6. For example, without limitation, the crystalline phase having the jeffbenite crystalline structure (or a portion thereof) may have the composition(s): (Mg,Fe,Mn,Zn)₃Al₂Si₃O₁₂, (Mg,Fe,Mn,Zn)_{3.1}(Zr,Ti,Sn)_{0.1}Al_{1.8}Si₃O₁₂, (Mg,Fe,Mn,Zn)_{3.2}(Zr,Ti,Sn)_{0.2}Al_{1.6}Si₃O₁₂, (Mg,Fe,Mn,Zn)_{3.3}(Zr,Ti,Sn)_{0.3}Al_{1.4}Si₃O₁₂, (Mg,Fe,Mn,Zn)_{3.4}(Zr,Ti,Sn)_{0.4}Al_{1.2}Si₃O₁₂, (Mg,Fe,Mn,Zn)_{3.5}(Zr,Ti,Sn)_{0.5}Al₁Si₃O₁₂, (Mg,Fe,Mn,Zn)_{3.6}(Zr,Ti,Sn)_{0.6}Al_{0.8}Si₃O₁₂, (Mg,Fe,Mn,Zn)_{3.7}(Zr,Ti,Sn)_{0.7}Al_{0.6}Si₃O₁₂, (Mg,Fe,Mn,Zn)_{3.8}(Zr,Ti,Sn)_{0.8}Al_{0.4}Si₃O₁₂, or (Mg,Fe,Mn,Zn)_{3.9}(Zr,Ti,Sn)_{0.9}Al_{0.2}Si₃O₁₂. In these embodiments, it should be understood that the Fe, Mn, Zn, Ti, and Sn components in the formulas are each optional and the composition may be formed without one or more of these elements. For example, the composition may be free of Fe, but may include Mn, Ti, and Sn, or be free of Sn but include Fe, Mn, Zn and Ti. As such, it should be understood that the above referenced formulas can be written without one or more of Fe, Mn, Zn, Ti, and Sn.

[0138] It should be understood that other substitutions and modifications to the crystalline phase comprising the jeffbenite structure are contemplated and possible. For example, in embodiments of the glass-ceramics described herein, the crystalline phase comprising the jeffbenite crystalline structure (or a portion thereof) may be modified by the addition of metal oxides to the composition as a source of divalent metal cations (expressed as “R²⁺”) in substitution for a portion of the magnesium in the jeffbenite crystalline structure. Examples of divalent metal cations include, without limitation, Ca²⁺, Mn²⁺, Fe²⁺, Zn²⁺, and the like. In these embodiments, the divalent metal cations may have an ionic radius of less than 1 angstrom (0.1 nm). Similarly, the crystalline phase comprising the jeffbenite crystalline structure (or a portion thereof) may be modified by the addition of metal oxides to the composition as a source of tetravalent metal cations (expressed as “R⁴⁺”) in substitution for a portion of the zirconium in the jeffbenite crystalline structure. Examples of tetravalent metal cations include Ti⁴⁺, Sn⁴⁺, Hf⁴⁺, and the like. In such embodiments, the crystalline phase comprising the jeffbenite crystalline structure (or a portion thereof) may have a composition according to the following formula: (Mg,R²⁺)_{3+x}(Zr,R⁴⁺)_xAl_{2-2x}Si₃O₁₂, where x is greater than or equal to 0 to less than or equal to 1. In embodiments, x may be greater than or equal to 0 to less than or equal to 0.6. For example, without limitation, the crystalline phase having the jeffbenite crystalline structure (or a portion thereof) may have the composition(s): (Mg,R²⁺)₃Al₂Si₃O₁₂, (Mg,R²⁺)_{3.1}(Zr,R⁴⁺)_{0.1}Al_{1.8}Si₃O₁₂, (Mg,R²⁺)_{3.2}(Zr,R⁴⁺)_{0.2}Al_{1.6}Si₃O₁₂, (Mg,R²⁺)_{3.3}(Zr,R⁴⁺)_{0.3}Al_{1.4}Si₃O₁₂, (Mg,R²⁺)_{3.4}(Zr,R⁴⁺)_{0.4}Al_{1.2}Si₃O₁₂, (Mg,R²⁺)_{3.5}(Zr,R⁴⁺)_{0.5}Al₁Si₃O₁₂, (Mg,R²⁺)_{3.6}(Zr,R⁴⁺)_{0.6}Al_{0.8}Si₃O₁₂, (Mg,R²⁺)_{3.7}(Zr,R⁴⁺)_{0.7}Al_{0.6}Si₃O₁₂, (Mg,R²⁺)_{3.8}(Zr,R⁴⁺)_{0.8}Al_{0.4}Si₃O₁₂, or (Mg,R²⁺)_{3.9}(Zr,R⁴⁺)_{0.9}Al_{0.2}Si₃O₁₂. In these embodiments, it should be understood that the R²⁺ and R⁴⁺ components in the formulas are each optional and the composition may be formed without one or the other of these

elements. As such, it should be understood that the above referenced formulas can be written without one or the other of R^{2+} and R^{4+} .

[0139] In embodiments, the one or more crystalline phases of the phase assemblage may comprise one or more accessory crystalline phases. The one or more accessory crystalline phases may be present in the glass-ceramic in an amount less than the primary crystalline phase. In embodiments, the one or more accessory crystalline phases may comprise tetragonal zirconia (ZrO_2), $ZrTiO_4$, or a combination thereof. However, it should be understood that other accessory crystalline phases may also be present in the resultant glass-ceramic. In embodiments, one or more of the accessory crystalline phases may enter the structure of the crystalline phase having the jeffbenite crystalline structure (e.g., there may be a secondary phase within the lattice of the jeffbenite crystalline phase).

[0140] In embodiments, the phase assemblage of the glass-ceramics described herein comprises greater than or equal to 25 wt. % of the one or more crystalline phases by weight of the glass-ceramic article (i.e., wt. %) and less than or equal to 75 wt. % of the glass phase, greater than or equal to 30 wt. % of the one or more crystalline phases and less than or equal to 70 wt. % of the glass phase, greater than or equal to 40 wt. % of the one or more crystalline phases and less than or equal to 60 wt. % of the glass phase, greater than or equal to 50 wt. % of the one or more crystalline phases and less than or equal to 50 wt. % of the glass phase, greater than or equal to 60 wt. % of the one or more crystalline phases and less than or equal to 40 wt. % of the glass phase, greater than or equal to 70 wt. % of the one or more crystalline phases and less than or equal to 30 wt. % of the glass phase, greater than or equal to 80 wt. % of the one or more crystalline phases and less than or equal to 20 wt. % of the glass phase, as determined according to Rietveld analysis of the XRD spectrum. It should be understood that the crystalline phase content or the glass content may be within a sub-range formed from any and all of the foregoing endpoints. In embodiments, the crystalline phase(s) and glass phase may be homogeneously distributed throughout the glass-ceramic. It should further be noted that at least some, most (>50 wt %), or essentially all of such crystalline phase may have a jeffbenite crystalline structure (e.g., as identified by XRD; tetragonal, stoichiometrically garnet, $Mg_3Al_2Si_3O_{12}$ and variations thereof).

[0141] SiO_2 may be the primary glass former in the precursor glass and glass-ceramic compositions described herein and may function to stabilize the network structure of the glass-ceramics. The concentration of SiO_2 in the precursor glass and glass-ceramic compositions should be sufficiently high (e.g., greater than or equal to 35 mol. %) to form the crystalline phase when the precursor glass is heat-treated to convert the precursor glass to a glass-ceramic. The amount of SiO_2 may be limited (e.g., to less than or equal to 65 mol. %) to control the melting point of the precursor glass or glass-ceramic composition, as the melting temperature of pure SiO_2 or high- SiO_2 glasses is undesirably high. Thus, limiting the concentration of SiO_2 may aid in improving the meltability and the formability of the precursor glass or glass-ceramic composition.

[0142] In embodiments, the precursor glass or glass-ceramic composition may comprise a positive amount of silica, such as an amount greater than tramp (0.05 mol. % or greater), such as from greater than or equal to 35 mol. %

to less than or equal to 65 mol. % SiO_2 . In embodiments, the precursor glass or glass-ceramic composition may comprise SiO_2 in an amount greater than or equal to 35 mol. % to less than or equal to 62 mol. %, greater than or equal to 35 mol. % to less than or equal to 59 mol. %, greater than or equal to 35 mol. % to less than or equal to 56 mol. %, greater than or equal to 35 mol. % to less than or equal to 53 mol. %, greater than or equal to 35 mol. % to less than or equal to 50 mol. %, greater than or equal to 35 mol. % to less than or equal to 47 mol. %, greater than or equal to 35 mol. % to less than or equal to 44 mol. %, greater than or equal to 35 mol. % to less than or equal to 41 mol. %, greater than or equal to 35 mol. % to less than or equal to 38 mol. %, greater than or equal to 38 mol. % to less than or equal to 65 mol. %, greater than or equal to 41 mol. % to less than or equal to 65 mol. %, greater than or equal to 42 mol. % to less than or equal to 65 mol. %, greater than or equal to 43 mol. % to less than or equal to 65 mol. %, greater than or equal to 44 mol. % to less than or equal to 65 mol. %, greater than or equal to 45 mol. % to less than or equal to 65 mol. %, greater than or equal to 46 mol. % to less than or equal to 65 mol. %, greater than or equal to 47 mol. % to less than or equal to 65 mol. %, greater than or equal to 48 mol. % to less than or equal to 65 mol. %, greater than or equal to 49 mol. % to less than or equal to 65 mol. %, greater than or equal to 50 mol. % to less than or equal to 65 mol. %, greater than or equal to 51 mol. % to less than or equal to 65 mol. %, greater than or equal to 52 mol. % to less than or equal to 65 mol. %, greater than or equal to 53 mol. % to less than or equal to 65 mol. %, greater than or equal to 54 mol. % to less than or equal to 65 mol. %, greater than or equal to 55 mol. % to less than or equal to 65 mol. %, greater than or equal to 56 mol. % to less than or equal to 65 mol. %, greater than or equal to 57 mol. % to less than or equal to 65 mol. %, greater than or equal to 58 mol. % to less than or equal to 65 mol. %, greater than or equal to 59 mol. % to less than or equal to 65 mol. %, greater than or equal to 60 mol. % to less than or equal to 65 mol. %, greater than or equal to 61 mol. % to less than or equal to 65 mol. %, or any and all sub-ranges formed from any of these endpoints. In some embodiments, the precursor glass or glass-ceramic composition may comprise from greater than or equal to 48 mol. % to less than or equal to 54 mol. % SiO_2 . In embodiments, the concentration of SiO_2 may be greater than or equal to 40 mol. %, 45 mol. %, or 50 mol. %. In embodiments, the concentration of SiO_2 may be less than or equal to 65 mol. %, 60 mol. % or 55 mol. %.

[0143] Like SiO_2 , Al_2O_3 may also stabilize the glass network and additionally provides improved mechanical properties and chemical durability to the glass-ceramics. The amount of Al_2O_3 may also be tailored to control the viscosity of the precursor glass or glass-ceramic composition. However, if the amount of Al_2O_3 is too high, the viscosity of the glass melt may increase. In embodiments, the precursor glass or glass-ceramic composition may comprise from greater than or equal to 5 mol. % to less than or equal to 20 mol. % Al_2O_3 . In embodiments, the concentration of Al_2O_3 in the precursor glass or glass-ceramic composition may be a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), such as greater than or equal to 5 mol. % to less than or equal to 20 mol. %, greater than or equal to 5 mol. % to less than or equal to 19 mol. %, greater than or equal to 5 mol. % to less than or equal to 18 mol. %, greater than or equal to 5 mol. % to less

than or equal to 17 mol. %, greater than or equal to 5 mol. % to less than or equal to 16 mol. %, greater than or equal to 5 mol. % to less than or equal to 15 mol. %, greater than or equal to 5 mol. % to less than or equal to 14 mol. %, greater than or equal to 5 mol. % to less than or equal to 13 mol. %, greater than or equal to 5 mol. % to less than or equal to 12 mol. %, greater than or equal to 5 mol. % to less than or equal to 11 mol. %, greater than or equal to 5 mol. % to less than or equal to 10 mol. %, greater than or equal to 5 mol. % to less than or equal to 9 mol. %, greater than or equal to 5 mol. % to less than or equal to 8 mol. %, greater than or equal to 5 mol. % to less than or equal to 7 mol. %, greater than or equal to 5 mol. % to less than or equal to 6 mol. %, greater than or equal to 6 mol. % to less than or equal to 20 mol. %, greater than or equal to 7 mol. % to less than or equal to 20 mol. %, greater than or equal to 8 mol. % to less than or equal to 20 mol. %, greater than or equal to 9 mol. % to less than or equal to 20 mol. %, greater than or equal to 10 mol. % to less than or equal to 20 mol. %, greater than or equal to 11 mol. % to less than or equal to 20 mol. %, greater than or equal to 12 mol. % to less than or equal to 20 mol. %, greater than or equal to 13 mol. % to less than or equal to 20 mol. %, greater than or equal to 14 mol. % to less than or equal to 20 mol. %, greater than or equal to 15 mol. % to less than or equal to 20 mol. %, greater than or equal to 16 mol. % to less than or equal to 20 mol. %, greater than or equal to 17 mol. % to less than or equal to 20 mol. %, greater than or equal to 18 mol. % to less than or equal to 20 mol. %, greater than or equal to 19 mol. % to less than or equal to 20 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the concentration of Al_2O_3 in the precursor glass or glass-ceramic composition may be from greater than or equal to 9 mol. % to less than or equal to 13 mol. %. In embodiments, the concentration of Al_2O_3 in the precursor glass or glass-ceramic composition may be greater than or equal to 5 mol. %, 7 mol. %, or 10 mol. %. In embodiments, the concentration of Al_2O_3 may be less than or equal to 20 mol. %, 15 mol. %, 12 mol. %, or 10 mol. %.

[0144] In embodiments, the precursor glass or glass-ceramic composition may comprise a positive amount of MgO, such as an amount greater than tramp (0.05 mol. % or greater), such as from greater than or equal to 10 mol. % to less than or equal to 45 mol. % MgO. In embodiments, the precursor glass or glass-ceramic composition may comprise from greater than or equal to 7 mol. % to less than or equal to 65 mol. % MgO. Additions of MgO may increase the elastic modulus of the glass of the precursor glass and resultant glass-ceramic. MgO may also displace Al_2O_3 in the glass network and create a more open network structure, improving ion mobility in the glass network. Without intending to be bound by theory, addition of MgO may increase the crystallinity of the glass-ceramic composition. In embodiments, the concentration of MgO in the precursor glass or glass-ceramic composition may be greater than or equal to 7 mol. % to less than or equal to 65 mol. %, greater than or equal to 7 mol. % to less than or equal to 60 mol. %, greater than or equal to 7 mol. % to less than or equal to 55 mol. %, greater than or equal to 7 mol. % to less than or equal to 50 mol. %, greater than or equal to 7 mol. % to less than or equal to 48 mol. %, greater than or equal to 7 mol. % to less than or equal to 46 mol. %, greater than or equal to 7 mol. % to less than or equal to 44 mol. %, greater than or equal to 7 mol. % to less than or equal to 42 mol. %, greater than or equal to 7 mol. % to less than or equal to 40

mol. %, greater than or equal to 7 mol. % to less than or equal to 38 mol. %, greater than or equal to 7 mol. % to less than or equal to 36 mol. %, greater than or equal to 7 mol. % to less than or equal to 34 mol. %, greater than or equal to 7 mol. % to less than or equal to 32 mol. %, greater than or equal to 7 mol. % to less than or equal to 30 mol. %, greater than or equal to 7 mol. % to less than or equal to 28 mol. %, greater than or equal to 7 mol. % to less than or equal to 26 mol. %, greater than or equal to 7 mol. % to less than or equal to 24 mol. %, greater than or equal to 7 mol. % to less than or equal to 22 mol. %, greater than or equal to 7 mol. % to less than or equal to 20 mol. %, greater than or equal to 7 mol. % to less than or equal to 18 mol. %, greater than or equal to 7 mol. % to less than or equal to 16 mol. %, greater than or equal to 7 mol. % to less than or equal to 14 mol. %, greater than or equal to 7 mol. % to less than or equal to 12 mol. %, greater than or equal to 7 mol. % to less than or equal to 10 mol. %, greater than or equal to 9 mol. % to less than or equal to 50 mol. %, greater than or equal to 10 mol. % to less than or equal to 65 mol. %, greater than or equal to 10 mol. % to less than or equal to 60 mol. %, greater than or equal to 10 mol. % to less than or equal to 55 mol. %, greater than or equal to 10 mol. % to less than or equal to 45 mol. %, greater than or equal to 10 mol. % to less than or equal to 42 mol. %, greater than or equal to 10 mol. % to less than or equal to 40 mol. %, greater than or equal to 10 mol. % to less than or equal to 38 mol. %, greater than or equal to 10 mol. % to less than or equal to 36 mol. %, greater than or equal to 10 mol. % to less than or equal to 34 mol. %, greater than or equal to 10 mol. % to less than or equal to 32 mol. %, greater than or equal to 10 mol. % to less than or equal to 30 mol. %, greater than or equal to 10 mol. % to less than or equal to 28 mol. %, greater than or equal to 10 mol. % to less than or equal to 26 mol. %, greater than or equal to 10 mol. % to less than or equal to 24 mol. %, greater than or equal to 10 mol. % to less than or equal to 22 mol. %, greater than or equal to 10 mol. % to less than or equal to 20 mol. %, greater than or equal to 10 mol. % to less than or equal to 18 mol. %, greater than or equal to 10 mol. % to less than or equal to 16 mol. %, greater than or equal to 10 mol. % to less than or equal to 14 mol. %, greater than or equal to 10 mol. % to less than or equal to 12 mol. %, greater than or equal to 10 mol. % to less than or equal to 10 mol. % to less than or equal to 50 mol. %, greater than or equal to 12 mol. % and less than or equal to 60 mol. %, greater than or equal to 12 mol. % to less than or equal to 45 mol. %, greater than or equal to 13 mol. % and less than or equal to 65 mol. %, greater than or equal to 13 mol. % to less than or equal to 50 mol. %, greater than or equal to 14 mol. % to less than or equal to 60 mol. %, greater than or equal to 14 mol. % to less than or equal to 45 mol. %, greater than or equal to 15 mol. % to less than or equal to 65 mol. %, greater than or equal to 15 mol. % to less than or equal to 50 mol. %, greater than or equal to 16 mol. % to less than or equal to 65 mol. %, greater than or equal to 16 mol. % to less than or equal to 45 mol. %, greater than or equal to 17 mol. % to less than or equal to 65 mol. %, greater than or equal to 17 mol. % to less than or equal to 50 mol. %, greater than or equal to 18 mol. % to less than or equal to 50 mol. %, greater than or equal to 18 mol. % to less than or equal to 60 mol. %, greater than or equal to 18 mol. % to less than or equal to 45 mol. %, greater than or equal to 19 mol. % to less than or equal to 65 mol. %, greater than or equal to 19 mol. % to less than or equal to 50 mol. %, greater than or equal to 20 mol. % to less than

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or equal to 65 mol. %, 60 mol. %, 55 mol. %, 50 mol. %, 45 mol. %, 40 mol. %, 35 mol. %, or 30 mol. %. Without intending to be bound by theory, as the concentration of MgO in the glass-ceramic composition increases, then the opacity of the glass-ceramic may increase. Likewise, as the concentration of MgO in the glass-ceramic composition decreases, then the transparency may be improved. However, if the concentration of MgO in the glass-ceramic composition is too low, then the glass-ceramic composition may become hazy.

[0145] In embodiments, the precursor glass or glass-ceramic composition may comprise Na₂O. Additions of Na₂O may lower the liquidus viscosity of the glass which, in turn, may aid in forming or shaping the precursor glass. Na₂O may also facilitate ion-exchange strengthening of the resultant glass-ceramic as most of the Na₂O present in the precursor glass is partitioned into the residual glass phase following heat treatment (e.g., ceramming). In embodiments, the concentration of Na₂O in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 15 mol. %, greater than or equal to 0 mol. % to less than or equal to 13 mol. %, greater than or equal to 0 mol. % to less than or equal to 11 mol. %, greater than or equal to 0 mol. % to less than or equal to 9 mol. %, greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 15 mol. %, greater than or equal to 1 mol. % to less than or equal to 13 mol. %, greater than or equal to 1 mol. % to less than or equal to 11 mol. %, greater than or equal to 1 mol. % to less than or equal to 9 mol. %, greater than or equal to 1 mol. % to less than or equal to 7 mol. %, greater than or equal to 1 mol. % to less than or equal to 5 mol. %, greater than or equal to 1 mol. % to less than or equal to 3 mol. %, greater than or equal to 2 mol. % to less than or equal to 15 mol. %, greater than or equal to 2 mol. % to less than or equal to 13 mol. %, greater than or equal to 2 mol. % to less than or equal to 11 mol. %, greater than or equal to 2 mol. % to less than or equal to 9 mol. %, greater than or equal to 2 mol. % to less than or equal to 7 mol. %, greater than or equal to 2 mol. % to less than or equal to 5 mol. %, greater than or equal to 2 mol. % to less than or equal to 3 mol. %, greater than or equal to 3 mol. % to less than or equal to 15 mol. %, greater than or equal to 3 mol. % to less than or equal to 15 mol. %, greater than or equal to 6 mol. % to less than or equal to 15 mol. %, greater than or equal to 7 mol. % to less than or equal to 15 mol. %, greater than or equal to 8 mol. % to less than or equal to 15 mol. %, greater than or equal to 9 mol. % to less than or equal to 15 mol. %, greater than or equal to 10 mol. % to less than or equal to 15 mol. %, greater than or equal to 11 mol. % to less than or equal to 15 mol. %, greater than or equal to 12 mol. % to less than or equal to 15 mol. %, greater than or equal to 13 mol. % to less than or equal to 15 mol. %, greater than or equal to 14 mol. % to less than or equal to 15 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the concentration of Na₂O in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, 5 mol. % or 10

mol. %. In embodiments, the concentration of Na_2O may be less than or equal to 15 mol. %, 10 mol. %, or 5 mol. %. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of Na_2O .

[0146] In embodiments, the precursor glass or glass-ceramic composition may comprise K_2O . Additions of K_2O may lower the liquidus viscosity of the glass which, in turn, may aid in forming or shaping the precursor glass. K_2O may also facilitate ion-exchange strengthening of the resultant glass-ceramic as most of the K_2O is partitioned into the glass phase after ceramming. In embodiments, the concentration of K_2O in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 15 mol. %, greater than or equal to 0 mol. % to less than or equal to 13 mol. %, greater than or equal to 0 mol. % to less than or equal to 11 mol. %, greater than or equal to 0 mol. % to less than or equal to 9 mol. %, greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 15 mol. %, greater than or equal to 1 mol. % to less than or equal to 13 mol. %, greater than or equal to 1 mol. % to less than or equal to 11 mol. %, greater than or equal to 1 mol. % to less than or equal to 9 mol. %, greater than or equal to 1 mol. % to less than or equal to 7 mol. %, greater than or equal to 1 mol. % to less than or equal to 5 mol. %, greater than or equal to 1 mol. % to less than or equal to 3 mol. %, greater than or equal to 2 mol. % to less than or equal to 15 mol. %, greater than or equal to 2 mol. % to less than or equal to 13 mol. %, greater than or equal to 2 mol. % to less than or equal to 11 mol. %, greater than or equal to 2 mol. % to less than or equal to 9 mol. %, greater than or equal to 2 mol. % to less than or equal to 7 mol. %, greater than or equal to 2 mol. % to less than or equal to 5 mol. %, greater than or equal to 2 mol. % to less than or equal to 3 mol. %, greater than or equal to 3 mol. % to less than or equal to 15 mol. %, greater than or equal to 4 mol. % to less than or equal to 15 mol. %, greater than or equal to 5 mol. % to less than or equal to 15 mol. %, greater than or equal to 5 mol. % to less than or equal to 5 mol. % to less than or equal to 15 mol. %, greater than or equal to 6 mol. % to less than or equal to 15 mol. %, greater than or equal to 7 mol. % to less than or equal to 15 mol. %, greater than or equal to 8 mol. % to less than or equal to 15 mol. %, greater than or equal to 9 mol. % to less than or equal to 15 mol. %, greater than or equal to 10 mol. % to less than or equal to 15 mol. %, greater than or equal to 11 mol. % to less than or equal to 15 mol. %, greater than or equal to 12 mol. % to less than or equal to 15 mol. %, greater than or equal to 13 mol. % to less than or equal to 15 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the concentration of K_2O in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, 5 mol. % or 10 mol. %. In embodiments, the concentration of K_2O may be less than or equal to 15 mol. %, 10 mol. %, or 5 mol. %. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of K_2O .

[0147] In embodiments, Na_2O (mol. %)+ K_2O (mol. %) in the precursor glass or glass-ceramic compositions described herein may be greater than or equal to 0 mol. % and less than

[illegible]

less than or equal to 11 mol. %, greater than or equal to 11 mol. % to less than or equal to 15 mol. %, greater than or equal to 11 mol. % to less than or equal to 13 mol. %, greater than or equal to 13 mol. % to less than or equal to 15 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the concentration of Na_2O (mol. %)+ K_2O (mol. %) in the precursor glass or glass-ceramic composition may be greater than or equal to 2 mol. %, 3 mol. %, 5 mol. % or 10 mol. %. In embodiments, the concentration of Na_2O (mol. %)+ K_2O (mol. %) may be less than or equal to 15 mol. %, 10 mol. %, or 5 mol. %. In embodiments, Na_2O (mol. %)/(Na_2O (mol. %)+ K_2O (mol. %)) may be greater than or equal to 0.2 or even greater than or equal to 0.3 to improve the ion-exchangeability of the resultant glass-ceramic.

[0148] In embodiments, the ratio of Na₂O (mol. %) to K₂O (mol. %) may be about 1:3, about 2:5, or about 1:1, such as an amount of Na₂O in mol. % that is at least 33% or more of the amount of K₂O in a particular composition as disclosed herein or vice versa (i.e. an amount of K₂O in mol. % that is at least 33% or more of the amount of Na₂O) as shown in the Examples, such as an amount of Na₂O in mol. % that is at least 40% or more of the amount of K₂O or vice versa, such as an amount of Na₂O in mol. % that is at least 50% or more of the amount of K₂O or vice versa, such as an amount of Na₂O in mol. % that is at least 60% or more of the amount of K₂O or vice versa, such as an amount of Na₂O in mol. % that is at least 70% or more of the amount of K₂O or vice versa, such as an amount of Na₂O in mol. % that is at least 80% or more of the amount of K₂O or vice versa, such as an amount of Na₂O in mol. % that is at least 90% or more of the amount of K₂O or vice versa. Without wishing to be bound by theory, it is believed that the ratio of Na₂O (mol. %) to K₂O (mol. %) may be adjusted to improve the transparency of the glass-ceramic composition.

[0149] In embodiments, the precursor glass or glass-ceramic composition may comprise ZrO_2 . Without wishing to be bound by theory, it is believed that ZrO_2 acts as a nucleating agent that facilitates the nucleation of the crystalline phase having the jeffbenite crystalline structure during heat treatment at ambient atmospheric pressure (i.e., ~100 kPa). In embodiments, the ZrO_2 may be tetragonal ZrO_2 . In embodiments, the concentration of ZrO_2 in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 7 mol. % ZrO_2 , greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 7 mol. %, greater than or equal to 1 mol. % to less than or equal to 6 mol. %, greater than or equal to 1 mol. % to less than or equal to 5 mol. %, greater than or equal to 1 mol. % to less than or equal to 4 mol. %, greater than or equal to 1 mol. % to less than or equal to 3 mol. %, greater than or equal to 1 mol. % to less than or equal to 2 mol. %, greater than 1.5 mol. % to less than or equal to 7 mol. %, greater than 1.5 mol. % to less than or equal to 6 mol. %, greater than 1.5 mol. % to less

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[0150] In embodiments, the precursor glass or glass-ceramic composition may comprise HfO_2 . Without wishing to be bound by theory, it is believed that HfO_2 acts as a nucleating agent that facilitates the formation of crystalline phases during heat treatment. In embodiments, HfO_2 may be used as a nucleating in addition to ZrO_2 or as a replacement for ZrO_2 . Also, HfO_2 may help reduce the liquidus of the precursor glass. In embodiments, the concentration of HfO_2 in the precursor glass or the glass-ceramic composition may be from greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. %, or greater), to less than or equal to 12 mol. %. In embodiments, the concentration of HfO_2 in the precursor glass or the glass-ceramic composition may be from greater than or equal to 0 mol. % to less than or equal to 12 mol. %, greater than or equal to 0 mol. % to less than or equal to 10 mol. %, greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 12 mol. %, greater than or equal to 1 mol. % to less than or equal to 10 mol. %, greater than or equal to 1 mol. % to less than or equal to 7 mol. %, greater than or equal to 1 mol. % to less than or equal to 6 mol. %, greater than or equal to 1 mol. % to less than or equal to 5 mol. %, greater than or equal to 1 mol. % to less than or equal to 4 mol. %, greater than or equal to 1 mol. % to less than or equal to 3 mol. %, greater than or equal to 1 mol. % to less than or equal to 2 mol. %, greater than or equal to 2 mol. % to less than or equal to 12 mol. %, greater than or equal to 2 mol. % to less than or equal to 10 mol. %, greater than or equal to 2 mol. % to less than or equal to 7 mol. %, greater than or equal to 2 mol. % to less than or equal to 6 mol. %, greater than or equal to 2 mol. % to less than or equal to 5 mol. %, greater than or equal to 2 mol. % to less than or equal to 4 mol. %, greater than or equal to 2 mol. % to less than or equal to 3 mol. %, greater than or equal to 2 mol. % to less than or equal to 1 mol. %, greater than or equal to 3 mol. % to less than or equal to 12 mol. %, greater than or equal to 3 mol. % to less than or equal to 10 mol. %, greater than or equal to 3 mol. % to less than or equal to 7 mol. %, greater than or equal to 3 mol. % to less than or equal to 6 mol. %, greater than or equal to 3 mol. % to less than or equal to 5 mol. %, greater than or equal to 3 mol. % to less than or equal to 4 mol. %, greater than or equal to 3 mol. % to less than or equal to 1 mol. %, greater than or equal to 4 mol. % to less than or equal to 12 mol. %, greater than or equal to 4 mol. % to less than or equal to 10 mol. %, greater than or equal to 4 mol. % to less than or equal to 7 mol. %, greater than or equal to 4 mol. % to less than or equal to 6 mol. %, greater than or equal to 4 mol. % to less than or equal to 5 mol. %, greater than or equal to 4 mol. % to less than or equal to 1 mol. %, greater than or equal to 5 mol. % to less than or equal to 12 mol. %, greater than or equal to 5 mol. % to less than or equal to 10 mol. %, greater than or equal to 5 mol. % to less than or equal to 7 mol. %, greater than or equal to 5 mol. % to less than or equal to 6 mol. %, greater than or equal to 5 mol. % to less than or equal to 1 mol. %, greater than or equal to 6 mol. % to less than or equal to 12 mol. %, greater than or equal to 6 mol. % to less than or equal to 10 mol. %, greater than or equal to 6 mol. % to less than or equal to 7 mol. %, greater than or equal to 6 mol. % to less than or equal to 1 mol. %, greater than or equal to 7 mol. % to less than or equal to 12 mol. %, greater than or equal to 7 mol. % to less than or equal to 10 mol. %, greater than or equal to 7 mol. % to less than or equal to 1 mol. %, greater than or equal to 8 mol. % to less than or equal to 12 mol. %, greater than or equal to 8 mol. % to less than or equal to 10 mol. %, greater than or equal to 8 mol. % to less than or equal to 1 mol. %, greater than or equal to 9 mol. % to less than or equal to 12 mol. %, greater than or equal to 9 mol. % to less than or equal to 10 mol. %, greater than or equal to 9 mol. % to less than or equal to 1 mol. %, greater than or equal to 10 mol. % to less than or equal to 12 mol. %, greater than or equal to 10 mol. % to less than or equal to 1 mol. %, greater than or equal to 11 mol. % to less than or equal to 12 mol. %, greater than or equal to 11 mol. % to less than or equal to 1 mol. %, greater than or equal to 12 mol. % to less than or equal to 1 mol. %.

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[0151] In embodiments, the precursor glass or glass-ceramic composition may comprise TiO_2 . Without wishing to be bound by theory, it is believed that TiO_2 acts as a nucleating agent that facilitates the formation of crystalline phases during heat treatment. Increasing concentrations of TiO_2 may also impart color to the precursor glass and resultant glass-ceramic. In embodiments, the concentration of TiO_2 in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 0.3 mol. % to less than or equal to 7 mol. %, greater than or equal to 0.3 mol. % to less than or equal to 6 mol. %, greater than or equal to 0.3 mol. % to less than or equal to 5 mol. %, greater than or equal to 0.3 mol. % to less than or equal to 4 mol. %, greater than or equal to 0.3 mol. % to less than or equal to 3 mol. %, greater than or equal to 0.3 mol. % to less than or equal to 2 mol. %, greater than or equal to 0.3 mol. % to less than or equal to 1 mol. %, greater than or equal to 0.5 mol. % to less than or equal to 7 mol. %, greater than or equal to 0.5 mol. % to less than or equal to 6 mol. %, greater than or equal to 0.5 mol. % to less than or equal

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[0152] In embodiments, ZrO_2 (mol. %)+ TiO_2 (mol. %) in the precursor glass or glass-ceramic compositions described herein may be greater than or equal to 2.0 mol. %, greater than or equal to 3.0 mol. %, or even greater than or equal to 4.0 mol. %. In embodiments, ZrO_2 (mol. %)/(ZrO₂ (mol. %)+ TiO_2 (mol. %)) in the precursor glass or glass-ceramic compositions described herein may be greater than or equal to 0.3. Without intending to be bound by theory, when ZrO_2 (mol. %)/(ZrO₂ (mol. %)+ TiO_2 (mol. %)) in the precursor glass or glass-ceramic compositions is less than 0.3 then the primary crystalline phase in the glass-ceramic may be a forsterite crystalline phase instead of a crystalline phase having a jeffbenite crystalline structure, which may be undesirable.

[0153] In embodiments, the precursor glass or glass-ceramic composition may comprise SnO_2 . SnO_2 primarily functions as a fining agent in the precursor glass composition. However, additions of SnO_2 may also aid the nucleating agents (such as TiO_2 and ZrO_2) in nucleating crystalline phases during heat treatment. In embodiments, the concentration of SnO_2 in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. % to less than or equal to 0.2 mol. %, greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 0.18 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.16 mol. %, greater than or equal to 0 mol. % to

less than or equal to 0.14 mol. %, greater than or equal to mol. % to less than or equal to 0.12 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.11 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.1 mol. %, greater than or equal to 0.08 mol. % to less than or equal to 0.2 mol. %, greater than or equal to 0.09 mol. % to less than or equal to 0.2 mol. %, greater than or equal to 0.1 mol. % to less than or equal to 0.2 mol. %, greater than or equal to 0.12 mol. % to less than or equal to 0.2 mol. %, greater than or equal to 0.13 mol. % to less than or equal to 0.2 mol. %, greater than or equal to 0.14 mol. % to less than or equal to 0.2 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the concentration of SnO_2 in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, 0.08 mol. %, or 0.09 mol. %. In embodiments, the concentration of SnO_2 may be less than or equal to 0.2 mol. %, 0.18 mol. %, or 0.16 mol. %. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of SnO_2 .

[0154] In embodiments, the precursor glass or glass-ceramic composition may comprise BaO. BaO may increase the refractive index of the residual glass of the glass-ceramic to better match the refractive index of the crystalline phase having the jeffbenite crystalline structure in the glass-ceramic. In embodiments, the concentration of BaO in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 8 mol. %, greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 8 mol. %, greater than or equal to 1 mol. % to less than or equal to 7 mol. %, greater than or equal to 1 mol. % to less than or equal to 6 mol. %, greater than or equal to 1 mol. % to less than or equal to 5 mol. %, greater than or equal to 1 mol. % to less than or equal to 4 mol. %, greater than or equal to 1 mol. % to less than or equal to 3 mol. %, greater than or equal to 1 mol. % to less than or equal to 2 mol. %, greater than or equal to 2 mol. % to less than or equal to 8 mol. %, greater than or equal to 2 mol. % to less than or equal to 7 mol. %, greater than or equal to 2 mol. % to less than or equal to 6 mol. %, greater than or equal to 2 mol. % to less than or equal to 5 mol. %, greater than or equal to 2 mol. % to less than or equal to 4 mol. %, greater than or equal to 2 mol. % to less than or equal to 3 mol. %, greater than or equal to 3 mol. % to less than or equal to 8 mol. %, greater than or equal to 3 mol. % to less than or equal to 7 mol. %, greater than or equal to 3 mol. % to less than or equal to 6 mol. %, greater than or equal to 3 mol. % to less than or equal to 5 mol. %, greater than or equal to 3 mol. % to less than or equal to 4 mol. %, greater than or equal to 4 mol. % to less than or equal to 8 mol. %, greater than or equal to 4 mol. % to less than or equal to 7 mol. %, greater than or equal to 4 mol. % to less than or equal to 6 mol. %, greater than or equal to 4 mol. % to less than or equal to 5 mol. %, greater than or equal to 5 mol. % to less than or equal to 8 mol. %, greater

than or equal to 5 mol. % to less than or equal to 7 mol. %, greater than or equal to 5 mol. % to less than or equal to 6 mol. %, greater than or equal to 6 mol. % to less than or equal to 8 mol. %, greater than or equal to 6 mol. % to less than or equal to 7 mol. %, greater than or equal to 7 mol. % to less than or equal to 8 mol. % or any and all sub-ranges formed from any of these endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of BaO.

[0155] In embodiments, the precursor glass or glass-ceramic composition may comprise ZnO. Additions of ZnO may increase the refractive index of the residual glass in the glass-ceramic to better match the refractive index of the crystalline phase having the jeffbenite crystalline structure in the glass-ceramic. While not wishing to be bound by theory, it is believed that additions of ZnO may result in the replacement of at least a portion of the Mg in the jeffbenite crystalline structure with Zn. ZnO may also help stabilize the precursor glass, prevent devitrification, and lower the liquidus viscosity. However, too much ZnO may disrupt the formation of the crystalline phase having the jeffbenite crystalline structure during ceramming. In embodiments, the concentration of ZnO in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 15 mol. %, greater than or equal to 0 mol. % to less than or equal to 14 mol. %, greater than or equal to 0 mol. % to less than or equal to 13 mol. %, greater than or equal to 0 mol. % to less than or equal to 12 mol. %, greater than or equal to 0 mol. % to less than or equal to 11 mol. %, greater than or equal to 0 mol. % to less than or equal to 10 mol. %, greater than or equal to 0 mol. % to less than or equal to 9 mol. %, greater than or equal to 0 mol. % to less than or equal to 8 mol. %, greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 15 mol. %, greater than or equal to 1 mol. % to less than or equal to 14 mol. %, greater than or equal to 1 mol. % to less than or equal to 13 mol. %, greater than or equal to 1 mol. % to less than or equal to 12 mol. %, greater than or equal to 1 mol. % to less than or equal to 11 mol. %, greater than or equal to 1 mol. % to less than or equal to 10 mol. %, greater than or equal to 1 mol. % to less than or equal to 9 mol. %, greater than or equal to 1 mol. % to less than or equal to 8 mol. %, greater than or equal to 1 mol. % to less than or equal to 7 mol. %, greater than or equal to 1 mol. % to less than or equal to 6 mol. %, greater than or equal to 1 mol. % to less than or equal to 5 mol. %, greater than or equal to 1 mol. % to less than or equal to 4 mol. %, greater than or equal to 1 mol. % to less than or equal to 3 mol. %, greater than or equal to 1 mol. % to less than or equal to 2 mol. %, greater than or equal to 2 mol. % to less than or equal to 15 mol. %, greater than or equal to 2 mol. % to less than or equal to 14 mol. %, greater than or equal to 2 mol. % to less than or equal to 13 mol. %, greater than or equal to 2 mol. % to less than or equal to 12 mol. %, greater than or equal to 2 mol. % to less than or equal to 11 mol. %, greater

greater than or equal to 2 mol. % to less than or equal to 10 mol. %, greater than or equal to 2 mol. % to less than or equal to 9 mol. %, greater than or equal to 2 mol. % to less than or equal to 8 mol. %, greater than or equal to 2 mol. % to less than or equal to 7 mol. %, greater than or equal to 2 mol. % to less than or equal to 6 mol. %, greater than or equal to 2 mol. % to less than or equal to 5 mol. %, greater than or equal to 2 mol. % to less than or equal to 4 mol. %, greater than or equal to 2 mol. % to less than or equal to 3 mol. %, greater than or equal to 3 mol. % to less than or equal to 15 mol. %, greater than or equal to 3 mol. % to less than or equal to 14 mol. %, greater than or equal to 3 mol. % to less than or equal to 13 mol. %, greater than or equal to 3 mol. % to less than or equal to 12 mol. %, greater than or equal to 3 mol. % to less than or equal to 11 mol. %, greater than or equal to 3 mol. % to less than or equal to 10 mol. %, greater than or equal to 3 mol. % to less than or equal to 9 mol. %, greater than or equal to 3 mol. % to less than or equal to 8 mol. %, greater than or equal to 3 mol. % to less than or equal to 7 mol. %, greater than or equal to 3 mol. % to less than or equal to 6 mol. %, greater than or equal to 3 mol. % to less than or equal to 5 mol. %, greater than or equal to 3 mol. % to less than or equal to 4 mol. %, greater than or equal to 4 mol. % to less than or equal to 15 mol. %, greater than or equal to 4 mol. % to less than or equal to 14 mol. %, greater than or equal to 4 mol. % to less than or equal to 13 mol. %, greater than or equal to 4 mol. % to less than or equal to 12 mol. %, greater than or equal to 4 mol. % to less than or equal to 11 mol. %, greater than or equal to 4 mol. % to less than or equal to 10 mol. %, greater than or equal to 4 mol. % to less than or equal to 9 mol. %, greater than or equal to 4 mol. % to less than or equal to 8 mol. %, greater than or equal to 4 mol. % to less than or equal to 7 mol. %, greater than or equal to 4 mol. % to less than or equal to 6 mol. %, greater than or equal to 4 mol. % to less than or equal to 5 mol. %, greater than or equal to 5 mol. % to less than or equal to 15 mol. %, greater than or equal to 5 mol. % to less than or equal to 14 mol. %, greater than or equal to 5 mol. % to less than or equal to 13 mol. %, greater than or equal to 5 mol. % to less than or equal to 12 mol. %, greater than or equal to 5 mol. % to less than or equal to 11 mol. %, greater than or equal to 5 mol. % to less than or equal to 10 mol. %, greater than or equal to 5 mol. % to less than or equal to 9 mol. %, greater than or equal to 5 mol. % to less than or equal to 8 mol. %, greater than or equal to 5 mol. % to less than or equal to 7 mol. %, greater than or equal to 5 mol. % to less than or equal to 6 mol. %, greater than or equal to 6 mol. % to less than or equal to 15 mol. %, greater than or equal to 6 mol. % to less than or equal to 14 mol. %, greater than or equal to 6 mol. % to less than or equal to 13 mol. %, greater than or equal to 6 mol. % to less than or equal to 12 mol. %, greater than or equal to 6 mol. % to less than or equal to 11 mol. %, greater than or equal to 6 mol. % to less than or equal to 10 mol. %, greater than or equal to 6 mol. % to less than or equal to 9 mol. %, greater than or equal to 6 mol. % to less than or equal to 8 mol. %, greater than or equal to 6 mol. % to less than or equal to 7 mol. %, greater than or equal to 7 mol. % to less than or equal to 15 mol. %, greater than or equal to 7 mol. % to less than or equal to 14 mol. %, greater than or equal to 7 mol. % to less than or equal to 13 mol. %, greater than or equal to 7 mol. % to less than or equal to 12 mol. %, greater than or equal to 7 mol. % to less than or equal to 11 mol. %, greater than or equal to 7 mol. % to less than or equal to 10 mol. %, greater than or equal to 7 mol. % to less than or equal to 9 mol. %, greater than or equal to 7 mol. % to less than or equal to 8 mol. %, greater than or equal to 7 mol. % to less than or equal to 7 mol. %, or any and all sub-ranges formed from any of these

endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of ZnO.

[0156] In embodiments, the precursor glass or glass-ceramic composition may comprise FeO. While not wishing to be bound by theory, it is believed that additions of FeO may result in the replacement of at least a portion of the Mg in the jeffbenite crystalline structure with Fe. FeO may also impart color to the precursor glass and glass-ceramic. In embodiments, the concentration of FeO in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 9 mol. %, greater than or equal to 0 mol. % to less than or equal to 8 mol. %, greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 9 mol. %, greater than or equal to 2 mol. % to less than or equal to 9 mol. %, greater than or equal to 3 mol. % to less than or equal to 9 mol. %, greater than or equal to 4 mol. % to less than or equal to 9 mol. %, greater than or equal to 5 mol. % to less than or equal to 9 mol. %, greater than or equal to 6 mol. % to less than or equal to 9 mol. %, greater than or equal to 7 mol. % to less than or equal to 9 mol. %, greater than or equal to 8 mol. % to less than or equal to 9 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of FeO.

[0157] In embodiments, the precursor glass or glass-ceramic composition may comprise CaO, SrO, or combinations thereof. Additions of CaO, SrO and combinations thereof may increase the amount of residual glass in the glass-ceramic. Without wishing to be bound by theory, including SrO in the glass-ceramic may increase the refractive index of the residual glass to better match the refractive index of the crystalline phase. In embodiments, the concentration of CaO, SrO, or a combination thereof may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 2.0 mol. %, greater than or equal to 0.2 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 0.4 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 0.6 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 0.8 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 1.0 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 1.2 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 1.4 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.7 mol. %, greater than or equal to 0.2 mol. % to less than or equal to 1.7 mol. %, greater than or equal to 0.4 mol. % to less than or equal to 1.7 mol. %, greater than or equal to 0.6 mol. % to less than or equal to 1.7 mol. %, greater than or equal to 0.8 mol. % to less than or equal to 1.7 mol. %, greater than or equal to 1.0 mol. % to less than or equal to 1.7 mol. %, greater than or equal to 1.2 mol. % to less than or equal to 1.7 mol. %, greater than or equal to 1.4 mol. % to less than or equal to 1.7 mol. %, greater than or equal to 1.7 mol. %, or any and all sub-ranges formed from any of these endpoints.

0 mol. % to less than or equal to 1.4 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.0 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.8 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.6 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.4 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.2 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the concentration of CaO may be greater than or equal to 0 mol. % to less than or equal to 10 mol. %, greater than or equal to 1 mol. % to less than or equal to 10 mol. %, greater than or equal to 2 mol. % to less than or equal to 10 mol. %, greater than or equal to 3 mol. % to less than or equal to 10 mol. %, greater than or equal to 4 mol. % to less than or equal to 10 mol. %, greater than or equal to 5 mol. % to less than or equal to 10 mol. %, greater than or equal to 6 mol. % to less than or equal to 10 mol. %, greater than or equal to 7 mol. % to less than or equal to 10 mol. %, greater than or equal to 8 mol. % to less than or equal to 10 mol. %, greater than or equal to 9 mol. % to less than or equal to 10 mol. %, greater than or equal to 0 mol. % to less than or equal to 9 mol. %, greater than or equal to 0 mol. % to less than or equal to 8 mol. %, greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, or any and all subranges formed from any of these endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of CaO. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of SrO. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of both CaO and SrO.

[0158] In embodiments, the precursor glass or glass-ceramic composition may comprise Cs_2O . Without wishing to be bound by theory, it is believed that additions of Cs_2O remain in the residual glass following ceramming and function to raise the index of refraction of the residual glass without causing crystallization. In embodiments, the concentration of Cs_2O in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of Cs_2O .

[0159] In embodiments, the precursor glass or glass-ceramic composition may comprise Li_2O . For example, without limitation, the precursor glass or glass-ceramic composition may comprise less than or equal to 3 mol. %, less than

or equal to 2.5 mol. %, less than or equal to 2 mol. %, less than or equal to 1.5 mol. %, less than or equal to 1 mol. %, less than or equal to 0.7 mol. %, less than or equal to 0.5 mol. %, less than or equal to 0.3 mol. %, or less than or equal to 0.1 mol. % Li_2O , but a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), or only tramp amounts at most (i.e. less than 0.05 mol. %). In embodiments, the precursor glass or glass-ceramic composition may be substantially free of Li_2O . Without intending to be bound by theory, when the precursor glass comprises less than or equal to 3 mol. % Li_2O the formation of beta quartz during heat treatment may be reduced.

[0160] In embodiments, the precursor glass or glass-ceramic composition may comprise Y_2O_3 . Without wishing to be bound by theory, Y_2O_3 may stabilize ZrO_2 included in the precursor glass or glass-ceramic composition. In embodiments, the concentration of Y_2O_3 may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 2.0 mol. %, greater than or equal to 0.2 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 0.4 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 0.6 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 0.8 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 1.0 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 1.2 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 1.4 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 1.6 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 1.8 mol. % to less than or equal to 2.0 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.8 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.6 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.4 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.0 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.8 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.6 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.4 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.2 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of Y_2O_3 .

[0161] In embodiments, the precursor glass or glass-ceramic composition may be substantially free from P_2O_5 . In some embodiments, the precursor glass or glass-ceramic composition may be free from P_2O_5 . Without wishing to be bound by theory, inclusion of P_2O_5 in the precursor glass or glass-ceramic composition may result in the formation of magnesium phosphate and reduced formation of the crystalline phase having the jeffbenite crystalline structure.

[0162] In embodiments, the precursor glass or glass-ceramic composition may comprise P_2O_5 . While not wishing to be bound by theory, it is believed that the addition of P_2O_5 may improve diffusivity of the glass-ceramic composition during ion exchange processes. In embodiments, the concentration of P_2O_5 in the precursor glass or the glass-ceramic composition may be from greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3.5 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. %

to less than or equal to 2.5 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.5 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.5 mol. %, greater than or equal to 0.5 mol. % to less than or equal to 4 mol. %, greater than or equal to 1 mol. % to less than or equal to 4 mol. %, greater than or equal to 1.5 mol. % to less than or equal to 4 mol. %, greater than or equal to 2 mol. % to less than or equal to 4 mol. %, greater than or equal to 2.5 mol. % to less than or equal to 4 mol. %, greater than or equal to 3 mol. % to less than or equal to 4 mol. %, greater than or equal to 3.5 mol. % to less than or equal to 4 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the P_2O_5 comprised in a glass-ceramic composition may remain in the residual glass phase of the glass-ceramic composition. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of P_2O_5 .

[0163] In embodiments, the precursor glass or glass-ceramic composition may comprise MnO_2 . While not wishing to be bound by theory, it is believed that additions of MnO_2 may result in the replacement of at least a portion of the Mg in the jeffbenite crystalline structure with Mn. MnO_2 may also impart color to the precursor glass and glass-ceramic. For example, without limitation, the addition of MnO_2 may impart a black color to the glass-ceramic. In embodiments, the concentration of MnO_2 may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 10 mol. %, greater than or equal to 0 mol. % to less than or equal to 9 mol. %, greater than or equal to 0 mol. % to less than or equal to 8 mol. %, greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1.0 mol. %, greater than or equal to 0.1 mol. % to less than or equal to 1 mol. %, greater than or equal to 0.2 mol. % to less than or equal to 1.0 mol. %, greater than or equal to 0.4 mol. % to less than or equal to 1.0 mol. %, greater than or equal to 0.6 mol. % to less than or equal to 1.0 mol. %, greater than or equal to 0.8 mol. % to less than or equal to 1.0 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.8 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.6 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.4 mol. %, greater than or equal to 0 mol. % to less than or equal to 0.2 mol. %, from greater than or equal to 1 mol. % to less than or equal to 10 mol. %, from greater than or equal to 2 mol. % to less than or equal to 10 mol. %, from greater than or equal to 3 mol. % to less than or equal to 10 mol. %, from greater than or equal to 4 mol. % to less than or equal to 10 mol. %, from greater than or equal to 5 mol. % to less than or equal to 10 mol. %, from greater than or equal to 6 mol. % to less than or equal to 10 mol. %, from greater than or equal to 7 mol. % to less than or equal to 10 mol. %, from greater than or equal to 8 mol. % to less than or equal to 10 mol. %, from greater than or equal to 9 mol. % to less than or equal to 10 mol. %, or any and all sub-ranges formed from

any of these endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of MnO_2 .

[0164] In embodiments, the precursor glass or glass-ceramic composition may comprise La_2O_3 . While not wishing to be bound by theory, it is believed that the addition of La_2O_3 may increase the refractive index of the residual glass of a glass-ceramic composition, which may improve transparency of the glass-ceramic composition. In embodiments, the concentration of La_2O_3 in the precursor glass or glass-ceramic composition may be from greater than or equal to 0 mol. % to less than or equal to 7 mol. %, greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 5 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 3 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 7 mol. %, greater than or equal to 1 mol. % to less than or equal to 6 mol. %, greater than or equal to 1 mol. % to less than or equal to 5 mol. %, greater than or equal to 1 mol. % to less than or equal to 4 mol. %, greater than or equal to 1 mol. % to less than or equal to 3 mol. %, greater than or equal to 1 mol. % to less than or equal to 2 mol. %, greater than or equal to 2 mol. % to less than or equal to 7 mol. %, greater than or equal to 2 mol. % to less than or equal to 6 mol. %, greater than or equal to 2 mol. % to less than or equal to 5 mol. %, greater than or equal to 2 mol. % to less than or equal to 4 mol. %, greater than or equal to 2 mol. % to less than or equal to 3 mol. %, greater than or equal to 2 mol. % to less than or equal to 7 mol. %, greater than or equal to 3 mol. % to less than or equal to 6 mol. %, greater than or equal to 3 mol. % to less than or equal to 5 mol. %, greater than or equal to 3 mol. % to less than or equal to 4 mol. %, greater than or equal to 4 mol. % to less than or equal to 7 mol. %, greater than or equal to 4 mol. % to less than or equal to 6 mol. %, greater than or equal to 4 mol. % to less than or equal to 5 mol. % to less than or equal to 7 mol. %, greater than or equal to 5 mol. % to less than or equal to 6 mol. %, greater than or equal to 6 mol. % to less than or equal to 7 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of La_2O_3 .

[0165] The articles formed from the precursor glass or glass-ceramics described herein may be any suitable thickness depending on the particular application of the glass-ceramic. Glass-ceramic sheet embodiments may have a thickness T of from greater than or equal to 0.2 mm to less than or equal to 10 mm. In embodiments, the glass-ceramic sheet embodiments may have a thickness T of 6 mm or less, 5 mm or less, 4 mm or less, 3 mm or less, 2 mm or less, 1.0 mm or less, 750 μ m or less, 500 μ m or less, or 250 μ m or less. In embodiments, the glass-ceramic sheet embodiments may have a thickness T of from greater than or equal to 200 μ m to less than or equal to 5 mm, greater than or equal to 500 μ m to less than or equal to 5 mm, greater than or equal to 200 μ m to less than or equal to 4 mm, greater than or equal to 200 μ m to less than or equal to 2 mm, greater than or equal to 400 μ m to less than or equal to 5 mm, or greater than or equal to 400 μ m to less than or equal to 2 mm. It should be

understood that the thickness of the precursor glass or glass-ceramic article may be within a sub-range formed from any and all of the foregoing endpoints. Alternatively, the glass sheets may be thicker than 10 mm, such as for use in certain armored windows or other uses for example. Alternatively, containers and tubes comprising the glass or glass-ceramics disclosed herein may have such thicknesses as wall thicknesses, and rod and spheres and other articles comprising the glass or glass-ceramics disclosed herein may have similar thicknesses for example.

[0166] In embodiments, the precursor glass or glass-ceramic compositions described herein are ion exchangeable to facilitate strengthening the precursor glass or glass-ceramic. In typical ion exchange processes, smaller metal ions in the glass-ceramic are replaced or “exchanged” with larger metal ions of the same valence within a layer that is close to the outer surface of the glass-ceramic. The replacement of smaller ions with larger ions creates a compressive stress within the layer of the glass-ceramic. In embodiments, the metal ions are monovalent metal ions (e.g., Li^+ , Na^+ , K^+ , and the like), and ion exchange is accomplished by immersing the glass-ceramic in a bath comprising at least one molten salt of the larger metal ion that is to replace the smaller metal ion in the glass-ceramic. The ion exchange process or processes that are used to strengthen the glass-ceramic may include, but are not limited to, immersion in a single bath or multiple baths of like or different compositions with washing and/or annealing steps between immersions. In embodiments, the glass-ceramics may be ion exchanged by exposure to molten KNO_3 salt, molten NaNO_3 salt, or a mixture of molten salts comprising KNO_3 and NaNO_3 . If Na_2O is present in the precursor glass or glass-ceramic, Na^+ for K^+ ion exchange may occur in a KNO_3 salt bath or a salt bath comprising KNO_3 in combination with NaNO_3 . If Li_2O is present in the precursor glass or glass-ceramic, Na^+ for Li^+ ion exchange may occur in a NaNO_3 salt bath or a salt bath comprising NaNO_3 in combination with KNO_3 . In embodiments, the glass-ceramics may be ion exchanged in a molten salt bath at a bath temperature of from greater than or equal to 350°C . to less than or equal to 500°C . For example, without limitation, the glass-ceramics may be ion exchanged at a bath temperature of greater than or equal to 350°C . to less than or equal to 530°C ., greater than or equal to 375°C . to less than or equal to 530°C ., greater than or equal to 400°C . to less than or equal to 530°C ., greater than or equal to 425°C . to less than or equal to 530°C ., greater than or equal to 450°C . to less than or equal to 530°C ., greater than or equal to 475°C . to less than or equal to 530°C ., greater than or equal to 500°C . to less than or equal to 530°C ., greater than or equal to 350°C . to less than or equal to 500°C ., greater than or equal to 375°C . to less than or equal to 500°C ., greater than or equal to 400°C . to less than or equal to 500°C ., greater than or equal to 425°C . to less than or equal to 500°C ., greater than or equal to 450°C . to less than or equal to 500°C ., greater than or equal to 475°C . to less than or equal to 500°C ., greater than or equal to 350°C . to less than or equal to 475°C ., greater than or equal to 350°C . to less than or equal to 450°C ., greater than or equal to 350°C . to less than or equal to 425°C ., greater than or equal to 350°C . to less than or equal to 400°C ., greater than or equal to 350°C . to less than or equal to 375°C ., or any and all sub-ranges formed from any of these endpoints. The ion exchange time may be from greater than or equal to 1 hour to less than or equal to 48

hours. In embodiments, the ion exchange process may develop a surface compressive layer in the glass precursors or glass-ceramic compositions. The development of this surface compression layer is beneficial for achieving a better crack resistance and higher flexural strength compared to non-ion-exchanged materials. The surface compression layer has a higher concentration of the ions exchanged into the glass-ceramic article in comparison to the concentration of the ions exchanged into the glass-ceramic article for the body (i.e., the area not including the surface compression) of the glass-ceramic article.

[0167] In embodiments, the precursor glass and/or glass-ceramics may be ion exchanged to achieve a depth of compression of about $30\text{ }\mu\text{m}$ or greater, about $40\text{ }\mu\text{m}$ or greater, about $50\text{ }\mu\text{m}$ or greater, about $60\text{ }\mu\text{m}$ or greater, about $70\text{ }\mu\text{m}$ or greater, about $80\text{ }\mu\text{m}$ or greater, about $90\text{ }\mu\text{m}$ or greater, or about $100\text{ }\mu\text{m}$ or greater. In embodiments, the depth of compression may be greater than or equal to 3% of the thickness of the article formed from the precursor glass and/or the glass-ceramics, greater than or equal to 5% of the thickness, greater than or equal to 10% of the thickness, greater than or equal to 15% of the thickness, greater than or equal to 20% of the thickness, or even greater than or equal to 22% of the thickness. The development of this surface compression layer is beneficial for achieving a better crack resistance and higher flexural strength compared to non-ion-exchanged materials. The surface compression layer has a higher concentration of the ion exchanged into the precursor glass and/or glass-ceramic article in comparison to the concentration of the ion exchanged into the article for the body (i.e., area not including the surface compression) of the article.

[0168] In embodiments, the precursor glass and/or glass-ceramics are ion exchanged to achieve a central tension greater than or equal to 10 MPa. In embodiments, the central tension may be greater than or equal to 10 MPa and less than or equal to 200 MPa, greater than or equal to 20 MPa and less than or equal to 200 MPa, greater than or equal to 30 MPa and less than or equal to 200 MPa, greater than or equal to 40 MPa and less than or equal to 200 MPa, greater than or equal to 50 MPa and less than or equal to 200 MPa, greater than or equal to 60 MPa and less than or equal to 200 MPa, greater than or equal to 70 MPa and less than or equal to 200 MPa, greater than or equal to 80 MPa and less than or equal to 200 MPa, greater than or equal to 90 MPa and less than or equal to 200 MPa, greater than or equal to 100 MPa and less than or equal to 200 MPa, greater than or equal to 110 MPa and less than or equal to 200 MPa, greater than or equal to 120 MPa and less than or equal to 200 MPa, greater than or equal to 130 MPa and less than or equal to 200 MPa, greater than or equal to 140 MPa and less than or equal to 200 MPa, greater than or equal to 150 MPa and less than or equal to 200 MPa, greater than or equal to 160 MPa and less than or equal to 200 MPa, greater than or equal to 170 MPa and less than or equal to 200 MPa, greater than or equal to 180 MPa and less than or equal to 200 MPa, greater than or equal to 190 MPa and less than or equal to 200 MPa, or any and all sub-ranges formed from any of these endpoints.

[0169] In embodiments, the precursor glass or glass-ceramic can have a surface compressive stress in a range from greater than or equal to 100 MPa to less than or equal to 1 GPa, greater than or equal to 100 MPa to less than or equal to 950 MPa, greater than or equal to 100 MPa to less than

or equal to 900 MPa, greater than or equal to 100 MPa to less than or equal to 850 MPa, greater than or equal to 100 MPa to less than or equal to 800 MPa, greater than or equal to 100 MPa to less than or equal to 750 MPa, greater than or equal to 100 MPa to less than or equal to 700 MPa, greater than or equal to 100 MPa to less than or equal to 650 MPa, greater than or equal to 100 MPa to less than or equal to 600 MPa, greater than or equal to 100 MPa to less than or equal to 550 MPa, greater than or equal to 100 MPa to less than or equal to 500 MPa, greater than or equal to 100 MPa to less than or equal to 450 MPa, greater than or equal to 100 MPa to less than or equal to 400 MPa, greater than or equal to 100 MPa to less than or equal to 350 MPa, greater than or equal to 100 MPa to less than or equal to 300 MPa, greater than or equal to 100 MPa to less than or equal to 250 MPa, greater than or equal to 100 MPa to less than or equal to 200 MPa, greater than or equal to 100 MPa to less than or equal to 150 MPa, 150 MPa to less than or equal to 500 MPa, greater than or equal to 150 MPa to less than or equal to 450 MPa, greater than or equal to 150 MPa to less than or equal to 400 MPa, greater than or equal to 150 MPa to less than or equal to 350 MPa, greater than or equal to 150 MPa to less than or equal to 300 MPa, greater than or equal to 150 MPa to less than or equal to 250 MPa, greater than or equal to 150 MPa to less than or equal to 200 MPa, 200 MPa to less than or equal to 500 MPa, greater than or equal to 200 MPa to less than or equal to 450 MPa, greater than or equal to 200 MPa to less than or equal to 400 MPa, greater than or equal to 200 MPa to less than or equal to 350 MPa, greater than or equal to 200 MPa to less than or equal to 300 MPa, greater than or equal to 200 MPa to less than or equal to 250 MPa, 250 MPa to less than or equal to 500 MPa, greater than or equal to 250 MPa to less than or equal to 450 MPa, greater than or equal to 250 MPa to less than or equal to 400 MPa, greater than or equal to 250 MPa to less than or equal to 350 MPa, greater than or equal to 250 MPa to less than or equal to 300 MPa, 300 MPa to less than or equal to 500 MPa, greater than or equal to 300 MPa to less than or equal to 450 MPa, greater than or equal to 300 MPa to less than or equal to 400 MPa, greater than or equal to 300 MPa to less than or equal to 350 MPa, 350 MPa to less than or equal to 500 MPa, greater than or equal to 350 MPa to less than or equal to 450 MPa, greater than or equal to 350 MPa to less than or equal to 400 MPa, 400 MPa to less than or equal to 500 MPa, greater than or equal to 400 MPa to less than or equal to 450 MPa, greater than or equal to 450 MPa to less than or equal to 500 MPa or any and all sub-ranges formed from any of these endpoints. In embodiments, the precursor glass or glass-ceramic can have a surface compressive stress of about 100 MPa or greater, about 150 MPa or greater, about 200 MPa or greater, about 250 MPa or greater, about 300 MPa or greater, about 350 MPa or greater, about 400 MPa or greater, about 450 MPa or greater, or about 500 MPa or greater.

[0170] In embodiments, the process for making the glass-ceramic includes melting a batch of constituent components to form the precursor glass. The molten precursor glass may be poured into a mold. In embodiments, the mold may comprise steel. The precursor glass may be annealed. Pucks of the precursor glass may be sectioned and then heat treated to form the glass-ceramic.

[0171] Alternatively, the precursor glasses described herein can be manufactured from molten precursor glass and

formed into sheets via processes, including but not limited to, slot draw, float, rolling, and other sheet-forming processes known in the art.

[0172] In embodiments, the processes for making the glass-ceramic includes heat treating (also referred to herein as “ceramming”) the precursor glass at one or more preselected temperatures for one or more preselected times to induce glass homogenization and crystallization (i.e., nucleation and growth) of one or more crystalline phases (e.g., having one or more compositions, amounts, morphologies, sizes or size distributions, etc.). The one or more preselected temperatures may be less than 1500 K throughout the heat treating process. It should be noted that temperature may refer to temperatures within a range, not necessarily a static singular temperature. Similarly, it should be noted that crystal nucleation and/or growth may occur over continuous or multiple discrete heat treatments that add together to achieve a desired crystal growth. Without wishing to be bound by theory, the nucleating agents may function as or form nucleation sites at which nucleation and growth of grains of the crystalline phases occur, including nucleation and growth of grains of the crystalline phases having the jeffbenite crystalline structure. The nucleation sites are positioned and oriented within the precursor glass such that grains of the resultant crystalline phases, including grains of the crystalline phases having the jeffbenite crystalline structure, are homogeneously distributed throughout the resultant glass-ceramic and grow in random orientations, resulting in a glass-ceramic with isotropic material properties. In embodiments, the heat treatment may include heating the precursor glass in a heat treatment furnace at a rate of 1-10° C./min until the furnace reaches a first temperature. The first temperature of the furnace may be from greater than or equal to 700° C. to less than or equal to 950° C. In embodiments, the first temperature of the furnace may be greater than or equal to 700° C. to less than or equal to 950° C., greater than or equal to 710° C. to less than or equal to 950° C., greater than or equal to 730° C. to less than or equal to 950° C., greater than or equal to 750° C. to less than or equal to 950° C., greater than or equal to 770° C. to less than or equal to 950° C., greater than or equal to 790° C. to less than or equal to 950° C., greater than or equal to 810° C. to less than or equal to 950° C., greater than or equal to 830° C. to less than or equal to 950° C., greater than or equal to 850° C. to less than or equal to 950° C., greater than or equal to 870° C. to less than or equal to 950° C., greater than or equal to 890° C. to less than or equal to 950° C., greater than or equal to 910° C. to less than or equal to 950° C., greater than or equal to 930° C. to less than or equal to 950° C., greater than or equal to 950° C. to less than or equal to 950° C., greater than or equal to 700° C. to less than or equal to 930° C., greater than or equal to 700° C. to less than or equal to 910° C., greater than or equal to 700° C. to less than or equal to 890° C., greater than or equal to 700° C. to less than or equal to 870° C., greater than or equal to 700° C. to less than or equal to 850° C., greater than or equal to 700° C. to less than or equal to 830° C., greater than or equal to 700° C. to less than or equal to 810° C., greater than or equal to 700° C. to less than or equal to 790° C., greater than or equal to 700° C. to less than or equal to 770° C., greater than or equal to 700° C. to less than or equal to 750° C., greater than or equal to 700° C. to less than or equal to 730° C., greater than or equal to 700° C. to less than or equal to 710° C., or any and all sub-ranges formed from any of these endpoints. Unless otherwise indicated, the temperature of a heat treatment or

ion exchange treatment refers to the temperature of the environment to which the article is exposed (such as the furnace for a heat treatment or the molten salt bath for an ion exchange treatment). In embodiments, the processes for making the glass-ceramic includes maintaining the precursor glass at the first temperature for a first time in a range from 0.25 hours to 6 hours. For example, without limitation, the precursor glass may be maintained at the first temperature for a first time in a range from greater than or equal to 0.25 hours to less than or equal to 6 hours, greater than or equal to 0.5 hours to less than or equal to 6 hours, greater than or equal to 0.75 hours to less than or equal to 6 hours, greater than or equal to 1 hour to less than or equal to 6 hours, greater than or equal to 1.25 hours to less than or equal to 6 hours, greater than or equal to 1.5 hours to less than or equal to 6 hours, greater than or equal to 1.75 hours to less than or equal to 6 hours, greater than or equal to 2 hours to less than or equal to 6 hours, greater than or equal to 2.25 hours to less than or equal to 6 hours, greater than or equal to 2.5 hours to less than or equal to 6 hours, greater than or equal to 2.75 hours to less than or equal to 6 hours, greater than or equal to 3 hours to less than or equal to 6 hours, greater than or equal to 3.25 hours to less than or equal to 6 hours, greater than or equal to 3.5 hours to less than or equal to 6 hours, greater than or equal to 3.75 hours to less than or equal to 6 hours, greater than or equal to 4 hours to less than or equal to 6 hours, greater than or equal to 4.25 to less than or equal to 6 hours, greater than or equal to 4.5 hours to less than or equal to 6 hours, greater than or equal to 4.75 hours to less than or equal to 6 hours, greater than or equal to 5 hours to less than or equal to 6 hours, greater than or equal to 5.25 hours to less than or equal to 6 hours, greater than or equal to 5.5 hours to less than or equal to 6 hours, greater than or equal to 5.75 hours to less than or equal to 6 hours, greater than or equal to 0.25 hours to less than or equal to 5.75 hours, greater than or equal to 0.25 hours to less than or equal to 5.5 hours, greater than or equal to 0.25 hours to less than or equal to 5.25 hours greater than or equal to 0.25 hours to less than or equal to 5 hours, greater than or equal to 0.25 hours to less than or equal to 4.75 hours, greater than or equal to 0.25 hours to less than or equal to 4.5 hours, greater than or equal to 0.25 hours to less than or equal to 4.25 hours, greater than or equal to 0.25 hours to less than or equal to 4 hours, greater than or equal to 0.25 hours to less than or equal to 3.75 hours, greater than or equal to 0.25 hours to less than or equal to 3.5 hours, greater than or equal to 0.25 hours to less than or equal to 3.25 hours, greater than or equal to 0.25 hours to less than or equal to 3 hours, greater than or equal to 0.25 hours to less than or equal to 2.75 hours, greater than or equal to 0.25 hours to less than or equal to 2.5 hours, greater than or equal to 0.25 hours to less than or equal to 2.25 hours, greater than or equal to 0.25 hours to less than or equal to 1 hours, greater than or equal to 0.25 hours to less than or equal to 0.75 hours, greater than or equal to 0.25 hours to less than or equal to 0.5 hours, or any and all sub-ranges formed from any of these endpoints. In embodiments, heat treating the precursor glass in the heat treatment furnace at the first temperature for the first time may facilitate both nucleating and growing the desired crystalline phases in the precursor glass to form the glass-ceramic. In other embodiments, heat treating the precursor glass in the heat treatment furnace at the first temperature for the first time may facilitate nucleating the desired crystalline phases in the precursor glass and

a second heat treatment step is implemented to grow the nucleated crystalline phases in the precursor glass to form the glass-ceramic.

[0173] For example, in embodiments the heat treatment may include a second step of heating the precursor glass in the heat treatment furnace at a rate of 1-10° C./min until the furnace reaches a second temperature. The second temperature may be different than the first temperature. The second temperature of the furnace may be from greater than or equal to 750° C. to less than or equal to 950° C. In embodiments, the second temperature may be greater than or equal to 750° C. to less than or equal to 950° C., greater than or equal to 770° C. to less than or equal to 950° C., greater than or equal to 790° C. to less than or equal to 950° C., greater than or equal to 810° C. to less than or equal to 950° C., greater than or equal to 830° C. to less than or equal to 950° C., greater than or equal to 850° C. to less than or equal to 950° C., greater than or equal to 870° C. to less than or equal to 950° C., greater than or equal to 890° C. to less than or equal to 950° C., greater than or equal to 910° C. to less than or equal to 950° C., greater than or equal to 930° C. to less than or equal to 950° C., greater than or equal to 750° C. to less than or equal to 930° C., greater than or equal to 750° C. to less than or equal to 910° C., greater than or equal to 750° C. to less than or equal to 890° C., greater than or equal to 750° C. to less than or equal to 870° C., greater than or equal to 750° C. to less than or equal to 850° C., greater than or equal to 750° C. to less than or equal to 830° C., greater than or equal to 750° C. to less than or equal to 810° C., greater than or equal to 750° C. to less than or equal to 790° C., greater than or equal to 750° C. to less than or equal to 770° C., or any and all sub-ranges formed from any of these endpoints. In embodiments, the processes for making the glass-ceramic includes maintaining the precursor glass at the second temperature for a second time in a range from greater than or equal to 0.25 hours to less than or equal to 6 hours. For example, without limitation, the precursor glass may be maintained at the second temperature for a second time in a range from greater than or equal to 0.25 hours to less than or equal to 6 hours, greater than or equal to 0.5 hours to less than or equal to 6 hours, greater than or equal to 0.75 hours to less than or equal to 6 hours, greater than or equal to 1 hour to less than or equal to 6 hours, greater than or equal to 1.25 hours to less than or equal to 6 hours, greater than or equal to 1.5 hours to less than or equal to 6 hours, greater than or equal to 1.75 hours to less than or equal to 6 hours, greater than or equal to 2 hours to less than or equal to 6 hours, greater than or equal to 2.25 hours to less than or equal to 6 hours, greater than or equal to 2.5 hours to less than or equal to 6 hours, greater than or equal to 2.75 hours to less than or equal to 6 hours, greater than or equal to 3 hours to less than or equal to 6 hours, greater than or equal to 3.25 hours to less than or equal to 6 hours, greater than or equal to 3.5 hours to less than or equal to 6 hours, greater than or equal to 3.75 hours to less than or equal to 6 hours, greater than or equal to 4 hours to less than or equal to 6 hours, greater than or equal to 4.25 to less than or equal to 6 hours, greater than or equal to 4.5 hours to less than or equal to 6 hours, greater than or equal to 4.75 hours to less than or equal to 6 hours, greater than or equal to 5 hours to less than or equal to 6 hours, greater than or equal to 5.25 hours to less than or equal to 6 hours, greater than or equal to 5.5 hours to less than or equal to 6 hours, greater than or equal to 5.75 hours to less than or equal to 6 hours, greater

than or equal to 0.25 hours to less than or equal to 5.75 hours, greater than or equal to 0.25 hours to less than or equal to 5.5 hours, greater than or equal to 0.25 hours to less than or equal to 5.25 hours, greater than or equal to 0.25 hours to less than or equal to 5 hours, greater than or equal to 0.25 hours to less than or equal to 4.75 hours, greater than or equal to 0.25 hours to less than or equal to 4.5 hours, greater than or equal to 0.25 hours to less than or equal to 4.25 hours, greater than or equal to 0.25 hours to less than or equal to 4 hours, greater than or equal to 0.25 hours to less than or equal to 3.75 hours, greater than or equal to 0.25 hours to less than or equal to 3.5 hours, greater than or equal to 0.25 hours to less than or equal to 3.25 hours, greater than or equal to 0.25 hours to less than or equal to 3 hours, greater than or equal to 0.25 hours to less than or equal to 2.75 hours, greater than or equal to 0.25 hours to less than or equal to 2.5 hours, greater than or equal to 0.25 hours to less than or equal to 2.25 hours, greater than or equal to 0.25 hours to less than or equal to 1 hours, greater than or equal to 0.25 hours to less than or equal to 0.75 hours, greater than or equal to 0.25 hours to less than or equal to 0.5 hours, or any and all sub-ranges formed from any of these endpoints. Heat treating the precursor glass with nucleated crystalline phases in the heat treatment furnace at the second temperature for the second time facilitates growing the desired crystalline phases in the precursor glass to form the glass-ceramic.

[0174] In embodiments, heat-treating the precursor glass may further comprise heating the precursor glass in the heat treatment furnace to one or more subsequent furnace temperatures, such as from greater than or equal to 750° C. to less than or equal to 950° C., and holding the precursor glass at each subsequent furnace temperature for a time in a range, such as from greater than or equal to 0.25 hours to less than or equal to 6 hours.

[0175] In embodiments, heat-treating the precursor glass may occur at ambient pressure. In embodiments, heat-treating the precursor glass may occur at ambient atmospheric pressure (e.g., 101.325 kPa). In embodiments, heat-treating the precursor glass may occur at about 100 kPa. In one or more embodiments, heat treating the precursor glass may occur at a pressure of less than or equal to 15 GPa. For example, without limitation, heat-treating the precursor glass may occur at a pressure of less than or equal to 15 GPa, less than or equal to 10 GPa, less than or equal to 5 GPa, or less than or equal to 1 GPa. According to the methods described herein, jeffbenite may be formed in a glass-ceramic article without the need to such pressurization.

[0176] Following heat-treating (i.e., following the nucleation and growth of the crystalline phase(s) in the precursor glass), at least some of the grains of the crystalline phase having the jeffbenite crystalline structure overlap and interlock within one another within the residual glass phase. The grains of the crystalline phase having the jeffbenite crystalline structure may overlap and interlock with one another within the glass-ceramic of the body, such as to a degree that the glass-ceramic has a fracture toughness as disclosed herein, such as about 0.75 MPa·m^{1/2} or other such values as disclosed herein. In other aspects of the present disclosure, overlap and interlock of the jeffbenite crystalline structure occurs within the glass-ceramic of the body, but to a degree that does not provide such fracture toughness. In still other aspects, crystals of the jeffbenite crystalline structure may be

present but so sparse so as not to overlap or interlock with one another to any degree whatsoever within the glass-ceramic of the body.

[0177] In embodiments, the resultant glass-ceramic may be transparent, translucent, or opaque. In embodiments, the glass-ceramics have an average transmittance of ≥85% of light over the wavelength range from about 400 nm to about 1,000 nm at an article thickness of 0.85 mm. In embodiments, the average transmittance for the glass-ceramic is about 85% or greater, about 86% or greater, about 87% or greater, about 88% or greater, about 89% or greater, about 90% or greater, about 91% or greater, about 92% or greater, about 93% or greater for light over the wavelength range of about 400 nm to about 1000 nm at an article thickness of 0.85 mm.

[0178] In embodiments, the resultant glass-ceramic may be colored. Colored glass-ceramic articles may comprise at least one colorant in a colorant package that functions to impart a desired color to the glass-ceramic. The colorant package may comprise at least one of Au, Ag, Cr₂O₃, transition metal oxides (e.g., CuO, NiO, Co₃O₄, TiO₂, Cr₂O₃), rare earth metal oxides (e.g., CeO₂), and/or combinations thereof as colorants in the colorant package and the precursor glass or glass-ceramic may include greater than or equal to 1×10⁶ mol. %, such as greater than 0.0005 mol. %, such as greater than 0.001 mol. %, such as greater than or equal to 0.01 mol. %, such as greater than or equal to 0.1 mol. %, such as greater than or equal to 0.2 mol. %, such as greater than or equal to 0.01 mol. %, and/or less than or equal to 10 mol. % of colorant (i.e., the sum of all colorants in the colorant package), such as less than 5 mol. %, such as less than 2 mol. %, such as less than 1 mol. %, such as less than 0.5 mol. %, such as less than 0.25 mol. % in some cases. For example, a yellow glass or glass-ceramic may comprise rare earth metal oxides, such as greater than about 0.2 mol. % and less than 1 mol. %; a black or gray glass may comprise NiO and Co₃O₄ in such quantities; green glass or glass-ceramic may likewise comprise Cr₂O₃; and pink, red, or orange glasses or glass-ceramics may comprise small amounts of gold, such as greater than 1×10⁶ mol. % and less than 0.5 mol. %, in alignment with disclosure of U.S. application Ser. No. 17/691,813. Other colorants may be used as well in such quantities, such as iron oxides for green or brown glasses or glass-ceramics for example, manganese oxides for amber or purple, selenium for red, antimony for white, uranium for glowing colors, copper for red, tin for white, lead for yellow, and other colorants. However, some such colorants, such as lead, antimony, and selenium may be less desirable than others for example. Further redox of colorants, UV-light treatment, and particulates may contribute to colors of the glasses, and Applicant hereby incorporates by reference U.S. Application Nos. 63/433,060 filed Dec. 16, 2022; 63/433,065 filed Dec. 16, 2022; and 63/433,119 filed Dec. 16, 2022 herein in their entireties.

[0179] In embodiments, the colored glass-ceramic may form a glass-ceramic sheet. The colored glass-ceramic may be semi-translucent (e.g., >1% total transmittance, >5%, >10%, and/or <95%, <92%, <90%, <85%, <80%, <70%) at some, most, and/or all frequencies in the visible spectrum (e.g., 4×10¹⁴ to 8×10¹⁴ Hz; 380 to 750 nanometers wavelengths) at thicknesses disclosed herein. Further, total transmittance of at least one 10 nm wide band within a wavelength range from 380 nm to 750 nm (e.g., 380 to 390 nm; 390 to 400 nm; 385 to 395 nm; and/or 740 to 750 nm) is less

than 90%, such as less than 85%, less than 80%, less than 70%, less than 60%, and/or less than 50%, through the glass of the sheet at the primary thickness (e.g., about 1 mm, about 0.6 mm, about 0.8 mm, about 1.2 mm, about 1.6 mm, about 2 mm, where about in this context refers to within 0.2 mm).

[0180] In embodiments, the colored glass-ceramics may be opaque, such as a solid color (e.g., white, black, forest green) that is essentially not transmissive to visible light (e.g., less than 1% total transmittance between 380 to 750 nanometers wavelengths). The colored glass-ceramics may also be translucent, such as having transmission, >95%, >96%, and/or <100%) at some, most, and/or all frequencies in the visible spectrum (e.g., 4×10^{14} to 8×10^{14} Hz; 380 to 750 nanometers wavelengths) at thicknesses disclosed herein.

[0181] In embodiments, the color of a glass-ceramic may be described using CIELAB color space coordinates L^* , a^* , and b^* . In embodiments, the value for L^* may be from 0 to 100, from 10 to 100, from 20 to 100, from 30 to 100, from 40 to 100, from 50 to 100, from 60 to 100, from 70 to 100, from 80 to 100, from 90 to 100, from 0 to 90, from 0 to 80, from 0 to 70, from 0 to 60, from 0 to 50, from 0 to 40, from 0 to 30, from 0 to 20, from 0 to 10, or any and all sub-ranges formed from these endpoints. In embodiments, the L^* coordinate may be greater than or equal to 50, greater than or equal to 60, greater than or equal to 70, greater than or equal to 80, and less than or equal to 100. In embodiments, the value for a^* may be from -128 to 128, from -120 to 128, from -110 to 128, from -100 to 128, from -90 to 128, from -80 to 128, from -70 to 128, from -60 to 128, from -50 to 128, from -40 to 128, from -30 to 128, from -20 to 128, from -10 to 128, from 0 to 128, from 10 to 128, from 20 to 128, from 30 to 128, from 40 to 128, from 50 to 128, from 60 to 128, from 70 to 128, from 80 to 128, from 90 to 128, from 100 to 128, from 110 to 128, from 120 to 128, or any and all sub-ranges formed from these endpoints. In embodiments, the a^* coordinate may be greater than or equal to -100, greater than or equal to -90, greater than or equal to -80, greater than or equal to -70, greater than or equal to -60, greater than or equal to -50, greater than or equal to -40, greater than or equal to -30, or greater than or equal to -20. In embodiments, the a^* coordinate may be less than or equal to 100, less than or equal to 90, less than or equal to 80, less than or equal to 70, less than or equal to 60, less than or equal to 50, less than or equal to 40, less than or equal to 30, or less than or equal to 20. In embodiments, the value for b^* may be from -128 to 128, from -120 to 128, from -110 to 128, from -100 to 128, from -90 to 128, from -80 to 128, from -70 to 128, from -60 to 128, from -50 to 128, from -40 to 128, from -30 to 128, from -20 to 128, from -10 to 128, from 0 to 128, from 10 to 128, from 20 to 128, from 30 to 128, from 40 to 128, from 50 to 128, from 60 to 128, from 70 to 128, from 80 to 128, from 90 to 128, from 100 to 128, from 110 to 128, from 120 to 128, or any and all sub-ranges formed from these endpoints. In embodiments, the b^* coordinate may be greater than or equal to -100, greater than or equal to -90, greater than or equal to -80, greater than or equal to -70, greater than or equal to -60, greater than or equal to -50, greater than or equal to -40, greater than or equal to -30, or greater than or equal to -20. In embodiments, the b^* coordinate may be less than or equal to 100, less than or equal to 90, less than or equal to 80, less than

or equal to 70, less than or equal to 60, less than or equal to 50, less than or equal to 40, less than or equal to 30, or less than or equal to 20.

[0182] In embodiments, the resultant glass-ceramic may have a density from greater than or equal greater than or equal to 2.65 g/cm^3 to less than or equal to 2.95 g/cm^3 . In embodiments, the resultant glass-ceramic may have a density from greater than or equal greater than or equal to 2.50 g/cm^3 to less than or equal to 3.70 g/cm^3 . For example, without limitation, the glass-ceramic may have a density greater than or equal to 2.50 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 2.55 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 2.60 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 2.65 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 2.70 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 2.75 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 2.80 g/cm^3 to less than or equal to 3.70 , greater than or equal to 2.85 g/cm^3 to less than or equal to 3.70 , greater than or equal to 2.90 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 3.00 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 3.10 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 3.20 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 3.30 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 3.40 g/cm^3 to less than or equal to 3.70 g/cm^3 , greater than or equal to 2.50 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 2.55 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 2.60 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 2.65 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 2.70 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 2.75 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 2.80 g/cm^3 to less than or equal to 3.60 , greater than or equal to 2.85 g/cm^3 to less than or equal to 3.60 , greater than or equal to 2.90 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 3.00 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 3.10 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 3.20 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 3.30 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 3.40 g/cm^3 to less than or equal to 3.60 g/cm^3 , greater than or equal to 2.65 g/cm^3 to less than or equal to 2.95 g/cm^3 , greater than or equal to 2.70 g/cm^3 to less than or equal to 2.95 g/cm^3 , greater than or equal to 2.75 g/cm^3 to less than or equal to 2.95 g/cm^3 , greater than or equal to 2.80 g/cm^3 to less than or equal to 2.95 g/cm^3 , greater than or equal to 2.85 g/cm^3 to less than or equal to 2.95 g/cm^3 , greater than or equal to 2.90 g/cm^3 to less than or equal to 2.95 g/cm^3 , greater than or equal to 2.75 g/cm^3 to less than or equal to 2.85 g/cm^3 , greater than or equal to 2.75 g/cm^3 to less than or equal to 2.80 g/cm^3 , or any and all sub-ranges formed from any of these endpoints. Without intending to be bound by theory, the presence of a crystalline phase having a jeffbenite crystalline structure in the glass-ceramic article may result in a glass-ceramic with a relatively high density when considering the relatively lightweight components of the glass-ceramic precursor.

[0183] In embodiments, the resultant glass-ceramic may comprise a crystalline phase having a jeffbenite crystalline structure. In embodiments, at least some of the grains of the crystalline phase having the jeffbenite crystalline structure in

the glass-ceramic, or even a majority of the grains, may have a dimension (e.g., measured from a sectioned/polished cut of glass-ceramic, where the “dimension” is a linear cross-sectional dimension, measured from opposite facing outermost surfaces of the grain through a geometric centroid of the grain along a surface of the sectioned/polished cut of glass-ceramic, such as longest cross-sectional dimension, shortest cross-sectional dimension, average cross-sectional dimension; unless otherwise specified, “dimension” in this context refers to the longest such cross-sectional dimension for a given grain; see grains shown in FIGS. 3-4, 6A-6F, and generally in FIG. 9) less than the wavelength of visible light. For example, without limitation, at least some of the grains of the crystalline phase having the jeffbenite crystalline structure, or even a majority of the grains, may have a dimension of less than or equal to 500 nm, 400 nm, 350 nm, 300 nm, 250 nm, 200 nm, 150 nm, or even less than or equal to 100 nm. In embodiments, at least some of the grains of the crystalline phase having the jeffbenite crystalline structure in the glass-ceramic, or even a majority of the grains, may have a dimension greater than or equal to 20 nm or even greater than or equal to 30 nm. For example, without limitation, at least some of the grains of the crystalline phase having the jeffbenite crystalline structure in the glass-ceramic, or even a majority, may have a dimension greater than or equal to 20 nm to less than or equal to 100 nm, greater than or equal to 30 nm to less than or equal to 100 nm, greater than or equal to 40 nm to less than or equal to 100 nm, greater than or equal to 50 nm to less than or equal to 100 nm, greater than or equal to 60 nm to less than or equal to 100 nm, greater than or equal to 70 nm to less than or equal to 100 nm, greater than or equal to 80 nm to less than or equal to 100 nm, greater than or equal to 90 nm to less than or equal to 100 nm, greater than or equal to 20 nm to less than or equal to 90 nm, greater than or equal to 30 nm to less than or equal to 90 nm, greater than or equal to 20 nm to less than or equal to 80 nm, greater than or equal to 30 nm to less than or equal to 80 nm, greater than or equal to 20 nm to less than or equal to 70 nm, greater than or equal to 30 nm to less than or equal to 70 nm, greater than or equal to 20 nm to less than or equal to 60 nm, greater than or equal to 30 nm to less than or equal to 60 nm, greater than or equal to 20 nm to less than or equal to 50 nm, greater than or equal to 30 nm to less than or equal to 50 nm, greater than or equal to 20 nm to less than or equal to 40 nm, greater than or equal to 30 nm to less than or equal to 40 nm, or any and all sub-ranges formed from any of these endpoints. In embodiments, at least some of the grains of the crystalline phase having the jeffbenite crystalline structure may be in an elongate form. In embodiments, at least some of the grains of the crystalline phase having the jeffbenite crystalline structure may comprise an acicular or tabular shape. In embodiments, the grains of the crystalline phase having the jeffbenite crystalline structure may comprise a bulk elastic modulus of about 170 GPa. In embodiments, the grains of the crystalline phase having the jeffbenite crystalline structure may comprise a hardness of about 1350 VHN.

[0184] In embodiments, the resultant glass-ceramic may comprise a phase assemblage in which at least some of the grains of the crystalline phases of the phase assemblage, or even a majority of the grains, have a dimension less than the wavelength of visible light. For example, without limitation, at least some of the grains of the crystalline phases in the phase assemblage, or even a majority, may have a dimension of less than or equal to 500 nm, 400 nm, 350 nm, 300 nm,

250 nm, 200 nm, 150 nm, or even less than or equal to 100 nm. In embodiments, at least some of the grains of the crystalline phases in the phase assemblage, or even a majority, may have a dimension greater than or equal to 20 nm or even greater than or equal to 30 nm. For example, without limitation, at least some of the grains of the crystalline phases in the glass-ceramic, or even a majority, may have a grain size greater than or equal to 20 nm to less than or equal to 100 nm, greater than or equal to 30 nm to less than or equal to 100 nm, greater than or equal to 40 nm to less than or equal to 100 nm, greater than or equal to 50 nm to less than or equal to 100 nm, greater than or equal to 60 nm to less than or equal to 100 nm, greater than or equal to 70 nm to less than or equal to 100 nm, greater than or equal to 80 nm to less than or equal to 100 nm, greater than or equal to 90 nm to less than or equal to 100 nm, greater than or equal to 20 nm to less than or equal to 90 nm, greater than or equal to 30 nm to less than or equal to 90 nm, greater than or equal to 20 nm to less than or equal to 80 nm, greater than or equal to 30 nm to less than or equal to 80 nm, greater than or equal to 20 nm to less than or equal to 70 nm, greater than or equal to 30 nm to less than or equal to 70 nm, greater than or equal to 20 nm to less than or equal to 60 nm, greater than or equal to 30 nm to less than or equal to 60 nm, greater than or equal to 20 nm to less than or equal to 50 nm, greater than or equal to 30 nm to less than or equal to 50 nm, greater than or equal to 20 nm to less than or equal to 40 nm, greater than or equal to 30 nm to less than or equal to 40 nm, or any and all sub-ranges formed from any of these endpoints.

[0185] In embodiments, the glass-ceramic may have an elastic modulus greater than or equal to 50 GPa and less than or equal to 200 GPa. In embodiments, the glass-ceramic may have an elastic modulus greater than or equal to 50 GPa, greater than or equal to 80 GPa, greater than or equal to 90 GPa, or even greater than or equal to 100 GPa. In embodiments, the glass-ceramic may have an elastic modulus less than or equal to 200 GPa or even less than or equal to 150 GPa. In embodiments, the glass-ceramic may have an elastic modulus greater than or equal to 50 GPa and less than or equal to 200 GPa, greater than or equal to 50 GPa and less than or equal to 175 GPa, greater than or equal to 60 GPa and less than or equal to 175 GPa, greater than or equal to 60 GPa and less than or equal to 150 GPa, greater than or equal to 70 GPa and less than or equal to 175 GPa, greater than or equal to 70 GPa and less than or equal to 150 GPa, greater than or equal to 80 GPa and less than or equal to 175 GPa, or even greater than or equal to 80 GPa and less than or equal to 150 GPa, or any and all sub-ranges formed from any of these endpoints.

[0186] In embodiments, the glass-ceramics exhibit a fracture toughness of about $0.75 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, about $0.85 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, about $1.0 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, about $1.1 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, $1.2 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, $1.3 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, $1.4 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, $1.5 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, $1.6 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, $1.7 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, $1.8 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, $1.9 \text{ MPa}\cdot\text{m}^{1/2}$ or greater, or about $2.0 \text{ MPa}\cdot\text{m}^{1/2}$. In embodiments, the fracture toughness is in the range from greater than or equal to $0.75 \text{ MPa}\cdot\text{m}^{1/2}$ to less than or equal to $2 \text{ MPa}\cdot\text{m}^{1/2}$, or any and all sub-ranges formed from any of these endpoints.

[0187] In one or more embodiments, the glass-ceramics have Vickers hardness. In some embodiments, a non-ion-exchanged glass-ceramic exhibits a Vickers hardness in the range from greater than or equal to 600 kgf/mm^2 to less than

or equal to 1400 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 1300 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 1200 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 1100 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 1000 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 900 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 875 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 850 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 825 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 800 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 775 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 750 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 725 kgf/mm², greater than or equal to 600 kgf/mm² to less than or equal to 700 kgf/mm², greater than or equal to 700 kgf/mm² to less than or equal to 900 kgf/mm², greater than or equal to 700 kgf/mm² to less than or equal to 875 kgf/mm², greater than or equal to 700 kgf/mm² to less than or equal to 850 kgf/mm², greater than or equal to 700 kgf/mm² to less than or equal to 825 kgf/mm², or greater than or equal to 700 kgf/mm² to less than or equal to 800 kgf/mm². In some embodiments, a Vickers hardness is 600 kgf/mm² or greater, 625 kgf/mm² or greater, 650 kgf/mm² or greater, 675 kgf/mm² or greater, 700 kgf/mm² or greater, 725 kgf/mm² or greater, 750 kgf/mm² or greater, 775 kgf/mm² or greater, 800 kgf/mm² or greater, 825 kgf/mm² or greater, 850 kgf/mm² or greater, 875 kgf/mm² or greater, or 900 kgf/mm² or greater, or any and all sub-ranges formed from any of these endpoints.

[0188] The resultant glass-ceramic may be provided as a sheet, which may then be reshaped or reformed by pressing, blowing, bending, sagging, vacuum forming, or other means into curved or bent pieces of uniform thickness. Reforming may be done before thermally treating or the forming step may also serve as a thermal treatment step in which both forming and thermal treating are performed substantially simultaneously.

[0189] The glass-ceramics and glass-ceramic articles described herein may be used for a variety of applications including, for example, for cover glass or glass backplane applications in consumer or commercial electronic devices including, for example, LCD and LED displays, computer monitors, and automated teller machines (ATMs); for touch screen or touch sensor applications, for portable electronic devices including, for example, mobile telephones, personal media players, and tablet computers; for integrated circuit applications including, for example, semiconductor wafers; for photovoltaic applications; for architectural glass applications; for automotive or vehicular glass applications; or for commercial or household appliance applications. In embodiments, a consumer electronic device (e.g., smart-phones, tablet computers, personal computers, ultrabooks, televisions, and cameras), an architectural glass, and/or an automotive glass may comprise a glass article as described herein.

[0190] An exemplary electronic device incorporating any of the glass-ceramic articles disclosed herein is shown in FIGS. 1 and 2. Specifically, FIGS. 1 and 2 show a consumer electronic device 100 including a housing 102 having a front surface 104, a back surface 106, and side surfaces 108; electrical components (not shown) that are at least partially

inside or entirely within the housing and including at least a controller, a memory, and a display 110 at or adjacent to the front surface of the housing; and a cover substrate 112 at or over the front surface of the housing such that it is over the display. In embodiments, at least a portion of at least one of the cover substrate 112 and the housing 102 may include any of the glass-ceramic articles disclosed herein.

[0191] Referring now to FIGS. 8A and 8B, an embodiment of a glass-ceramic article 200 formed from glass-ceramics described herein is depicted. In this embodiment, the glass-ceramic article is in the form of a glass sheet and may comprise a body 201 disposed between a first surface 202 opposite a second surface 204. The body 201 comprises or consists essentially of a glass-ceramic as described herein. The first surface 202 and the second surface 204 may be major surfaces of the glass sheet that face away from one another. In embodiments, the first surface 202 and the second surface 204 may be generally planar and spaced apart from one another by a thickness T defined between the first surface 202 and the second surface 204. The first surface 202 and the second surface 204 are delimited by at least one edge surface 206 forming a perimeter that generally defines a shape of the glass-ceramic article 200. In the embodiment depicted in FIGS. 8A and 8B, the glass-ceramic article is rectangular in shape and comprises a length L and a width W. In this embodiment, the width W of the glass sheet is defined as the distance along the first surface 202 orthogonal to the thickness T and between opposing edges. The length L is defined as the distance along the first surface 202 orthogonal to the thickness T and the width W and between opposing edges. In embodiments, the width W may be greater than or equal to the thickness T and the length L may be greater than or equal to the width W. However, it should be understood that other shapes of the glass-ceramic article are contemplated and possible, including, without limitation, squares, circles, and other regular or irregular geometric shapes. For example, a glass-ceramic article may also include a rod, a fiber, a boule, curved sheet, tube, bowl, lens, vial, bottle, or other container.

[0192] In embodiments, the thickness T of the glass-ceramic article 200 may be as described herein. The length L and the width W of the glass-ceramic article may be selected according to the specific application in which the glass-ceramic article 200 is employed. In embodiments, the length L and the width W of the glass-ceramic article may be greater than or equal to 5 mm, such as greater than or equal to 10 mm, greater than or equal to 15 mm, greater than or equal to 20 mm, greater than or equal to 25 mm, and greater than or equal to 30 mm. For example, and without limitation, the length L of the glass-ceramic article may be greater than or equal to 30 mm to less than or equal to 1 m, greater than or equal to 30 mm to less than or equal to 75 cm, greater than or equal to 30 mm to less than or equal to 50 cm, greater than or equal to 30 mm to less than or equal to 25 cm, greater than or equal to 30 mm to less than or equal to 20 cm, greater than or equal to 30 mm to less than or equal to 15 cm, greater than or equal to 30 mm to less than or equal to 10 cm, greater than or equal to 30 mm to less than or equal to 5 cm, or any and all sub-ranges formed from any of these endpoints. The width W of the glass-ceramic article may be greater than or equal to 30 mm to less than or equal to 1 m, greater than or equal to 30 mm to less than or equal to 75 cm, greater than or equal to 30 mm to less than or equal to 50 cm, greater than or equal to 30 mm to less than or equal to 25 cm,

greater than or equal to 30 mm to less than or equal to 20 cm, greater than or equal to 30 mm to less than or equal to 15 cm, greater than or equal to 30 mm to less than or equal to 10 cm, greater than or equal to 30 mm to less than or equal to 5 cm, or any and all sub-ranges formed from any of these endpoints. In embodiments where the glass-ceramic article is a glass sheet, as depicted in FIGS. 8A and 8B, the surfaces of the glass sheet may have an area greater than or equal to 25 mm². In embodiments where the glass-ceramic article is a glass sheet, the glass-ceramic in the body 201 of the glass sheet may have a volume greater than or equal to 25 mm³.

[0193] In embodiments, the body 201 of the glass-ceramic article 200 has an average transmittance of at least 75% for light in a wavelength range from 400 nm to 800 nm (inclusive of endpoints) at an article thickness of 0.85 mm such that the glass-ceramic article is transparent. In embodiments, the body 201 of the glass-ceramic article 200 has an average transmittance of greater than or equal to 20% to less than 75% for light in a wavelength range from 400 nm to 800 nm (inclusive of endpoints) at an article thickness of 0.85 mm such that the glass-ceramic article is translucent. In embodiments, the body of the glass-ceramic article has an average transmittance less than 20% when measured at normal incidence for light in a wavelength range from 400 nm to 800 nm (inclusive of endpoints) at an article thickness of 0.85 mm such that the glass-ceramic article is opaque. In embodiments, the body 201 of the glass-ceramic article 200 is at least partially translucent such that at least 20% of light having a wavelength of 400 nm to 800 nm directed into the thickness of the article is transmitted through the body.

Jeffbenite Characterization Method

[0194] In embodiments, a glass-ceramic article may be tested by X-ray diffraction to determine whether the glass-ceramic article comprises a jeffbenite crystalline phase, as described in detail herein. The test uses the following equipment and software: X-ray Diffractometer—Bruker-AXS D8 Endeavor equipped with a Cu radiation and a Lynx Eye detector, a Rocklabs Whisper Series Ring Mill, Backfill sample holders (Malvern Panalytical PW1770/10 Powder Sample Preparation Kit and PW18XX sample holder), 6" by 6" weigh papers, glass slides, a spatula, and data analysis software (MDI Jade, Bruker Topas, and Powder Diffraction File Database PDF-4).

[0195] A sample of a glass-ceramic article is received as small pieces of glass-ceramic. The pieces of glass-ceramic are broken up in the ring mill so that about 3 grams of the glass-ceramic may be obtained. The 3 gram sample of glass-ceramic is ground to a fine powder using the Rocklabs ring mill for a time of about 30 seconds.

[0196] A PW18XX sample holder is filled with the fine powder as described herein. The sample holder ring is clamped to a preparation table. The fine powder is spread in the sample holder ring so that the fine powder is heaped in a conical shape inside the holder ring. The fine powder is firmly pressed down in the holder ring using a glass slide. Any excess powder is scraped back into the holder ring using the glass slide. Additional fine powder may be added as necessary to fill the sample holder. This process is repeated until a densely packed powder specimen is obtained. Excess fine powder above the rim of the holder ring is removed using the edge of the glass slide. The bottom plate is placed onto the holder ring and clamped into

position. The complete sample holder is removed from the preparation table and loaded into the X-ray diffractometer.

[0197] A job file for the sample is created in XRD Commander. The job file includes the position of the sample in the X-ray diffractometer and identifying information for the sample. After the job file is created the X-ray diffractometer scans the sample.

[0198] After the sample is scanned by the X-ray diffractometer, the data obtained from the X-ray diffractometer is analyzed using MDI Jade software. A file containing the data from the X-ray diffractometer is opened in MDI Jade. The "Phases" window is used to identify indicator peaks in the data. Table 1 includes indicator peaks for at least some phases that may be present in glass-ceramics comprising jeffbenite. The PDF number for each phase is typed into the "PDF Recall" field. FIG. 14 depicts the location of the PDF Recall field in the Phases window of MDI Jade.

TABLE 1

Peaks (Angstroms)	Phase	Card Number
6.2, 3.2, 2.7	Jeffbenite	69-138
2.95, 2.54, 1.81	Tetragonal ZrO ₂	50-1089
3.87, 2.45	Forsterite	34-189

[0199] A large number of solid solutions are possible for some phases. The lattice parameter may be scaled so that the card data matches the sample data. To scale the lattice parameters, the name of the phase is selected in the phases tab and ascending size arrows on the right of the main display are selected. The location of the arrows is depicted in FIG. 15. The left and right mouse buttons may be used to increase and decrease the lattice constraints respectively.

[0200] The phases listed in Table 1 may be found in glass-ceramic samples comprising jeffbenite. If there are unidentified peaks in the data, the composition of the glass-ceramic and the peak locations may be used to search the PDF-4 database. After all the phases are identified and the lattice is adjusted to match the sample data, the "Assign Phases" functionality on the "Peaks" tab is used. The software lists the peaks and assigns phases to each peak. Some peaks may not have an assigned phase. This is illustrated by peak number 1 in the list of peaks depicted in FIG. 16. If a peak does not include an assigned phase, the peak is evaluated to determine whether it is a real peak or an artifact, either from the instrument or from the peak search algorithm. If the peak is a real peak, the PDF-4 database is searched manually to match the missing peaks and the chemistry of the sample. While the automated search match programs are robust, the high degree of solid solution in some glass-ceramic articles may be problematic for the automated matching program. Thus, an operator may have to undertake the steps described hereinabove.

[0201] A Rietveld Analysis is performed to determine lattice constants and cell volume using Topas. The file "Jeffbenite with hkl phase and ZrO2.pro" is opened in Topas. An initial refinement is run by clicking the red run arrow, as shown in FIG. 17. When the refinement converges, the parameters are kept. Then, the scan data from the X-ray diffractometer is loaded into Topas by clicking on the file name at the top of the file tree in the left hand window, selecting "Replace Scan Data," navigating to the folder where the data resides, and double clicking on the file containing the data to load the data. FIG. 18 shows the

location of “Replace Scan Data” in Topas. The file tree under the scan may be expanded so that all of the phases are visible. The model is set to represent the phases present in the current sample. This is done by clicking on the phase in the file tree then checking or unchecking the “Use Phase” box in the structure window. If the phase is present, the box is checked, and if the phase is not present, the box is unchecked. The location of the “Use Phase” box is depicted in FIG. 19. The Jeffbenite HKL phase is used in this step if there are no matching structures for the experimental jeffbenite in the database. The lattice parameters and cell volume in the Jeffbenite HKL phase are the best match and are used to characterize jeffbenite in the samples.

[0202] Once all of the structures and phases are correctly identified for the sample, the red run arrow is clicked, and Topas runs through the refinement until it converges on a solution. After the solution converges, the Rwp value, difference plot and Modeled Phase Data are analyzed. The Rwp value, Difference Plot, and Modeled Phase Data are depicted in FIG. 20. The Rwp value in the upper right window should be less than 5. In the Difference Plot, a curve between experimental data and modeled data (within oval) and stick patterns for the phases is shown. If there are peaks present in the difference curve, there is likely a phase missing. If there is a phase missing, the phase identification step of the analysis is repeated. The Modeled Phase Data is examined to determine if each phase looks reasonable. Reasonable looking phases stay below the background, do not have negative data points, and have peak widths that match experimental data. If the modeled phase data looks unreasonable, then the model is likely stuck in a local minimum. If the model is stuck in a local minimum, the Rietveld Analysis is restarted in Topas. If the Rwp value, Difference Plot, and Modeled Phase Data are acceptable, then the lattice constraints and cell volume for jeffbenite are correct, and the lattice constraints and cell volume for the sample are recorded.

[0203] In embodiments, the glass-ceramic article may have lattice parameters that correspond to jeffbenite and indicate a jeffbenite crystalline phase is present in the glass-ceramic article. For example, in embodiments, the glass-ceramic article may have an “a” lattice parameter greater than or equal to 6.5 Å and less than or equal to 6.7 Å or even greater than or equal to 6.6 Å and less than or equal to 6.7 Å. In embodiments, the glass-ceramic article may have a “c” lattice parameter greater than or equal to 18.0 Å and less than or equal to 18.5 Å, greater than or equal to 18.1 Å and less than or equal to 18.5 Å, greater than or equal to 18.2 Å and less than or equal to 18.5 Å, greater than or equal to 18.0 Å and less than or equal to 18.4 Å, greater than or equal to 18.1 Å and less than or equal to 18.4 Å, or even greater than or equal to 18.2 and less than or equal to 18.4 Å. In embodiments, the glass-ceramic article may have a lattice volume greater than or equal to 775 Å³ and less than or equal to 825 Å³, greater than or equal to 775 Å³ and less than or equal to 810 Å³, greater than or equal to 790 Å³ and less than or equal to 825 Å³, or even greater than or equal to 790 Å³ and less than or equal to 810 Å³. In embodiments, the glass-ceramic article may have a peak position greater than or equal to 2.55 Å and less than or equal to 2.75 Å, greater than or equal to 2.55 Å and less than or equal to 2.70 Å, greater than or equal to 2.60 Å and less than or equal to 2.75 Å, greater than or equal to 2.60 Å and less than or equal

to 2.70 Å, greater than or equal to 2.65 Å and less than or equal to 2.75 Å, or even greater than or equal to 2.65 Å and less than or equal to 2.70 Å.

[0204] In embodiments, the glass-ceramic article may have an X-ray diffraction (XRD) spectrum that corresponds to jeffbenite and indicates a jeffbenite crystalline phase is present in the glass-ceramic article. For example, in embodiments, the glass-ceramic article may have an XRD spectrum including a first peak between 2-theta angles of 30° to 32°; a second peak between 2-theta angles of 33° to 35°; a third peak between 2-theta angles of 40° to 42°; and a fourth peak and a fifth peak between 2-theta angles of 55° to 58°, wherein the first, second, third, fourth, and fifth peaks correspond to jeffbenite.

[0205] Without being bound to any theory, it is believed that the ability to produce, at temperatures and pressures disclosed herein (e.g., standard atmospheric pressure, about 101.325 kPa), what may otherwise be perceived as a high-pressure crystalline phase (e.g., nucleation and/or growth at 2.5 GPa) having jeffbenite crystalline structure, may be associated with alumina content of the precursor glass. More specifically, this ability may correspond with batching an amount of alumina less than stoichiometrically ideal for natural jeffbenite crystals, i.e. $Mg_xAl_ySi_3O_{12}$ where $x=3$ and $y=2$ (“nominal jeffbenite”), such as batching y greater than 0 but only approaching $y=2$, such as y preferably in a range of at least 0.5 to no more than 1.5, which may achieve a more desirable modulus for handling and forming (e.g., elastic modulus greater than or equal to 85 MPa), where x changes proportionally and zirconia or titania may be exchanged for extra magnesia, for example. In at least some crystals where locally available y of alumina is less than 2 in a mixture (e.g., amorphous glass, glass precursor to glass-ceramic), it is contemplated that at least some of the corresponding crystals (after nucleation and growth therein) contain half the aluminum stoichiometrically ideal for natural jeffbenite crystals, where the other half (i.e. not present) would therefore not need to be squashed under high pressures into the smallest octahedral site M2, and the present aluminum simply may go in the larger M3 site, e.g. $Mg_{3.5}(Zr,Ti)_0.5AlSi_3O_{12}$, where constituents in the parenthetical are in the alternative for the respective example crystal. Such an arrangement of constituents with a lesser amount of alumina in the batch may facilitate jeffbenite stability at lower pressures and temperatures, such as at perhaps even room temperature (e.g., 27° C.). With the above said, Applicants found very small amounts of a crystalline phase of jeffbenite crystalline structure when batching constituents at amounts stoichiometrically ideal for natural jeffbenite crystals, i.e. $Mg_3Al_2Si_3O_{12}$, and ceramming at atmospheric pressure. But, batching less than $y=1.5$ and/or greater than $y=0.5$, such as about half the aluminum content of nominal jeffbenite and heat treating at pressures disclosed herein (e.g., atmospheric) produced glass-ceramic with predominantly jeffbenite structure of the crystalline phase, as shown with examples disclosed herein.

[0206] In embodiments, the phase assemblage of the glass-ceramics described herein may comprise greater than or equal to 10 wt. % to less than or equal to 85 wt. % of the one or more crystalline phases by weight of the glass-ceramic article, such as crystalline phase comprising jeffbenite crystalline structure. In embodiments, the amount of the one or more crystalline phases by weight of the glass-ceramic article may be greater than or equal to 10 wt. %,

line phases may comprise tetragonal zirconia (ZrO_2), ZrTiO_4 , or a combination thereof. In embodiments, the amount of the one or more accessory crystalline phases in the one or more crystalline phases by total weight of the one or more crystalline phases may be greater than or equal to 0 wt. % to less than or equal to 10 wt. %. In embodiments, the amount of the one or more accessory crystalline phases in the one or more crystalline phases by total weight of the one or more crystalline phases may be greater than or equal to 0 wt. %, greater than or equal to 3 wt. %, or even greater than or equal to 5 wt. %. In embodiments, the amount of the one or more accessory crystalline phases in the one or more crystalline phases by total weight of the one or more crystalline phases may be less than or equal to 10 wt. %, less than or equal to 8 wt. %, less than or equal to 6 wt. %, less than or equal to 4 wt. %, or even less than or equal to 2 wt. %. In embodiments, the amount of the one or more accessory crystalline phases in the one or more crystalline phases by total weight of the one or more crystalline phases may be greater than or equal to 0 wt. % to less than or equal to 10 wt. %, greater than or equal to 0 wt. % to less than or equal to 8 wt. %, greater than or equal to 0 wt. % to less than or equal to 6 wt. %, greater than or equal to 0 wt. % to less than or equal to 4 wt. %, greater than or equal to 0 wt. % to less than or equal to 2 wt. %, greater than or equal to 3 wt. % to less than or equal to 10 wt. %, greater than or equal to 3 wt. % to less than or equal to 8 wt. %, greater than or equal to 3 wt. % to less than or equal to 6 wt. %, greater than or equal to 3 wt. % to less than or equal to 4 wt. %, greater than or equal to 0 wt. % to less than or equal to 10 wt. %, greater than or equal to 5 wt. % to less than or equal to 8 wt. %, or even greater than or equal to 5 wt. % to less than or equal to 6 wt. %, or any and all sub-ranges formed from any of these endpoints.

[0210] While not wishing to be bound by theory, it is believed that grain size may be increased by decreasing the concentration of the nucleating agent. In embodiments, at least some grains of the at least one crystalline phase comprising the jeffbenite crystalline structure may have a largest dimension greater than or equal to 20 nm to less than or equal to 500 nm. In embodiments, at least some grains of the at least one crystalline phase comprising the jeffbenite crystalline structure may have a largest dimension greater than or equal to 20 nm, greater than or equal to 40 nm, or even greater than or equal to 60 nm. In embodiments, at least some grains of the at least one crystalline phase comprising the jeffbenite crystalline structure may have a largest dimension less than or equal to 500 nm, less than or equal to 450 nm, less than or equal to 400 nm, less than or equal to 350 nm, less than or equal to 300 nm, less than or equal to 250 nm, or even less than or equal to 200 nm. In embodiments, at least some grains of the at least one crystalline phase comprising the jeffbenite crystalline structure may have a largest dimension greater than or equal to 20 nm to less than or equal to 500 nm, greater than or equal to 20 nm to less than or equal to 450 nm, greater than or equal to 20 nm to less than or equal to 400 nm, greater than or equal to 20 nm to less than or equal to 350 nm, greater than or equal to 20 nm to less than or equal to 300 nm, greater than or equal to 20 nm to less than or equal to 250 nm, greater than or equal to 20 nm to less than or equal to 200 nm, greater than or equal to 40 nm to less than or equal to 500 nm, greater than or equal to 40 nm to less than or equal to 450 nm, greater than or equal to 40 nm to less than or equal to 400 nm,

greater than or equal to 40 nm to less than or equal to 350 nm, greater than or equal to 40 nm to less than or equal to 300 nm, greater than or equal to 40 nm to less than or equal to 250 nm, greater than or equal to 40 nm to less than or equal to 200 nm, greater than or equal to 60 nm to less than or equal to 500 nm, greater than or equal to 60 nm to less than or equal to 450 nm, greater than or equal to 60 nm to less than or equal to 400 nm, greater than or equal to 60 nm to less than or equal to 350 nm, greater than or equal to 60 nm to less than or equal to 300 nm, greater than or equal to 60 nm to less than or equal to 250 nm, or even greater than or equal to 60 nm to less than or equal to 200 nm, or any and all sub-ranges formed from any of these endpoints.

[0211] In embodiments, the precursor glass or glass-ceramic composition may comprise from greater than or equal to 2.5 mol. % to less than or equal to 20 mol. % Al_2O_3 . In embodiments, the concentration of Al_2O_3 in the precursor glass or glass-ceramic composition may be a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), such as greater than or equal to 2.5 mol. % to less than or equal to 20 mol. %, greater than or equal to 2.5 mol. % to less than or equal to 18 mol. %, greater than or equal to 2.5 mol. % to less than or equal to 16 mol. %, greater than or equal to 2.5 mol. % to less than or equal to 14 mol. %, greater than or equal to 2.5 mol. % to less than or equal to 12 mol. %, greater than or equal to 4 mol. % to less than or equal to 20 mol. %, greater than or equal to 4 mol. % to less than or equal to 18 mol. %, greater than or equal to 4 mol. % to less than or equal to 16 mol. %, greater than or equal to 4 mol. % to less than or equal to 14 mol. %, greater than or equal to 4 mol. % to less than or equal to 12 mol. %, greater than or equal to 6 mol. % to less than or equal to 20 mol. %, greater than or equal to 6 mol. % to less than or equal to 18 mol. %, greater than or equal to 6 mol. % to less than or equal to 16 mol. %, greater than or equal to 6 mol. % to less than or equal to 14 mol. %, greater than or equal to 6 mol. % to less than or equal to 12 mol. %, greater than or equal to 8 mol. % to less than or equal to 20 mol. %, greater than or equal to 8 mol. % to less than or equal to 18 mol. %, greater than or equal to 8 mol. % to less than or equal to 16 mol. %, greater than or equal to 8 mol. % to less than or equal to 14 mol. %, or even greater than or equal to 8 mol. % to less than or equal to 12 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the concentration of Al_2O_3 in the precursor glass or glass-ceramic composition may be greater than or equal to 2.5 mol. %, 4 mol. %, 6 mol. %, or 8 mol. %. In embodiments, the concentration of Al_2O_3 may be less than or equal to 20 mol. %, 18 mol. %, 16 mol. %, 14 mol. %, or 12 mol. %.

[0212] In embodiments, the precursor glass or glass-ceramic composition may comprise B_2O_3 . Addition of B_2O_3 may increase the elastic modulus of the glass of the precursor glass and resultant glass-ceramic. B_2O_3 may also generally decrease diffusivity during ion exchange. However, while not wishing to be bound by theory, it is believed that B_2O_3 is present in the crystalline phase on the glass-ceramics described herein, thereby leading to relatively higher diffusivity during ion exchange as compared to a glass-ceramic having B_2O_3 present in the glass phase. In embodiments, the concentration of B_2O_3 in the precursor glass or glass-ceramic composition may be greater than or equal to 0 mol. %, such as a positive amount, such as an amount greater than tramp (0.05 mol. % or greater), to less than or

equal to 10 mol. %, greater than or equal to 0 mol. % to less than or equal to 8 mol. %, greater than or equal to 0 mol. % to less than or equal to 6 mol. %, greater than or equal to 0 mol. % to less than or equal to 4 mol. %, greater than or equal to 0 mol. % to less than or equal to 2 mol. %, greater than or equal to 0 mol. % to less than or equal to 1 mol. %, greater than or equal to 1 mol. % to less than or equal to 10 mol. %, greater than or equal to 1 mol. % to less than or equal to 8 mol. %, greater than or equal to 1 mol. % to less than or equal to 6 mol. %, greater than or equal to 1 mol. % to less than or equal to 4 mol. %, greater than or equal to 1 mol. % to less than or equal to 2 mol. %, greater than or equal to 2 mol. % to less than or equal to 10 mol. %, greater than or equal to 2 mol. % to less than or equal to 8 mol. %, greater than or equal to 2 mol. % to less than or equal to 6 mol. %, greater than or equal to 2 mol. % to less than or equal to 4 mol. %, greater than or equal to 4 mol. % to less than or equal to 10 mol. %, greater than or equal to 4 mol. % to less than or equal to 8 mol. %, greater than or equal to 4 mol. % to less than or equal to 6 mol. %, greater than or equal to 4 mol. % to less than or equal to 2 mol. %, or even greater than or equal to 8 mol. % to less than or equal to 10 mol. %, or any and all sub-ranges formed from any of these endpoints. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of B_2O_3 .

[0213] Referring once more to FIGS. 10, 30, and 31, according to an aspect of the present disclosure, at thicknesses disclosed herein, such as at 0.6 mm, glass-ceramics disclosed herein transmit light, having total transmittance across the visible spectrum between 380 to 700 nm wavelength greater than 30%, such as greater than 40%, such as greater than 50%. Some glass-ceramics are translucent, as labeled herein, and shown for example with GC23 in FIG. 31, having greater than 80% total transmittance over most of the visible spectrum for example. According to an aspect, glass-ceramics disclosed herein transmit light such that a band of light (e.g., 10 nm wide, 50 nm wide, 100 nm wide) between 380 to 700 nm wavelength has total transmittance greater than 70%, such as greater than 80%, greater than 85%, about 90% at the above thicknesses, as shown in FIG. 31. Similarly, the glass-ceramic includes band (e.g., 10 nm wide, 50 nm wide, 100 nm wide) between 700 nm to 1400 nm with total transmittance greater than 70%, such as greater than 80%, greater than 85%, about 90%. According to an aspect, at 400 nm wavelength at thicknesses disclosed herein, such as 0.6 mm, the glass-ceramic exhibits less than 30% diffuse transmission, such as less than 25%, such as less than 20%, such as less than 10%, about 5%. According to an aspect, at 700 nm wavelength at thicknesses disclosed herein, such as 0.6 mm, the glass-ceramic exhibits a scatter ratio of less than 30%, such as less than 25%, such as less than 20%, such as less than 10%, about 5%.

[0214] In embodiments, the resultant glass-ceramic may be colored. Colored glass-ceramic articles may comprise at least one colorant in a colorant package that functions to impart a desired color to the glass-ceramic. The colorant package may comprise at least one of Au, Ag, Cr_2O_3 , transition metal oxides (e.g., CuO, NiO, CoO, Co_3O_4 , TiO_2 , Cr_2O_3 , V_2O_5 , and MnO), rare earth metal oxides (e.g., CeO_2), and/or combinations thereof as colorants in the colorant package. In embodiments, the concentration of a colorant (e.g., any one of the above oxides individually, but

where a mixture may include more than one of such colorants present having such concentrations) in the precursor glass or the glass-ceramic composition may be greater than or equal to 1×10^{-6} mol. % (where x in such a use means x or 'multiplied by'), greater than or equal to 1×10^{-5} mol. %, greater than or equal to 1×10^{-4} mol. %, greater than or equal to 0.001 mol. %, greater than or equal to 0.01 mol. %, or even greater than or equal to 0.1 mol. %; and/or may be less than or equal to 10 mol. %, less than or equal to 5 mol. %, less than or equal to 3 mol. %, less than or equal to 2 mol. %, less than or equal to 1 mol. %, less than or equal to 0.5 mol. %, less than or equal to 0.1 mol. %, or even less than or equal to 0.01 mol. % or any and all sub-ranges formed from any of these endpoints, for example the concentration of a colorant in the precursor glass or the glass-ceramic composition may be greater than or equal to 1×10^6 mol. % to less than or equal to 2 mol. %.

[0215] In embodiments, the color of a glass-ceramic may be described using CIELAB color space coordinates L^* , a^* , and b^* , which may be at least partially controlled by the amount of colorants in the batch for example. In embodiments, the value for L^* may be greater than or equal to 0, greater than or equal to 20, greater than or equal to 40, greater than or equal to 60, or even greater than or equal to 80, 85, 90, 95, 96, 97, 98, or even greater than or equal to 99 and/or the value of L^* may be less than or equal to 100, less than or equal to 80, less than or equal to 60, less than or equal to 40, or even less than or equal to 20, 99, 98, 97, 96, 95, 90, or even less than or equal to 85, or any and all sub-ranges formed from any of these endpoints. In embodiments, the value for a^* and b^* of CIELAB coordinates may each or either be greater than or equal to -128, greater than or equal to -120, greater than or equal to -100, greater than or equal to -80, greater than or equal to -60, greater than or equal to -40, greater than or equal to -20, greater than or equal to 0, greater than or equal to 20, greater than or equal to 40, greater than or equal to 60, greater than or equal to 80, greater than or equal to 100, or even greater than or equal to 120, and/or the value for a^* and/or b^* may be less than or equal to 127, less than or equal to 120, less than or equal to 100, less than or equal to 80, less than or equal to 60, less than or equal to 40, less than or equal to 20, less than or equal to 0, less than or equal to -20, less than or equal to -40, less than or equal to -60, less than or equal to -80, less than or equal to -100, or even less than or equal to -120, or any combination thereof; such as a^* greater than or equal to -128 to less than or equal to 127 and b^* greater than or equal to -128 to less than or equal to 100, and L^* greater than 20 and less than 99 or any and all sub-ranges formed from any of these endpoints.

[0216] In embodiments, the precursor glass or glass-ceramic composition may include other components, such as a chemical fining agent. Such fining agents include, but are not limited to, SnO_2 , As_2O_3 , Sb_2O_3 , F, Cl and Br. In some embodiments, the concentrations of the chemical fining agents are kept at a level of 3, 2, 1, or 0.5, >0 mol. %. In embodiments, the precursor glass or glass-ceramic composition may be substantially free of a chemical fining agent.

[0217] In embodiments, the precursor glass or glass-ceramic compositions described herein allow for a relatively quick ion exchange. Without being bound by any particular theory, Applicants contemplate that the particularly high rate of ion-exchanging may be at least in part due to local tension within residual glass produced as the crystals form and

contract, the tension drawing open the molecular network for faster exchange. For example, in embodiments, the glass-ceramic composition may have a diffusivity during a Na^+ for K^+ ion exchange process in a 450°C . salt bath greater than or equal to $200\ \mu\text{m}^2/\text{hour}$, greater than or equal to $250\ \mu\text{m}^2/\text{hour}$, greater than or equal to $300\ \mu\text{m}^2/\text{hour}$, greater than or equal to $350\ \mu\text{m}^2/\text{hour}$, greater than or equal to $400\ \mu\text{m}^2/\text{hour}$, greater than or equal to $450\ \mu\text{m}^2/\text{hour}$, or even greater than or equal to $500\ \mu\text{m}^2/\text{hour}$. In embodiments, the glass-ceramic composition may have a diffusivity during a Na^+ for K^+ ion exchange process in a 550°C . salt bath greater than or equal to $1500\ \mu\text{m}^2/\text{hour}$, greater than or equal to $1550\ \mu\text{m}^2/\text{hour}$, greater than or equal to $1600\ \mu\text{m}^2/\text{hour}$, greater than or equal to $1650\ \mu\text{m}^2/\text{hour}$, greater than or equal to $1700\ \mu\text{m}^2/\text{hour}$, or even greater than or equal to $1750\ \mu\text{m}^2/\text{hour}$.

[0218] As used herein, “depth of layer” (DOL) refers to the depth within a glass article at which an ion of metal oxide diffuses into the glass article where the concentration of the ion reaches a minimum value. DOL values given herein are calculated from diffusivity measurements but may alternatively be measure by surface stress meter (FSM). In embodiments, the precursor glass and/or glass-ceramic may be ion exchanged to achieve a depth of layer greater than or equal to 5%, greater than or equal to 10%, greater than or equal to 20%, or even greater than or equal to 30% of the thickness of the article formed from the precursor glass and/or the glass-ceramic.

[0219] It should be understood that various shapes of the glass-ceramic article are contemplated and possible. For example, a glass-ceramic article may not include a rod, a fiber, a boule, curved sheet, tube, bowl, lens, vial, bottle, or other container.

[0220] In embodiments, the precursor glass or glass-ceramic composition may be powdered to allow for formation of a sintered body.

[0221] In embodiments, the glass-ceramic article may include or be incorporated into a laminate or layered product. Porous or powdered forms of the glass-ceramic may be incorporated into a composite material, such as a polymer composite.

[0222] According to an aspect of the present disclosure, Applicants contemplate that the glasses and glass-ceramics disclosed herein may be formed as a sheet having major surfaces and edges extending therebetween and defining a perimeter of the sheet (see generally body **201** in FIG. **8A**, first surface **202**, second surface **204** in FIG. **8B**, and edge surface **206**). According to an aspect, thickness (see generally thickness **T** in FIG. **8B**) varies very little between the major surfaces of the sheet, such that a difference in maximum and minimum values of thickness is close to zero for the full sheet, or as close as possible thereto while still being measurable under ASTM F657. Such high levels of geometric consistency may be achievable for sheets of small surface area, such as with careful polishing. However, according to an aspect, the major surfaces of at least some sheets formed from the glasses and glass-ceramics disclosed herein have at least some total thickness variation therebetween, such as greater than $0.1\ \mu\text{m}$, greater than $0.2\ \mu\text{m}$, greater than $0.5\ \mu\text{m}$, greater than $1\ \mu\text{m}$, and/or less than $1\ \text{mm}$, such as less than $500\ \mu\text{m}$, such as less than $200\ \mu\text{m}$, for sheets with geometries in alignment with the discussion above with respect to FIGS. **8A** and **8B**. For example, such sheets may have total thickness variation greater than $0.1\ \mu\text{m}$

and/or less than $1\ \text{mm}$ (or other such values as just disclosed, and as may be controlled by polishing) and also, in the same sheet, one or more of the following three types of geometric attributes, depending upon how the glass or glass-ceramic is formed or cut: (1) surface(s) with a total area of greater than $25\ \text{mm}^2$ and no more than $10,000\ \text{m}^2$; (2) a body with volume greater than $25\ \text{mm}^3$ and no more than $100\ \text{m}^3$; and (3) a surface-to-surface dimension extending through a geometric centroid of the article that is greater than $5\ \text{mm}$ and less than $5\ \text{km}$.

[0223] According to an aspect, the sheet may be circular, having a single continuous edge forming a perimeter between major surfaces thereof. However, according to an aspect the sheet may have corners (or vertices) partitioning the perimeter to form different sides of the sheet, such as four-sided sheet as shown in FIG. **8A**. The corners of the glass or glass-ceramic sheet may be a sharp geometric intersection of two lines corresponding to sides of the perimeter, forming a crisp point or vertex, which may be achieved nearly to the atomic level by very careful polishing. Or according to an aspect, at least one or more corners of a sheet or other glass or glass-ceramic article as disclosed herein may be dulled, such as rounded or fillet-shaped. As used herein “dulled” is meant to refer to geometry that is less than ideally sharp (or in terms of geometry, curvature does not include a discontinuity at such as corner or vertex). Such dulled corners may be rounded to have a constant radius of curvature, or may be rounded with a changing radius. In still other aspects, dulled corners of the sheet or other glass or glass-ceramic article may be dulled by beveling (e.g., chamfering). Dulling the corners of the sheet may make handling the sheet more manageable, for example. Similarly, according to an aspect, full edges between major surfaces of the sheet or other glass or glass-ceramic article form sharp/ideal right angles relative to the respective major surfaces and the side edges are nearly perfectly straight and fully at right angles to the major surfaces. However, according to an aspect, the edges may dulled as well, such as rounded, beveled, or filleted for example.

[0224] Similarly, according to an aspect, glass or glass-ceramic articles other than a sheet (e.g., tube, container, sphere, glass filling industrial glass-melting furnace tank, rolled sheet or ribbon) may be cut, polished, or formed to have such geometric attributes, such as the above disclosed total thickness variations in combination with one or more of the following attributes: (1) surface(s) with a total area of greater than $5\ \text{cm}^2$ and no more than $1000\ \text{m}^2$; (2) a body with volume greater than $50\ \text{mm}^3$ and no more than $1\ \text{m}^3$; and (3) a dimension extending from surface-to-surface through a geometric centroid of the article is greater than $5\ \text{cm}$ and less than $5\ \text{m}$. With that said, an aspect of the present disclosure also includes articles of the presently disclosed glass or glass-ceramic, e.g., powder, microspheres, having a surface area less than $25\ \text{mm}^2$ and/or a volume less than $25\ \text{mm}^3$. Similarly, other glass or glass-ceramic articles, such as lenses or kitchenware of complex geometry may purposely include thickness variation greater than $1\ \text{mm}$, but may have any combination of other geometric attributes disclosed above.

[0225] In embodiments, glasses or glass-ceramics may be coated with conductive carbon. According to an aspect, glasses or glass-ceramics disclosed herein may be coated with organic matter (such as other than 100% pure conductive carbon), such as carbon-based polymeric chains, and/or

another coating, such as oxides, nitrides, sulfides, metals, etc. The coating may comprise polymers for example, such as epoxy, acrylic, urethane, silicone, phenolic resin, polycarbonate, or others. The coating may comprise a thickness greater than 20 nm (as measured by cross-sectioning for example), such as greater than 50 nm, such as greater than 100 nm, and/or no more than 50 μm in many such as uses. Still other coatings may conceivably be thinner than 20 nm, such as those deposited by atomic layer deposition. According to an aspect, a method of making an article may comprise coating the glass or glass-ceramics disclosed herein, such as by sputtering, atomic layer deposition, spraying, vapor deposition, dipping, etc. The coating may provide a functional benefit to an article comprising the glasses or glass-ceramics, regardless of whether the article is in the form of a sheet. For example, coating may serve to improve scratch resistance, provide anti-reflective properties, change a coefficient of friction, etc.

[0226] According to an aspect of the present disclosure, referring again to FIGS. 1 and 2, a device **100** comprises the glass or, more preferably, the glass-ceramics disclosed herein. The glass or glass-ceramic may form most of the cover substrate **112** and/or the housing **102** in terms of volume thereof. For example, the cover substrate **112** may comprise a transparent glass-ceramic disclosed herein and the housing **102** may comprise an opaque and/or colored glass-ceramic disclosed herein. The cover substrate **112** or other article may be coated, such as with an anti-reflective coating (i.e. reduces reflection in visible), an anti-scratch coating (i.e. improves resistance to scratch), or another coating. The cover substrate **112** or other article may include decoration, such as paint or dye. Likewise the housing **102**, which may comprise the glass or glass-ceramic disclosed herein, may be coated or decorated. The cover substrate **112** or housing **102** may be adhered or otherwise bonded to another part of the device **100**, such as the electronics, such as the display **110**. According to an aspect, the glass or, more preferably, the glass-ceramics disclosed herein is in relatively close proximity to at least some of the electrical components, such as within 10 cm to the controller, memory, and/or display, such as within 1 cm, within 0.5 cm, 10 mm, and/or adjacent thereto or within direct contact thereof. According to an aspect, the device **100** comprises materials other than the glass or glass-ceramic, such as metal (e.g., circuitry, encasements, fasteners) and/or plastic and the glass or glass-ceramic disclosed herein is coupled thereto, either directly, such as with an adhesive or fastener for example, or indirectly, such as by way of other elements, such as an intermediate layer, guide, frame, or other feature. According to an aspect, Applicants contemplate the glass or glass-ceramics disclosed herein may be flexed or under a load in such a device **100** or assembly, where an external force is applied from a different element of the device onto the glass or glass-ceramic and the glass or glass-ceramic is at least partially deformed in strain or bending therefrom. According to an aspect, the device **100** includes an antenna (see generally squat rectangle on side surface **108** in FIG. 2), a conductive coil, a magnet, wires, a circuit, an electronic speaker and/or receiver (see generally round feature on top part of front surface **104** in FIG. 2), a port coupled to the housing **102** (see generally elongate on front surface **104**), an on/off switch (see generally circle surrounding square rectangle on bottom side surface **108**), a battery, a light source, and/or other elements, where the glass or glass-

ceramic disclosed herein may be proximate thereto and provide support therefor, such as protection from external environment, where the glass or glass-ceramic serves as a cover substrate **112**, a housing **102**, and/or as another element of the device **100**, such as a board of a printed circuit, or an internal partition, or other purposes.

[0227] According to an aspect of the present disclosure, prior to ceramming, the glass disclosed herein may be fined during melting and/or during manufacturing, such as with fining agents such as arsenic, antimony oxides, sodium sulfate, redox oxide, and sodium chloride or others as disclosed herein, removing many bubbles therefrom, such that resulting solidified glass is relatively free of blisters (i.e. trapped gas bubbles having a diameter greater than about $\frac{1}{16}$ inch, or >1.5 mm) and seeds (diameter less than about $\frac{1}{16}$ inch, or <1.5 mm). According to an aspect, articles (e.g., cover sheets, containers, windows, display glass sheets) disclosed herein may have fewer than 100 blisters, such as fewer than 10 blisters, such as fewer than 2 blisters, such as none. Articles disclosed herein may have fewer than 100 seeds having a diameter less than 1.5 mm and greater than 20 μm , such as fewer than 10 such seeds, such as fewer than 2 such seeds, such as none, and/or at least one. However, other aspects of the present disclosure include foamed articles or porous articles that may have many more seeds and blisters. While the glass and/or glass-ceramics disclosed herein may be relatively free of blisters and seeds, Applicants contemplate the glass and/or glass-ceramics may include at least one detectable seed, such as a seed having a diameter of less than 50 μm , such as less than 20 μm but greater than 20 nm, and/or detectable using microscopy for example. Similarly, glasses and glass-ceramics disclosed herein may be melted in a furnace, such as where batch constituents as disclosed above are thoroughly melted and mixed to form the glasses disclosed herein. As such, the glasses and/or glass-ceramics disclosed herein may be free of unmelted batch particles larger than 50 μm in largest cross-sectional dimension (e.g., length, width, height, diameter). However, Applicants contemplate that in some instances, glasses and glass-ceramics as disclosed herein may include at least one unmelted batch particle, such as one having a largest cross-sectional dimension less than 50 μm , such as less than 20 μm and/or greater than 20 nm.

[0228] According to an aspect of the present disclosure, a stack of the glass or more preferably the glass-ceramic disclosed herein (e.g., sheets thereof) may comprise an interleaf material (e.g., paper, polymer) to help partition layers of the glass or glass-ceramic from one another during shipping and handling. The stack may be housed in a crate comprising a wooden pallet or other frame at least partially surrounding the stack. Sheets within the stack may be packaged close to one another, such as within 1 cm of another sheet. The stack may comprise at least 2 of the sheets, such as at least 10.

EXAMPLES

[0229] In order that various embodiments be more readily understood, reference is made to the following examples, which are intended to illustrate various embodiments of the glass-ceramics described herein.

[0230] Table 2 shows example precursor glass compositions AA-DH (in terms of mol. %).

TABLE 2-continued

La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00

[illegible][illegible][illegible]

TABLE 2-continued

Batched Comp. (mol. %)	BQ	BR	BS	BT	BU	BV	BW
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SiO ₂	54.46	50.66	50.66	48.01	43.73	48.68	46.60
Al ₂ O ₃	9.18	8.43	8.43	8.61	8.99	10.07	11.25
MgO	23.37	28.87	28.87	33.64	39.55	32.02	33.48
Na ₂ O	4.65	4.25	4.25	3.18	2.36	3.24	2.81
K ₂ O	4.75	4.26	4.26	3.11	2.17	3.11	1.61
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.10	2.33	2.33	2.23	2.37	2.79	2.74
TiO ₂	0.37	1.08	1.08	1.12	0.73	0.00	1.41
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.12	0.11	0.11	0.10	0.10	0.10	0.09
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SiO ₂	45.05	42.81	42.40	42.86	54.46	54.46	54.43
Al ₂ O ₃	9.66	12.88	11.34	11.45	9.18	9.18	9.18
MgO	29.26	35.74	36.76	37.06	23.37	23.37	23.36
Na ₂ O	2.95	4.20	3.79	0.00	4.65	4.65	4.10
K ₂ O	1.44	0.00	0.00	2.87	4.75	4.75	5.33
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	8.04	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	2.39	2.85	3.52	4.29	3.10	3.10	3.11
TiO ₂	1.11	1.44	2.11	1.39	0.37	0.37	0.37
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.10	0.08	0.08	0.08	0.12	0.12	0.12
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SiO ₂	50.47	45.94	53.07	52.38	50.37	54.46	50.47
Al ₂ O ₃	9.67	10.94	9.34	9.43	9.61	9.18	9.67
MgO	29.52	33.33	25.33	26.34	29.21	23.37	29.52
Na ₂ O	3.68	2.98	4.35	4.23	3.68	4.65	3.68
K ₂ O	1.70	1.31	3.72	3.18	1.70	4.75	1.70
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	4.10	4.65	3.56	3.80	3.80	3.10	4.10
TiO ₂	0.77	0.77	0.55	0.55	1.54	0.37	0.77
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.08	0.12	0.08

TABLE 2-continued

Batched Comp. (mol. %)	CL	CM	CN	CO	CP	CQ	CR
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SiO ₂	46.60	53.07	52.38	50.37	43.50	42.96	47.47
Al ₂ O ₃	11.25	9.34	9.43	9.61	11.38	11.36	11.17
MgO	33.48	25.33	26.34	29.21	36.80	36.79	27.78
Na ₂ O	2.81	4.35	4.23	3.68	3.76	3.78	3.84
K ₂ O	1.61	3.72	3.18	1.70	0.00	0.00	0.00
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	5.69
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	2.74	3.56	3.80	3.80	3.01	3.26	2.84
TiO ₂	1.41	0.55	0.55	1.54	1.50	1.80	1.17
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.09	0.08	0.08	0.08	0.04	0.04	0.04
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SiO ₂	46.17	57.23	54.37	51.08	47.18	53.76	53.76
Al ₂ O ₃	11.00	3.12	9.16	10.41	11.86	11.84	10.52
MgO	29.39	18.96	23.33	28.07	32.98	18.78	21.19
Na ₂ O	3.22	10.27	4.65	3.46	2.58	6.11	5.28
K ₂ O	0.00	0.00	4.74	3.45	1.82	5.75	5.25
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	5.97	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.04	8.78	1.43	3.06	2.78	2.81	3.07
TiO ₂	1.17	1.59	2.21	0.36	0.71	0.82	0.80
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.04	0.04	0.12	0.10	0.09	0.13	0.13
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SiO ₂	39.99	52.39	48.51	52.21	53.50	42.91	50.90
Al ₂ O ₃	6.71	9.45	11.88	9.40	10.40	8.54	9.80
MgO	46.60	26.40	26.36	26.25	21.20	37.05	29.8
Na ₂ O	0.00	3.76	3.76	3.74	5.20	2.92	2.70
K ₂ O	0.00	3.74	3.30	3.72	5.20	0.00	2.70
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	4.01	3.07	3.78	3.66	3.00	4.29	2.10
TiO ₂	2.70	0.00	0.81	0.94	0.80	1.43	0.00
P ₂ O ₅	0.00	1.11	1.51	0.00	0.00	0.00	0.00
SnO ₂	0.00	0.08	0.09	0.08	0.10	0.00	0.10
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.60	2.86	0.00

TABLE 2-continued

HfO ₂	0.00	0.00	0.00	0.00	0.00	0.00	1.90
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Batched Comp. (mol. %)	DG		DH				
SiO ₂	50.80		57.67				
Al ₂ O ₃	9.80		11.72				
MgO	29.80		15.07				
Na ₂ O	2.70		7.03				
K ₂ O	2.70		4.69				
CaO	0.00		0.00				
SrO	0.00		0.00				
BaO	0.00		0.00				
ZnO	0.00		0.00				
FeO	0.00		0.00				
Li ₂ O	0.00		0.00				
Y ₂ O ₃	1.00		0.00				
ZrO ₂	2.10		3.70				
TiO ₂	0.00		0.00				
P ₂ O ₅	0.00		0.00				
SnO ₂	0.10		0.13				
MnO ₂	0.00		0.00				
La ₂ O ₃	0.00		0.00				
HfO ₂	0.00		0.00				
Ta ₂ O ₅	1.00		0.00				

[0231] Individual samples were formed by melting a batch of constituent components to form the precursor glass having the indicated composition. The molten precursor glass was then poured into a steel mold and cooled to form pucks. Pucks of the precursor glass were sectioned and then heat

treated to form the glass-ceramic. Samples of the glass-ceramic were approximately 1 cm thick (unless otherwise specified).

[0232] Properties of the glass-ceramics were determined, including the crystalline phases, the appearance of the sample, the % volume decrease upon crystallization of the sample (i.e., the shrinkage), the precursor glass density, the glass-ceramic (GC) density, the % increase in density, the elastic modulus, the Shear Modulus, Poisson's ratio, fracture toughness, and Vickers hardness. The crystalline phases of the glass-ceramics were determined by the "Jeffbenite Characterization Method" described hereinabove. The ceramic schedule for achieving the glass-ceramics GC1-GC97 and the respective properties of the glass-ceramics are shown in Tables 3-6.

[0233] When describing the appearance of the samples, the term "white" refers to glass-ceramics that were white and opaque. The term "opal" refers to glass-ceramics that were white and slightly translucent. The term "transparent opal" refers to glass-ceramics that were white but more translucent. Additionally, some samples listed in Tables 3-6 were transparent, such as the glass-ceramics GC22 and GC23 formed from Batched Comp. CX and Batched Comp. CY as shown in FIG. 30.

[0234] Tables 3-6 include several samples of glass-ceramics that include jeffbenite. Tables 3-6 also includes comparative examples, where the crystalline phase does not include jeffbenite. For example, some compositions in Tables 3-6 (i.e., Batched Comp. AE, BI, BN, etc.) include ZrO₂ as the crystalline phase and no jeffbenite is present.

TABLE 3

Glass-ceramic	GC1	GC2	GC3	GC4	GC5	GC6
Batched Comp.	AA	AA	AA	AB	AB	AV
Ceram Schedule (Temperature ° C.-Hours)	780-4, 850-4	850-4	800-4, 925-4	725-4, 850-4	800-4, 925-4	800-4, 900-4
Crystalline Phase	Jeffbenite	Jeffbenite	Jeffbenite, ZrO ₂	Jeffbenite	—	Jeffbenite, Mg-Titanate
Appearance	Transparent	—	Opaque	Translucent	Opaque	White
Glass-ceramic	GC7	GC8	GC9	GC10	GC11	GC12
Batched Comp.	AX	BA	BB	BE	CG	CH
Ceram Schedule (Temperature ° C.-Hours)	780-4, 850-4	780-4, 850-4	780-4, 850-4	740-4, 790-4	780-4, 850-4	780-4, 850-4
Crystalline Phase	Jeffbenite, ZrO ₂	Jeffbenite, ZrO ₂	Jeffbenite	Jeffbenite, ZrO ₂	Jeffbenite	Jeffbenite
Appearance	Translucent	Opal	—	Opal to Translucent	Translucent, Opal	Translucent, Opal
Glass-ceramic	GC13	GC14	GC15	GC16	GC17	GC18
Batched Comp.	CI	CP	CQ	CR	CS	CT
Ceram Schedule (Temperature ° C.-Hours)	750-4, 825-4	780-4, 850-4	780-4, 850-4	780-4, 850-4	780-4, 850-4	780-4, 850-4
Crystalline Phase	Jeffbenite	Jeffbenite, ZrO ₂	Jeffbenite, ZrO ₂	None (Glass only)	None (Glass only)	ZrO ₂
Appearance	Translucent	Opaque	Opaque	—	—	—

TABLE 3-continued

Glass-ceramic	GC19	GC20	GC21	GC22	GC23	GC24
Batched Comp.	CU	CV	CW	CX	CY	CZ
Ceram Schedule	725-4,	750-4,	800-4,	780-4,	780-4,	780-4,
(Temperature ° C.-	850-2	850-4	850-4	850-4	850-4	850-4
Hours)						
Crystalline Phase	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite
Appearance	—	—	—	Transparent	Transparent	—
Glass-ceramic	GC25	GC26	GC98	GC99	GC100	GC101
Batched Comp.	DA	DB	DD	DE	DF	DG
Ceram Schedule	780-4,	780-4,	780-4,	780-4,	780-4,	780-4,
(Temperature ° C.-	850-4	850-4	850-4	850-4	850-4	850-4
Hours)						
Crystalline Phase	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite,	Jeffbenite,
					ZrO ₂	ZrO ₂
Appearance	—	—	Transparent	Opaque	Opaque	Opaque
			with Haze			

TABLE 4

Glass-ceramic	GC27	GC28	GC29	GC30	GC31	GC32
Batched Comp.	AF	AG	AH	AK	AL	AM
Ceram Schedule	750-4,	765-6,	765-4,	800-4,	800-4,	780-4,
(Temperature ° C.-	800-4	825-2	800-6	900-4	900-4	850-4
Hours)						
Crystalline Phase	—	Jeffbenite,	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite
		ZrO ₂				
Appearance	—	Translucent	Hazy	Transparent	Translucent	Translucent
			Transparent	Areas		
% Volume Decrease	-2.66	-7.74	-19.11	-14.66	-11.73	-48.41
on Crystallization						
Glass-ceramic	GC33	GC34	GC35	GC36	GC37	GC38
Batched Comp.	AN	AP	AU	AW	BC	BD
Ceram Schedule	800-4,	750-4,	—	—	780-4,	740-4,
(Temperature ° C.-	900-4	800-4			850-4	790-4
Hours)						
Crystalline Phase	Jeffbenite,	Jeffbenite	—	—	Jeffbenite,	Jeffbenite,
	Y ₂ ZrO ₅				ZrO ₂	Mg ₂ Zr ₅ O ₁₂
Appearance	Opaque	White	—	—	—	Translucent
		Opaque				
% Volume Decrease	-17.09	-10.86	-17.3	-9.92	-22.49	-14.03
on Crystallization						

TABLE 5

Glass-ceramic	GC39	GC40	GC41	GC42	GC43	GC44
Batched Comp.	BH	BI	BJ	BK	BL	BS
Ceram Schedule	780-4,	780-4,	780-4,	780-4,	780-4,	775-4,
(Temperature ° C.-	900-4	850-4	900-4	900-4	850-4	850-4
Hours)						
Crystalline Phase	—	ZrO ₂	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite
Appearance	—	Translucent	Opaque,	Opaque,	Opaque,	Opaque,
			White	White	White	White
Glass Density	2.76	2.74	2.722	2.78	2.898	2.677
(g/cm ³)						
Glass-ceramic	2.96	2.74	2.85	2.97	2.97	—
Density (g/cm ³)						

TABLE 5-continued

Glass-ceramic	GC45	GC46	GC47	GC48	GC49	GC50
Batched Comp.	CE	CF	CJ	CK	CL	CM
Ceram Schedule (Temperature ° C.- Hours)	780-4, 850-4	780-4, 850-4	725-4, 850-2	780-4, 850-4	800-4, 900-4	780-4, 850-4
Crystalline Phase	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite
Appearance	Opaque	Opaque	Translucent	Opaque, Opal	Opaque, White	Translucent, Opal
Glass Density (g/cm ³)	2.753	2.827	2.65	2.753	2.75	2.686
Glass-ceramic Density (g/cm ³)	2.884	3.021	2.712	2.902	2.926	2.767

Glass-ceramic	GC51	GC52
Batched Comp.	CN	CO
Ceram Schedule (Temperature ° C.- Hours)	780-4, 850-4	780-4, 850-4
Crystalline Phase	Jeffbenite	Jeffbenite
Appearance	Translucent, Opal	—
Glass Density (g/cm ³)	2.705	2.753
Glass-ceramic Density (g/cm ³)	2.796	2.886

TABLE 6

Glass-ceramic	GC53	GC54	GC55	GC56	GC57	GC58
Batched Comp.	AA	AA	AE	AE	AB	AF
Ceram Schedule (Temperature ° C.- Hours)	725-4, 850-2	780-4, 900-4	750-4, 800-4	750-4, 825-4	750-4, 850-4	750-4, 825-4
Crystalline Phase	Jeffbenite, ZrO ₂	—	ZrO ₂	—	—	Jeffbenite
Appearance	Translucent	—	Translucent	—	—	Opaque
% Volume Decrease on Crystallization	-5.16	—	-6.99	—	-3.6	—
Glass Density (g/cm ³)	2.6409	—	2.721	—	—	—
Glass-ceramic Density (g/cm ³)	2.72	2.6843	—	2.823	2.818	2.841
% Increase	2.64	—	—	—	—	—
Glass-ceramic Mohs' Hardness	7.5	7.5	—	7.5	7.5	7.5
Elastic Modulus (GPa)	90.4	—	93.4	—	—	112.9
Shear Modulus (GPa)	37.5	—	37.3	—	—	45.2
Poisson's Ratio	0.21	—	0.252	—	—	0.248
Fracture Toughness (MPa√m)	0.85	—	—	—	—	—
Vickers Hardness (Kgf/mm ²)	680.9	—	—	—	—	—
Refractive Index	—	—	—	—	—	—

TABLE 6-continued

Glass-ceramic	GC59	GC60	GC61	GC62	GC63	GC64
Batched Comp.	AF	AI	AJ	AO	AQ	AR
Ceram Schedule (Temperature ° C.- Hours)	750-4, 850-4	800-4, 900-4	800-4, 900-4	780-4, 850-4	780-4, 850-4	780-4, 850-4
Crystalline Phase	—	Jeffbenite, Enstatite	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite
Appearance	—	White Opal	Opaque White	Translucent Opal	Black	Translucent
% Volume Decrease on Crystallization	-6.89	-10.43	-7.02	-22.13	-6.98	-4.89
Glass Density (g/cm ³)	—	2.76	2.762	2.92	2.86	2.8
Glass-ceramic Density (g/cm ³)	2.948	2.95	2.947	2.98	3.06	2.98
% Increase	—	6.7	—	2.05	6.99	6.43
Glass-ceramic Mohs' Hardness	7.5	—	—	—	—	—
Elastic Modulus (GPa)	—	120.9	120.9	—	132.8	124.9
Shear Modulus (GPa)	—	49.0	49.0	—	53.6	50.3
Poisson's Ratio	—	0.23	0.234	—	0.24	0.24
Fracture Toughness (MPa√m)	—	1.11	1.11	—	—	—
Vickers Hardness (Kg/mm ²)	—	—	827	—	—	918
Refractive Index	—	—	—	—	—	—
Glass-ceramic	GC65	GC66	GC67	GC68	GC69	GC70
Batched Comp.	AR	AS	AS	AY	AZ	BF
Ceram Schedule (Temperature ° C.- Hours)	780-4, 850-4	780-4, 850-5	780-4, 850-5	780-4, 875-4	780-4, 850-4	750-4, 800-4
Crystalline Phase	Jeffbenite	Jeffbenite, ZrO ₂	Jeffbenite	Jeffbenite, Unknown Phase, ZrO ₂	Jeffbenite, ZrO ₂	Jeffbenite
Appearance	Translucent Opal	White Opal	Opaque White	Grey-white	Translucent to Transparent	Translucent
% Volume Decrease on Crystallization	—	—	—	—	-5.4	-1.41
Glass Density (g/cm ³)	2.855	—	2.803	2.86	—	2.718
Glass-ceramic Density (g/cm ³)	3.06	—	2.983	3.1	—	—
% Increase	—	—	—	8.22	—	—
Glass-ceramic Mohs' Hardness	—	—	—	—	—	—
Elastic Modulus (GPa)	132.8	—	124.9	—	—	96.8
Shear Modulus (GPa)	53.6	—	50.3	—	—	38.7
Poisson's Ratio	0.239	—	0.241	—	—	0.25
Fracture Toughness (MPa√m)	1.02	—	1.04	—	—	—
Vickers Hardness (Kg/mm ²)	940	819	832	—	961	—
Refractive Index	—	—	—	—	—	—

TABLE 6-continued

Glass-ceramic	GC71	GC72	GC73	GC74	GC75	GC76
Batched Comp.	BF	BF	BG	BG	BG	BH
Ceram Schedule (Temperature ° C.- Hours)	750-4, 825-4	750-4, 850-4	750-4, 850-4	750-4, 800-4	750-4, 825-4	770-4, 820-4
Crystalline Phase	Jeffbenite	—	Jeffbenite, ZrO ₂	—	—	Jeffbenite
Appearance	—	—	Opaque, White	—	—	Opaque, White
% Volume Decrease on Crystallization	—	-6.67	-6.47	-4.75	—	-4.15
Glass Density (g/cm ³)	2.718	2.718	2.771	2.771	2.771	2.76
Glass-ceramic Density (g/cm ³)	2.802	2.841	2.951	—	2.782	2.88
% Increase	—	—	—	—	—	—
Glass-ceramic Mohs' Hardness	7.5	7.5	7.5	—	7.5	—
Elastic Modulus (GPa)	—	—	—	—	—	—
Shear Modulus (GPa)	—	—	—	—	—	—
Poisson's Ratio	—	—	—	—	—	—
Fracture Toughness (MPa√m)	—	—	—	—	—	—
Vickers Hardness (Kgf/mm ²)	—	—	—	—	—	—
Refractive Index	—	—	—	—	—	—

Glass-ceramic	GC77	GC78	GC79	GC80	GC81	GC82
Batched Comp.	BH	BM	BN	BO	BP	BQ
Ceram Schedule (Temperature ° C.- Hours)	760-4, 800-4	780-4, 900-4	750-4, 800-4	750-4, 800-4	725-4, 850-4	725-4, 850-4
Crystalline Phase	—	Jeffbenite	ZrO ₂	Jeffbenite	Jeffbenite	Jeffbenite
Appearance	—	Opaque, White	Translucent	Translucent	Translucent	Translucent
% Volume Decrease on Crystallization	-1.78	—	—	—	—	—
Glass Density (g/cm ³)	2.76	2.741	—	—	2.648	2.647
Glass-ceramic Density (g/cm ³)	—	2.925	—	—	2.718	—
% Increase	—	—	—	—	—	—
Glass-ceramic Mohs' Hardness	—	—	—	—	—	—
Elastic Modulus (GPa)	—	117.7	95.6	97.2	90.4	—
Shear Modulus (GPa)	—	47.9	38.3	39.0	37.5	—
Poisson's Ratio	—	0.23	0.249	0.247	0.206	—
Fracture Toughness (MPa√m)	—	1.03	0.92	1	0.846	—
Vickers Hardness (Kgf/mm ²)	—	—	716.5	709.7	680.9	—
Refractive Index	—	—	1.5683	—	—	1.5587

TABLE 6-continued

Glass-ceramic	GC83	GC84	GC85	GC86	GC87	GC88
Batched Comp.	BQ	BR	BR	BT	BU	BV
Ceram Schedule (Temperature ° C.- Hours)	775-4, 850-4	775-4, 850-4	800-4, 950-4	780-4, 850-4	780-4, 850-4	780-4, 850-4
Crystalline Phase	—	Jeffbenite	—	Jeffbenite	Jeffbenite	Jeffbenite
Appearance	—	Opaque, White	—	Opaque, White	Opaque, White	Opaque, White
% Volume Decrease on Crystallization	—	—	—	—	—	—
Glass Density (g/cm ³)	2.647	2.677	2.677	—	—	2.721
Glass-ceramic Density (g/cm ³)	—	2.755	2.765	—	—	2.83
% Increase	—	—	—	—	—	—
Glass-ceramic Mohs [*] Hardness	—	—	—	—	—	7.5
Elastic Modulus (GPa)	—	97.8	99.4	108.2	121.5	107.8
Shear Modulus (GPa)	—	40.0	40.7	44.1	49.2	44.0
Poisson's Ratio	—	0.221	0.221	0.225	0.235	0.226
Fracture Toughness (MPa√m)	—	0.839	0.85	0.914	1.036	0.99
Vickers Hardness (Kgf/mm ²)	—	720.2	731.1	799.4	750.2	780.2
Refractive Index	1.5581	—	—	—	—	1.5901
Glass-ceramic	GC89	GC90	GC91	GC92	GC93	GC94
Batched Comp.	BW	BX	BY	BZ	CA	CB
Ceram Schedule (Temperature ° C.- Hours)	800-4, 900-4	750-4, 800-4	780-4, 850-4	780-4, 850-4	780-4, 850-4	725-4, 850-2
Crystalline Phase	Jeffbenite	—	Jeffbenite	Jeffbenite	Jeffbenite	Jeffbenite
Appearance	Opaque, White	—	Opaque, White	Translucent Opal	Translucent Opal	Translucent
% Volume Decrease on Crystallization	—	—	—	—	—	—
Glass Density (g/cm ³)	2.766	2.916	2.808	2.853	2.855	—
Glass-ceramic Density (g/cm ³)	2.949	2.976	2.99	3.077	3.095	—
% Increase	—	—	—	—	—	—
Glass-ceramic Mohs [*] Hardness	7.5	7.5	7.5	7.5	7.5	—
Elastic Modulus (GPa)	121.4	119.7	126.2	134.2	133.5	93.4
Shear Modulus (GPa)	49.1	48.1	51.0	54.2	53.9	38.1
Poisson's Ratio	0.236	0.245	0.237	0.239	0.239	0.227
Fracture Toughness (MPa√m)	1.07	1.0475	1.09	1.05	1.04	0.83
Vickers Hardness (Kgf/mm ²)	826.8	—	819.3	918	961.2	—
Refractive Index	1.6185	1.6099	1.6288	1.646	1.6435	1.5018

TABLE 6-continued

Glass-ceramic	GC95	GC96	GC97
Batched Comp.	CC	CD	DH
Ceram Schedule (Temperature ° C.- Hours)	750-5, 850-4	750-5, 850-4	780-4, 850-4
Crystalline Phase Appearance	Jeffbenite Translucent	Jeffbenite Translucent	Jeffbenite Opaque
% Volume Decrease on Crystallization	—	—	—
Glass Density (g/cm ³)	—	—	—
Glass-ceramic Density (g/cm ³)	—	—	—
% Increase	—	—	—
Glass-ceramic Mohs' Hardness	—	—	—
Elastic Modulus (GPa)	—	—	—
Shear Modulus (GPa)	—	—	—
Poisson's Ratio	—	—	—
Fracture Toughness (MPa√m)	—	—	—
Vickers Hardness (Kg/mm ²)	—	—	—
Refractive Index	1.5581	1.5579	—

[0235] The lattice parameters of glass-ceramics formed from precursor glass compositions were determined according to the “Jeffbenite Characterization Method” described hereinabove. The batched composition, ceram schedule, corresponding glass-ceramic, and lattice parameters are given in Table 7.

P₂O₅, which may improve diffusivity during ion exchange processes. The XRD spectra for glass-ceramics GC25 and GC26 of Table 3 exemplify that jeffbenite may be formed in glass-ceramic articles including P₂O₅.

[0238] FIG. 3 depicts the microstructure of the glass-ceramic GC53. FIG. 4 depicts the microstructure of the

TABLE 7

Batched Comp.	Ceram Schedule (Temperature ° C.- Hours)	Glass- ceramic	Jeffbenite a Lattice (Å)	Jeffbenite c Lattice (Å)	Jeffbenite Lattice Vol. (Å ³)	(024) Peak Position (Å)
CU	725-4, 850-2	GC19	6.638	18.333	807.8	2.689
AA	850-4	GC2	6.626	18.254	801.5	2.681
AB	725-4, 850-2	—	6.634	18.300	805.4	2.686
BF	750-4, 825-4	GC71	6.608	18.227	795.9	2.675
CV	750-4, 850-4	GC20	6.588	18.143	787.5	2.665
CW	800-4, 850-4	GC21	6.610	18.276	798.5	2.677
CZ	780-4, 850-4	GC24	6.593	18.231	792.3	2.671

[0236] The XRD spectra for glass-ceramics were obtained according to the “Jeffbenite Characterization Method” described hereinabove. The XRD spectra for several glass-ceramic articles are included in the figures. The XRD spectrum for glass-ceramic GC19 of Table 7 is depicted in FIG. 21. The XRD spectrum for glass-ceramic GC2 of Table 7 is depicted in FIG. 22. The XRD spectrum for the glass-ceramic formed from heat treating Batched Comp. AB as shown in Table 7 is depicted in FIG. 23. The XRD spectrum for glass-ceramic GC71 of Table 7 is depicted in FIG. 24. The XRD spectrum for glass-ceramic GC20 of Table 7 is depicted in FIG. 25. The XRD spectrum for glass-ceramic GC21 of Table 7 is depicted in FIG. 26. The XRD spectrum for glass-ceramic GC24 of Table 7 is depicted in FIG. 27.

[0237] The XRD spectrum for glass-ceramic GC25 of Table 3 is depicted in FIG. 28. The XRD spectrum for glass-ceramic GC26 of Table 3 is depicted in FIG. 29. Glass-ceramics GC25 and GC26 of Table 3 both included

glass-ceramic GC4. FIG. 3 also depicts ZrO₂ crystals (lighter colored areas) in the microstructure of the glass-ceramic.

[0239] A sample of the glass-ceramic composition of Batched Comp. AB was analyzed by high temperature X-ray diffraction (XRD). The sample was held for one hour at each of the following temperatures: 800° C., 825° C., 850° C., 875° C., 900° C., 925° C., 950° C., 975° C., and 1000° C. The XRD spectrum of the sample was obtained in the last fifteen minutes during which the sample was held at a specific temperature. The XRD spectra are depicted in FIG. 5. The high temperature XRD analysis indicates the ranges of stability of various crystalline phases in the sample. As depicted in FIG. 5, the sample included a crystalline phase having a jeffbenite crystalline structure and a tetragonal ZrO₂ crystalline phase.

[0240] The surface of the glass-ceramic GC53 was observed by scanning electron microscopy (SEM). The sample appeared to be translucent. To enable imaging by

SEM, the surface of the sample was etched in 0.5% HF for 10 seconds. Then, a conductive carbon coating was evaporated on the sample to reduce charging. SEM images were taken of the surface of the sample using a Hitachi SU70, 5 kv scanning electron microscope. SEM images were obtained at magnifications from 5K to 150K. FIGS. 6A-6F depict the surface of the sample. The magnification of the SEM micrograph increases from FIG. 6A to FIG. 6F. The grain size of the crystals in the sample were determined to be about 30 nm.

[0241] Referring now to FIG. 7, a sample of the precursor glass composition of Batched Comp. AP was heat treated at 780° C. for 4 hours and 850° C. for 4 hours to form a glass-ceramic. The resultant glass-ceramic was then analyzed by X-ray diffraction resulting in the diffraction pattern depicted in FIG. 7. The X-ray diffraction pattern depicted in FIG. 7 shows that compositions in which a portion of MgO is substituted for ZnO (in this case 21.5% of the MgO was replaced with ZnO) may still result in the formation of a crystalline phase having a jeffbenite crystalline structure upon heat treatment without producing other crystalline phases.

[0242] Samples formed from the composition of Batched Comp. AA were heat treated at 725° C. for 4 hours and 850° C. for 4 hours to form glass-ceramics. Samples formed from the composition of Batched Comp. AA were heat treated at 775° C. for 4 hours and 850° C. for 4 hours to form glass-ceramics. Polished parts having dimensions of 25 mm×25 mm×0.6 mm were prepared from the resulting glass-ceramics. These samples of the glass-ceramics were then ion exchanged in a bath of 100 wt % KNO₃ for 1 hour, 2 hours, 4 hours, 8 hours, 16 hours, and 32 hours at 450° C. The samples were then analyzed to determine the maximum surface compressive stress and maximum central tension as a function of ion exchange time. The depth of layer DOL was also calculated as a percentage of the thickness of the glass-ceramic from diffusivity measurements. The results are reported in Table 8.

[0243] As indicated in Table 8, maximum central tensions of up to 187.08 MPa and maximum surface compressive stresses of 690.57 MPa were achieved with different ion exchange times, indicating that the stress profile in the glass-ceramics could be tailored to meet different performance criteria.

[0244] The transmittance of a polished, about 0.6 mm thick sample of the glass-ceramic composition of Batched Comp. AA, cerammed at 725° C. for 4 hours and at 850° C. for 4 hours, was measured using the following procedure. The transmittance measurements were performed on a Lambda 950 UV-Vis-NIR Spectrophotometer manufactured by PerkinElmer Inc. (Waltham, Massachusetts USA). In the present examples, the spectrophotometer used the following instrument settings: 150 mm integrating sphere; a data interval of 2 nm; a detector changeover (InGaAs to PMT) at 860 nm; a lamp change at 340 nm; a tungsten-halogen source; an InGaAs spectral band width of servo; an InGaAs gain of 15; an InGaAs averaging time of 0.4 seconds; a PMT spectral band width of 3.5 nm; a PMT averaging time of 0.2 seconds; and a beam mode of Double. For total transmittance (Total Tx), the sample was fixed at the integrating sphere entry point. For diffuse transmittance (Diffuse Tx), the Spectralon® reference reflectance disk over the sphere exit port was removed to allow on-axis light to exit the sphere and enter a light trap. A zero offset measurement was made, with no sample, of the diffuse portion to determine efficiency of the light trap. To correct diffuse transmittance measurements, the zero offset contribution is subtracted from the sample measurement using the equation: Diffuse Tx=Diffuse Measured-(Zero Offset*(Total Tx/100)). The scatter ratio was measured for all wavelengths as: (% Diffuse Tx/% Total Tx). The axial transmittance was also measured. The axial transmittance measurement is a measurement using the on-axis light that escapes the integrating sphere through the exit port. The total transmittance of the glass-ceramic of Example A is depicted in FIG. 10. The diffuse transmittance of the glass-ceramic is depicted in FIG. 11. The axial transmittance of the glass-ceramic is depicted in FIG. 12, and the scatter ratio of the glass-ceramic is depicted in FIG. 13.

[0245] The transmittance of five glass-ceramic articles were measured using the procedure described hereinabove. The batch composition, ceram schedule, corresponding glass-ceramic, and thickness are listed in Table 9.

TABLE 8

Batched Comp.	Ceram Schedule (Temperature ° C.-Hours)	Ion Exchange Time (hours)	Maximum Central Tension (MPa)	Surface Compressive Stress (MPa)	DOL (% thickness)
AA	725-4, 850-4	1	51.62	350.43	10
AA	725-4, 850-4	2	84.04	189.28	14
AA	725-4, 850-4	4	102.1	268.21	20
AA	725-4, 850-4	8	155.04	241.3	28
AA	725-4, 850-4	16	187.08	309.39	40
AA	725-4, 850-4	32	157.98	260.74	56
AA	775-4, 850-4	1	47.15	336.61	—
AA	775-4, 850-4	2	70.4	468.29	—
AA	775-4, 850-4	4	93.53	690.57	—
AA	775-4, 850-4	8	135.8	296.62	—
AA	775-4, 850-4	16	177.5	253.39	—
AA	775-4, 850-4	32	153.98	330.68	—

TABLE 9

Batched Comp.	Ceram Schedule (Temperature ° C.-Hours)	Glass-ceramic	Thickness (mm)
CM	780-4, 850-4	GC50	0.62
CN	780-4, 850-4	GC51	0.58
DH	780-4, 850-4	GC97	0.63
CX	780-4, 850-4	GC22	0.63
CY	780-4, 850-4	GC23	0.63

[0246] The total transmittance of each of the glass-ceramics of Table 9 is depicted in FIG. 31. The diffuse transmittance of each of the glass-ceramics of Table 9 is depicted in FIG. 32. The scatter ratio of each the glass-ceramics of Table 9 is depicted in FIG. 33. The scatter ratio given in FIG. 33 was calculated using the following equation: % Scatter Ratio=(% Diffuse Tx/% Total Tx)*100.

[0247] The color coordinates of each of the glass-ceramics listed in Table 9 were determined using the following procedure. Color measurements were performed on a Lambda 950 UV-Vis-NIR Spectrophotometer manufactured

by PerkinElmer Inc. (Waltham, Massachusetts USA), as previously described. Color coordinates are generally calculated using a weighting and summation of the object spectral transmittance, the human eye “standard observer” spectral functions, and the illuminant power spectral distribution. The color coordinates for each sample were calculated from transmittance data taken using three illuminants, Illuminant CIE D65, Illuminant CIE A, and Illuminant CIE F2. Measurements were taken using 2° and 10° observer angles, and a wavelength range of 770 nm to 380 nm (with a 2 nm interval) was used. A first set of color coordinates were calculated using the total transmittance data, and a second set of color coordinates were calculated using diffuse transmittance data from each glass-ceramic article listed in Table 9. The color measurements made using total transmittance are included below in Table 10. The color measurements made using diffuse transmittance are included below in Table 11. Tables 10 and 11 includes CIE L*A*B* color space data as well as CIE Yxy color space data. It is noted that the glass-ceramics in Tables 10 and 11 appeared transparent.

TABLE 10

Color Calculated from Total Transmittance Data											
Glass-Ceramic	Illuminant and Observer Angle (°)	L*	A*	B*	X	Y	Z	x	y	z	
GC50	F2 2	89.23	-0.37	17.82	73.84	74.65	36.61	0.3989	0.4033	0.1978	
GC50	A 2	89.49	3.99	15.82	84.83	75.21	20.38	0.4702	0.4169	0.1129	
GC50	D65 2	88.64	-0.46	15.79	69.56	73.41	60.72	0.3415	0.3604	0.2981	
GC50	F2 10	88.97	0.31	18.03	76.65	74.10	37.06	0.4081	0.3946	0.1973	
GC50	A 10	89.30	4.22	16.35	85.49	74.81	19.84	0.4746	0.4153	0.1102	
GC50	D65 10	88.21	0.56	15.82	69.00	72.51	59.02	0.3441	0.3616	0.2943	
GC51	F2 2	89.59	-0.48	15.48	74.55	75.43	38.65	0.3952	0.3999	0.2049	
GC51	A 2	89.82	3.31	13.68	85.23	75.91	21.39	0.4669	0.4159	0.1172	
GC51	D65 2	89.09	-0.63	13.70	70.38	74.36	63.95	0.3373	0.3563	0.3064	
GC51	F2 10	89.37	0.11	15.67	77.42	74.95	39.19	0.4042	0.3913	0.2046	
GC51	A 10	89.65	3.52	14.16	85.94	75.56	20.87	0.4712	0.4143	0.1144	
GC51	D65 10	88.72	0.25	13.75	69.87	73.57	62.26	0.3397	0.3577	0.3027	
GC97	F2 2	93.13	-0.57	9.32	82.26	83.26	47.82	0.3856	0.3903	0.2242	
GC97	A 2	93.25	1.73	8.14	92.80	83.54	26.05	0.4585	0.4128	0.1287	
GC97	D65 2	92.84	-0.79	8.24	78.12	82.60	78.60	0.3264	0.3451	0.3284	
GC97	F2 10	92.99	-0.21	9.46	85.52	82.95	48.69	0.3938	0.3820	0.2242	
GC97	A 10	93.15	1.87	8.46	93.71	83.32	25.56	0.4626	0.4113	0.1262	
GC97	D65 10	92.62	-0.27	8.30	77.70	82.10	76.90	0.3283	0.3468	0.3249	
GC22	F2 2	94.95	-0.47	5.63	86.50	87.50	53.51	0.3802	0.3846	0.2352	
GC22	A 2	95.01	0.90	4.88	96.84	87.65	28.87	0.4539	0.4108	0.1353	
GC22	D65 2	94.78	-0.66	4.98	82.44	87.09	87.60	0.3206	0.3387	0.3407	

Color Calculated from Total Transmittance Data										
Glass-Ceramic	Illuminant and Observer Angle (°)	L*	A*	B*	X	Y	Z	x	y	z
GC22	F2 10	94.87	−0.25	5.72	89.98	87.30	54.61	0.3880	0.3765	0.2355
GC22	A 10	94.95	0.99	5.08	97.87	87.51	28.42	0.4578	0.4093	0.1329
GC22	D65 10	94.64	−0.34	5.03	82.10	86.78	85.96	0.3222	0.3405	0.3373
GC23	F2 2	95.34	−0.43	4.54	87.43	88.42	55.04	0.3787	0.3829	0.2384
GC23	A 2	95.38	0.67	3.91	97.67	88.53	29.62	0.4525	0.4102	0.1373
GC23	D65 2	95.20	−0.60	4.01	83.42	88.09	90.03	0.3190	0.3368	0.3442
GC23	F2 10	95.27	−0.24	4.62	90.97	88.26	56.22	0.3864	0.3749	0.2388
GC23	A 10	95.34	0.75	4.08	98.74	88.42	29.19	0.4564	0.4087	0.1349
GC23	D65 10	95.09	−0.34	4.06	83.10	87.84	88.41	0.3204	0.3387	0.3409

Color Calculated from Diffuse Transmittance Data

Glass-Ceramic	Illuminant and Observer Angle (°)	Colorimetric Data								
		L*	A*	B*	X	Y	Z	x	y	z
GC50	F2 2	35.91	-1.02	-30.66	8.77	8.96	14.51	0.2719	0.2780	0.4500
GC50	A 2	35.10	-14.39	-29.52	7.67	8.55	7.24	0.3269	0.3644	0.3087
GC50	D65 2	37.59	-1.57	-26.75	9.18	9.86	23.02	0.2183	0.2344	0.5473
GC50	F2 10	36.61	-2.83	-30.05	9.28	9.33	15.09	0.2753	0.2769	0.4478
GC50	A 10	35.65	-14.73	-29.48	7.99	8.83	7.33	0.3309	0.3656	0.3035
GC50	D65 10	38.67	-4.18	-25.59	9.40	10.47	23.09	0.2189	0.2436	0.5375
GC51	F2 2	40.40	-1.48	-25.98	11.19	11.49	15.64	0.2920	0.2999	0.4081
GC51	A 2	39.68	-12.25	-25.04	10.39	11.06	7.89	0.3540	0.3770	0.2690
GC51	D65 2	41.76	-2.18	-22.76	11.43	12.35	24.92	0.2347	0.2535	0.5118
GC51	F2 10	40.96	-2.85	-25.53	11.80	11.84	16.23	0.2960	0.2970	0.4071
GC51	A 10	40.13	-12.46	-25.06	10.74	11.33	7.96	0.3578	0.3772	0.2650
GC51	D65 10	42.64	-4.22	-21.85	11.65	12.92	24.94	0.2353	0.2610	0.5037
GC97	F2 2	18.98	3.75	-30.73	2.93	2.74	6.31	0.2443	0.2289	0.5268
GC97	A 2	18.41	-9.30	-28.59	2.36	2.61	3.02	0.2954	0.3265	0.3782
GC97	D65 2	20.52	4.67	-26.78	3.24	3.12	9.84	0.1999	0.1927	0.6074
GC97	F2 10	19.72	1.58	-30.23	3.11	2.92	6.63	0.2455	0.2307	0.5238
GC97	A 10	18.95	-10.04	-28.75	2.47	2.74	3.10	0.2974	0.3291	0.3735
GC97	D65 10	21.62	1.80	-25.84	3.34	3.41	10.01	0.1994	0.2035	0.5971

TABLE 11-continued

Color Calculated from Diffuse Transmittance Data										
Glass-Ceramic	Illuminant and Observer Angle (°)	L*	A*	B*	X	Y	Z	x	y	z
GC22	F2 2	5.09	4.63	-21.31	0.68	0.56	1.60	0.2379	0.1982	0.5639
GC22	A 2	4.90	-1.90	-18.93	0.54	0.54	0.74	0.2974	0.2974	0.4052
GC22	D65 2	5.80	6.50	-18.96	0.77	0.64	2.46	0.1986	0.1658	0.6356
GC22	F2 10	5.49	3.43	-21.30	0.72	0.61	1.70	0.2372	0.2007	0.5621
GC22	A 10	5.16	-2.47	-19.43	0.56	0.57	0.77	0.2962	0.2999	0.4039
GC22	D65 10	6.44	4.92	-18.71	0.80	0.71	2.54	0.1967	0.1762	0.6271
GC23	F2 2	3.34	3.94	-19.83	0.47	0.37	1.26	0.2229	0.1765	0.6006
GC23	A 2	3.18	-1.42	-17.40	0.35	0.35	0.57	0.2727	0.2770	0.4504
GC23	D65 2	3.90	5.54	-17.73	0.55	0.43	1.92	0.1884	0.1490	0.6626
GC23	F2 10	3.66	2.96	-19.95	0.50	0.41	1.34	0.2219	0.1810	0.5971
GC23	A 10	3.39	-1.88	-17.99	0.36	0.38	0.60	0.2717	0.2807	0.4476
GC23	D65 10	4.42	4.28	-17.69	0.57	0.49	1.99	0.1867	0.1608	0.6526

[0248] In addition to the foregoing examples and disclosure concerning glass-ceramic comprising jeffbenite, as part of exploring the presently disclosed technology of precursor glass compositions and glass-ceramics formed therefrom, the following tables provide additional compositions that

were melted by Applicants as described herein, which correspond to information disclosed herein as well as other samples disclosed and further described herein. The additional compositions are presented in mol. % on an oxide basis and are listed in Table 12.

TABLE 12

[illegible]

TABLE 12-continued

[illegible]

TABLE 12-continued

ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	2.75	2.75	3.10	3.10	2.33	2.33	2.72	3.15	2.70	2.24	2.68
TiO ₂	0.71	0.71	0.37	0.37	1.08	1.08	0.00	0.00	0.69	1.38	1.38
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.09	0.09	0.12	0.12	0.11	0.11	0.09	0.09	0.09	0.09	0.09
Composition (mol. %)	Ex. 67	Ex. 68	Ex. 69	Ex. 70	Ex. 71	Ex. 72	Ex. 73	Ex. 74	Ex. 75	Ex. 76	Ex. 77
SiO ₂	44.04	46.60	46.66	46.64	46.52	46.62	46.43	44.36	45.05	45.02	45.05
Al ₂ O ₃	7.90	11.25	11.24	11.23	11.21	11.26	11.22	11.15	9.66	9.65	9.66
MgO	44.12	33.48	33.54	33.43	33.47	33.50	33.45	36.54	37.31	29.25	29.26
Na ₂ O	0.00	2.81	2.78	2.82	2.86	2.80	2.82	2.80	2.95	2.95	2.9
K ₂ O	0.00	1.61	0.00	0.00	0.00	0.00	0.00	1.60	1.44	1.44	1.44
CaO	0.00	0.00	1.59	0.00	0.00	0.00	0.00	0.00	0.00	8.09	0.00
SrO	0.00	0.00	0.00	1.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	1.64	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	1.58	0.00	0.00	0.00	0.00	8.0
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	1.68	0.00	0.00	0.00	0.00
ZrO ₂	2.55	2.74	2.71	2.75	2.78	2.73	2.84	2.27	2.39	2.39	2.39
TiO ₂	1.31	1.41	1.39	1.41	1.43	1.40	1.46	0.00	1.11	1.11	1.11
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.18	0.00	0.00	0.00
SnO ₂	0.09	0.09	0.09	0.09	0.09	0.09	0.10	0.09	0.10	0.10	0.10
Composition (mol. %)	Ex. 78	Ex. 79	Ex. 80	Ex. 81	Ex. 82	Ex. 83	Ex. 84	Ex. 85	Ex. 86	Ex. 87	Ex. 88
SiO ₂	45.67	45.05	48.01	48.01	43.73	43.73	48.81	46.20	47.47	45.45	46.21
Al ₂ O ₃	10.05	9.66	8.61	8.61	8.99	8.99	10.82	9.93	10.40	8.68	9.93
MgO	33.89	29.26	33.64	33.64	39.55	39.55	31.27	35.98	33.60	38.14	31.43
Na ₂ O	3.00	2.95	3.18	3.18	2.36	2.36	3.72	2.87	3.29	2.82	2.87
K ₂ O	0.00	1.44	3.11	3.11	2.17	2.17	1.54	0.69	1.15	1.36	0.69
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.55
FeO	0.00	8.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	1.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.89	2.39	2.23	2.23	2.37	2.37	2.07	2.31	2.19	2.08	2.31
TiO ₂	1.80	1.11	1.12	1.12	0.73	0.73	1.67	1.93	1.80	1.38	1.93
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
Composition (mol. %)	Ex. 89	Ex. 90	Ex. 91	Ex. 92	Ex. 93	Ex. 94	Ex. 95	Ex. 96	Ex. 97	Ex. 98	Ex. 99
SiO ₂	46.21	44.62	44.62	43.79	43.79	42.91	42.91	42.82	42.40	43.40	42.81
Al ₂ O ₃	9.93	14.20	14.20	12.40	12.40	10.70	10.70	11.45	11.34	13.02	12.88
MgO	31.43	42.90	42.90	44.70	44.70	46.50	46.50	37.06	36.76	36.19	35.74
Na ₂ O	2.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.79	0.00	4.20
K ₂ O	0.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CaO	4.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.88	0.00	2.99	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	2.31	0.00	0.00	1.79	1.79	3.58	3.58	3.56	3.52	2.90	2.85
TiO ₂	1.93	0.00	0.00	0.00	0.00	0.00	0.00	2.15	2.11	1.42	1.44
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.08	0.08	0.08
Composition (mol. %)	Ex. 100	Ex. 101	Ex. 102	Ex. 103	Ex. 104	Ex. 105	Ex. 106	Ex. 107	Ex. 108	Ex. 109	
SiO ₂	44.72	47.88	44.74	43.11	43.08	44.40	43.03	40.00	40.01	44.61	
Al ₂ O ₃	14.94	11.58	14.92	12.23	12.24	14.06	12.21	6.67	6.64	8.38	
MgO	31.29	31.76	31.26	36.63	36.61	31.86	27.70	46.55	33.29	38.48	
Na ₂ O	0.00	2.81	4.53	2.93	0.00	4.42	2.90	0.00	0.00	2.93	
K ₂ O	0.00	0.00	0.00	0.00	2.94	0.00	0.00	0.00	0.00	0.00	

TABLE 12-continued

[illegible]

TABLE 12-continued

ZrO ₂	4.19	4.27	4.01	3.32	4.37	3.74	4.32	4.49	2.97	4.28
TiO ₂	1.43	1.44	1.90	1.85	1.44	1.52	1.41	0.49	1.50	1.42
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.09	0.08	0.08	0.08	0.04	0.04	0.04
Composition (mol. %)	Ex. 140	Ex. 141	Ex. 142	Ex. 143	Ex. 144	Ex. 145	Ex. 146	Ex. 147	Ex. 148	Ex. 149
SiO ₂	42.81	44.72	10.20	43.49	45.89	42.84	43.44	43.50	42.90	42.91
Al ₂ O ₃	11.45	14.90	3.07	13.02	11.00	12.86	10.13	10.12	8.55	8.54
MgO	34.20	31.35	85.02	36.15	36.54	37.13	36.28	36.25	37.07	37.05
Na ₂ O	2.88	0.00	0.69	0.00	0.00	0.00	2.90	2.92	2.90	2.92
K ₂ O	0.00	4.47	0.00	2.94	1.93	1.44	0.00	0.00	0.00	0.00
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	2.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	2.90	0.00	2.87	0.00
ZrO ₂	4.30	3.00	0.68	2.89	3.69	4.30	2.91	2.88	4.27	4.29
TiO ₂	1.44	1.52	0.34	1.46	0.91	1.39	1.45	1.43	1.44	1.43
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.04	0.04	0.01	0.04	0.04	0.04	0.00	0.00	0.00	0.00
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.90	0.00	2.86
Composition (mol. %)	Ex. 150	Ex. 151	Ex. 152	Ex. 153	Ex. 154	Ex. 155	Ex. 156	Ex. 157	Ex. 158	Ex. 159
SiO ₂	44.47	48.68	46.60	45.05	42.81	42.40	42.86	40.85	41.02	43.41
Al ₂ O ₃	14.12	10.07	11.25	9.66	12.88	11.34	11.45	8.86	9.60	13.01
MgO	31.82	32.02	33.48	29.26	35.74	36.76	37.06	31.93	34.27	29.01
Na ₂ O	2.94	3.24	2.81	2.95	4.20	3.79	0.00	0.00	0.00	0.00
K ₂ O	1.47	3.11	1.61	1.44	0.00	0.00	2.87	0.00	0.00	0.00
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	8.04	0.00	0.00	0.00	13.57	10.96	7.26
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.94
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.67	2.79	2.74	2.39	2.85	3.52	4.29	3.41	3.42	3.62
TiO ₂	1.50	0.00	1.41	1.11	1.44	2.11	1.39	1.34	0.70	0.71
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.00	0.10	0.09	0.10	0.08	0.08	0.08	0.04	0.04	0.04
Composition (mol. %)	Ex. 160	Ex. 161	Ex. 162	Ex. 163	Ex. 164	Ex. 165	Ex. 166	Ex. 167	Ex. 168	Ex. 169
SiO ₂	43.54	43.44	43.48	43.40	43.23	44.36	42.27	44.38	42.53	40.53
Al ₂ O ₃	13.02	13.04	13.06	13.05	12.98	14.07	9.84	14.02	13.50	8.13
MgO	28.97	31.92	31.94	32.03	31.75	30.39	29.55	22.93	21.93	32.39
Na ₂ O	2.88	2.89	2.86	2.85	2.85	2.97	0.00	3.00	2.84	0.00
K ₂ O	0.00	0.00	0.00	0.00	0.00	1.48	0.00	1.48	0.00	0.00
CaO	0.00	4.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	4.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	4.32	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	7.21	0.00	0.00	0.00	4.80	1.48	9.85	8.91	8.50	13.50
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	2.82	0.00	5.68	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.62	4.37	3.29	3.24	4.35	3.71	4.22	3.72	3.53	4.08
TiO ₂	0.72	0.00	1.03	1.07	0.00	1.51	1.41	1.50	1.44	1.34
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
Composition (mol. %)	Ex. 170	Ex. 171	Ex. 172	Ex. 173	Ex. 174	Ex. 175	Ex. 176	Ex. 177	Ex. 178	Ex. 179
SiO ₂	41.60	41.57	43.40	44.87	44.02	45.06	44.31	44.06	43.47	43.50
Al ₂ O ₃	10.71	10.72	10.14	11.33	9.88	10.93	11.01	10.95	11.41	11.38
MgO	27.26	27.29	28.34	26.28	28.91	26.87	27.06	26.90	31.78	36.80
Na ₂ O	4.15	2.77	2.87	3.11	3.05	3.08	3.10	3.09	3.79	3.76
K ₂ O	0.00	1.37	1.41	1.36	1.34	1.35	1.36	1.35	0.00	0.00
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

TABLE 12-continued

BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	11.05	11.05	9.43	8.68	8.51	8.61	8.67	8.62	5.02	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.81	3.81	3.09	3.13	3.07	2.84	3.64	2.59	3.01	3.01
TiO ₂	1.37	1.38	1.27	1.21	1.18	1.20	0.80	2.40	1.47	1.50
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
Composition (mol. %)	Ex. 180	Ex. 181	Ex. 182	Ex. 183	Ex. 184	Ex. 185	Ex. 186	Ex. 187	Ex. 188	Ex. 189
SiO ₂	43.48	42.96	42.44	44.27	45.43	47.47	45.44	46.17	46.56	47.24
Al ₂ O ₃	11.38	11.36	11.39	11.08	11.38	11.17	11.34	11.00	11.63	10.61
MgO	28.01	36.79	29.76	29.73	28.16	27.78	29.45	29.39	27.86	28.23
Na ₂ O	3.79	3.78	3.77	2.83	3.46	3.84	3.53	3.22	4.25	3.83
K ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	8.80	0.00	6.98	7.63	7.28	5.69	5.99	5.97	5.70	5.61
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	2.98	3.26	3.48	3.16	3.07	2.84	3.04	3.04	2.85	3.35
TiO ₂	1.51	1.80	2.13	1.26	1.18	1.17	1.17	1.17	1.10	1.10
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
Composition (mol. %)	Ex. 190	Ex. 191	Ex. 192	Ex. 193	Ex. 194	Ex. 195	Ex. 196	Ex. 197	Ex. 198	Ex. 199
SiO ₂	45.12	57.23	54.46	54.45	54.48	54.47	54.46	54.42	52.13	51.48
Al ₂ O ₃	10.72	3.12	9.18	9.18	9.18	9.18	9.18	9.18	10.69	9.74
MgO	29.91	18.96	23.36	23.36	23.37	23.37	23.37	18.39	25.02	26.67
Na ₂ O	3.53	10.27	2.85	4.75	6.55	8.45	4.65	4.72	4.47	4.03
K ₂ O	0.00	0.00	6.56	4.68	2.81	0.94	4.75	4.69	3.55	3.98
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	5.91	0.00	0.00	0.00	0.00	0.00	0.00	4.95	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.60	8.78	3.10	3.10	3.11	3.11	3.10	3.11	3.22	3.35
TiO ₂	1.17	1.59	0.37	0.37	0.37	0.37	0.37	0.41	0.79	0.63
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.04	0.04	0.12	0.12	0.12	0.12	0.12	0.13	0.13	0.12
Composition (mol. %)	Ex. 200	Ex. 201	Ex. 202	Ex. 203	Ex. 204	Ex. 205	Ex. 206	Ex. 207	Ex. 208	Ex. 209
SiO ₂	51.70	48.46	52.64	46.17	46.94	46.54	53.14	52.88	53.97	53.96
Al ₂ O ₃	9.15	10.23	8.21	11.52	11.74	9.20	0.00	3.80	9.01	9.00
MgO	27.58	24.14	26.18	34.42	27.28	34.24	22.98	24.51	27.93	27.93
Na ₂ O	4.09	4.23	4.18	3.62	3.65	3.22	0.00	1.96	4.02	4.05
K ₂ O	4.07	2.31	4.24	0.51	0.55	1.80	0.00	1.96	0.97	0.97
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	6.05	0.00	0.00	6.04	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	17.84	9.67	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	2.91	3.32	4.10	3.64	3.67	4.13	5.94	5.11	3.21	3.97
TiO ₂	0.39	1.12	0.31	0.00	0.00	0.76	0.00	0.00	0.76	0.00
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.12	0.13	0.12	0.12	0.13	0.12	0.11	0.11	0.12	0.12
Composition (mol. %)	Ex. 210	Ex. 211	Ex. 212	Ex. 213	Ex. 214	Ex. 215	Ex. 216	Ex. 217	Ex. 218	Ex. 219
SiO ₂	48.63	51.33	50.70	51.26	47.96	47.88	46.08	46.28	46.18	46.28
Al ₂ O ₃	8.10	8.55	9.41	10.40	11.31	10.18	10.11	11.23	10.10	11.21
MgO	34.92	31.48	32.18	31.55	35.17	36.05	38.30	37.39	34.27	32.40
Na ₂ O	2.87	3.59	3.02	3.01	2.47	2.49	1.90	1.88	1.86	1.85

TABLE 12-continued

K ₂ O	0.32	0.45	0.77	0.77	0.00	0.00	0.00	0.00	0.00	0.00
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.02	4.98
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	4.00	3.81	3.82	2.89	2.24	3.32	3.49	2.37	3.49	2.40
TiO ₂	1.04	0.68	0.00	0.00	0.74	0.00	0.00	0.73	0.00	0.76
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.12	0.12	0.12	0.12	0.12	0.08	0.12	0.12	0.08	0.12
Composition (mol. %)	Ex. 220	Ex. 221	Ex. 222	Ex. 223	Ex. 224	Ex. 225	Ex. 226	Ex. 227	Ex. 228	Ex. 229
SiO ₂	46.95	47.87	47.32	48.24	48.82	47.68	48.62	54.46	54.43	54.47
Al ₂ O ₃	7.78	8.87	9.01	10.12	11.33	9.60	10.74	9.18	9.18	9.17
MgO	37.33	34.19	36.80	33.65	32.87	36.35	33.17	23.37	23.36	23.36
Na ₂ O	2.47	3.48	2.47	3.49	3.51	2.49	3.51	4.65	4.10	5.20
K ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.75	5.33	4.17
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	4.01	4.08	3.30	3.36	3.39	3.80	3.88	3.10	3.11	3.14
TiO ₂	1.40	1.43	1.03	1.05	0.00	0.00	0.00	0.37	0.37	0.37
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.12	0.12	0.12
Composition (mol. %)	Ex. 230	Ex. 231	Ex. 232	Ex. 233	Ex. 234	Ex. 235	Ex. 236	Ex. 237	Ex. 238	Ex. 239
SiO ₂	54.38	54.45	54.35	44.75	46.13	46.74	45.99	45.46	44.69	54.32
Al ₂ O ₃	9.17	9.19	9.16	8.43	10.58	9.86	11.25	11.23	11.02	9.20
MgO	23.35	23.40	23.31	34.10	36.24	36.53	33.60	33.74	30.99	23.43
Na ₂ O	5.80	6.34	7.11	2.93	2.89	2.38	2.95	2.96	3.02	6.36
K ₂ O	3.63	2.94	2.34	0.00	0.00	0.56	1.49	1.37	1.33	2.92
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	4.17	0.00	0.00	0.00	0.00	3.84	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.18	3.21	3.25	3.98	3.34	2.73	3.27	3.78	3.70	3.30
TiO ₂	0.37	0.36	0.36	1.52	0.75	1.11	1.37	1.38	1.33	0.39
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.12	0.12	0.12	0.12	0.08	0.08	0.08	0.08	0.08	0.08
Composition (mol. %)	Ex. 240	Ex. 241	Ex. 242	Ex. 243	Ex. 244	Ex. 245	Ex. 246	Ex. 247	Ex. 248	Ex. 249
SiO ₂	52.06	49.82	46.60	46.68	49.18	46.22	49.02	44.73	44.65	33.74
Al ₂ O ₃	9.70	10.19	11.31	11.32	9.30	8.89	9.32	14.92	10.87	8.21
MgO	26.08	28.70	33.51	33.45	29.78	33.33	29.90	31.33	37.71	27.09
Na ₂ O	5.52	4.69	2.79	2.78	4.27	3.53	1.95	4.48	1.73	1.59
K ₂ O	2.51	2.10	0.00	0.00	0.48	0.20	0.47	0.00	1.01	0.90
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	1.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	2.35	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	2.31	2.59	2.32	0.00	0.00	0.00
ZrO ₂	3.43	3.56	2.70	2.69	3.46	3.86	3.47	2.99	3.05	27.63
TiO ₂	0.62	0.85	1.39	1.40	1.13	1.29	1.12	1.48	0.90	0.78
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.09	0.09	0.08	0.08	0.08	0.06

TABLE 12-continued

[illegible]

TABLE 12-continued

[illegible]

TABLE 12-continued

B ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PbO	0.00	0.00	0.00	0.00	0.00	2.01	0.00	0.00	0.00	0.00
ThO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.23	0.24	0.24	0.47
Composition (mol. %)	Ex. 310	Ex. 311	Ex. 312	Ex. 313	Ex. 314	Ex. 315	Ex. 316	Ex. 317	Ex. 318	Ex. 319
SiO ₂	51.28	51.25	46.43	46.25	45.94	46.86	46.68	47.04	49.15	48.98
Al ₂ O ₃	9.79	9.80	11.00	10.97	10.94	11.71	11.68	11.74	9.43	9.40
MgO	29.87	29.76	33.32	33.25	33.33	32.09	32.02	32.16	30.56	30.49
Na ₂ O	3.72	3.73	2.97	2.98	2.98	3.00	3.00	2.99	3.76	3.77
K ₂ O	1.72	1.72	1.30	1.31	1.31	1.31	1.32	1.31	1.76	1.76
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.54	3.65	4.13	4.39	4.65	4.17	4.43	3.91	3.73	3.99
TiO ₂	0.00	0.00	0.77	0.77	0.77	0.77	0.78	0.77	1.53	1.54
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Cs ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
B ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PbO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ThO ₂	0.47	0.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Composition (mol. %)	Ex. 320	Ex. 321	Ex. 322	Ex. 323	Ex. 324	Ex. 325	Ex. 326	Ex. 327	Ex. 328	Ex. 329
SiO ₂	48.93	49.41	49.24	43.91	44.25	48.85	53.07	51.71	52.38	51.34
Al ₂ O ₃	9.37	9.46	9.43	9.37	12.64	10.06	9.34	9.53	9.43	9.58
MgO	30.28	30.50	30.43	28.46	33.37	30.66	25.33	27.34	26.34	27.78
Na ₂ O	3.78	3.77	3.78	3.08	3.01	3.49	4.35	4.02	4.23	4.01
K ₂ O	1.77	1.76	1.77	1.49	1.32	1.57	3.72	2.71	3.18	2.44
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	7.83	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	4.25	4.24	4.50	4.65	4.55	4.51	3.56	3.99	3.80	4.14
TiO ₂	1.54	0.77	0.77	1.12	0.78	0.77	0.55	0.62	0.55	0.62
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
Composition (mol. %)	Ex. 330	Ex. 331	Ex. 332	Ex. 333	Ex. 334	Ex. 335	Ex. 336	Ex. 337	Ex. 338	Ex. 339
SiO ₂	46.71	50.49	50.45	50.50	50.37	50.27	50.46	46.19	50.28	50.38
Al ₂ O ₃	8.82	9.69	9.71	9.73	9.61	9.57	9.61	12.36	9.01	9.35
MgO	26.43	29.43	29.47	29.38	29.21	29.12	29.36	30.48	30.41	29.92
Na ₂ O	3.09	3.39	3.09	2.80	3.68	3.69	3.68	4.43	3.36	3.38
K ₂ O	1.90	2.03	2.30	2.60	1.70	1.71	1.70	1.85	2.02	2.02
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	7.37	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	4.40	4.11	4.12	4.10	3.80	4.01	3.95	3.59	4.08	4.09
TiO ₂	1.20	0.77	0.77	0.81	1.54	1.55	1.16	1.01	0.77	0.77
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
Composition (mol. %)	Ex. 340	Ex. 341	Ex. 342	Ex. 343	Ex. 344	Ex. 345	Ex. 346	Ex. 347	Ex. 348	Ex. 349
SiO ₂	50.59	50.70	50.81	50.92	50.47	50.57	52.34	52.39	52.28	52.37
Al ₂ O ₃	10.04	10.39	10.74	11.09	9.75	9.77	9.38	9.41	9.39	9.38
MgO	28.94	28.44	27.94	27.43	29.45	29.35	26.22	26.29	26.24	26.23
Na ₂ O	3.40	3.42	3.43	3.45	2.51	2.21	3.25	3.74	3.74	2.94

TABLE 12-continued

K ₂ O	2.04	2.05	2.06	2.07	2.90	3.17	4.21	3.73	3.72	4.61
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	4.13	4.15	4.16	4.18	4.14	4.14	3.83	3.82	3.45	3.68
TiO ₂	0.78	0.78	0.78	0.79	0.78	0.78	0.79	0.63	1.18	0.79
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.00	0.00	0.00	0.00	0.00	0.00
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Cs ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
B ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PbO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ThO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SeO ₂	0.00	0.00	0.00	0.00	0.11	0.11	0.11	0.11	0.11	0.11
Composition (mol. %)	Ex. 350	Ex. 351	Ex. 352	Ex. 353	Ex. 354	Ex. 355	Ex. 356	Ex. 357	Ex. 358	Ex. 359
SiO ₂	52.09	52.17	50.77	50.80	50.76	53.07	50.09	50.15	50.15	50.12
Al ₂ O ₃	9.43	9.47	9.49	9.51	9.44	9.94	9.72	9.69	9.67	9.69
MgO	24.17	26.29	28.92	28.84	26.67	25.16	29.66	29.60	29.72	31.76
Na ₂ O	4.26	4.25	2.70	2.71	2.72	3.70	3.70	3.71	3.72	3.74
K ₂ O	3.20	1.60	2.70	1.32	1.33	1.76	1.71	1.71	1.72	1.68
CaO	2.24	0.00	0.00	0.00	2.23	0.00	0.00	0.00	0.00	0.00
SrO	0.00	1.63	0.00	1.38	1.39	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.77	3.77	3.82	3.83	3.85	3.93	3.12	2.12	0.00	0.00
TiO ₂	0.79	0.78	1.55	1.56	1.56	2.39	1.01	1.01	1.02	0.00
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	1.00	2.01	4.00	3.01
SnO ₂	0.04	0.04	0.04	0.04	0.04	0.04	0.00	0.00	0.00	0.00
Composition (mol. %)	Ex. 360	Ex. 361	Ex. 362	Ex. 363	Ex. 364	Ex. 365	Ex. 366	Ex. 367	Ex. 368	Ex. 369
SiO ₂	54.46	50.47	46.60	53.07	52.38	50.37	54.46	50.47	46.60	53.07

TABLE 12-continued

P ₂ O ₅	0.00	0.00	2.91	3.24	2.84	2.64	3.09	2.87	0.00	0.00
SnO ₂	0.08	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.08
Composition (mol. %)	Ex. 380	Ex. 381	Ex. 382	Ex. 383	Ex. 384	Ex. 385	Ex. 386	Ex. 387	Ex. 388	Ex. 389
SiO ₂	52.55	52.72	52.39	51.90	52.16	51.65	51.39	50.70	50.19	50.05
Al ₂ O ₃	9.44	9.44	9.45	8.75	9.41	9.72	9.70	10.44	10.55	10.52
MgO	26.37	26.51	26.40	27.70	26.28	28.16	28.14	27.94	28.21	28.17
Na ₂ O	3.73	3.73	3.76	3.72	3.74	3.54	3.46	4.27	4.29	4.23
K ₂ O	3.72	3.71	3.74	3.71	3.73	1.66	1.61	2.09	2.17	2.12
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.70	3.80	3.07	3.04	3.82	3.82	3.84	3.70	3.41	4.05
TiO ₂	0.39	0.00	0.00	0.00	0.78	0.00	0.00	0.77	0.77	0.78
P ₂ O ₅	0.00	0.00	1.11	1.10	0.00	1.37	1.78	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.41	0.08
Composition (mol. %)	Ex. 390	Ex. 391	Ex. 392	Ex. 393	Ex. 394	Ex. 395	Ex. 396	Ex. 397	Ex. 398	Ex. 399
SiO ₂	44.71	44.23	43.87	49.73	49.36	48.91	59.92	57.17	55.57	54.81
Al ₂ O ₃	9.54	9.50	9.39	8.64	8.58	8.53	8.11	8.62	11.06	10.10
MgO	28.96	28.73	28.35	26.37	26.09	25.83	20.51	21.97	18.66	21.10
Na ₂ O	3.04	3.06	3.09	2.72	2.74	2.77	4.14	4.36	5.75	5.26
K ₂ O	1.47	1.48	1.49	1.33	1.34	1.35	4.12	4.47	5.71	5.23
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	7.94	7.85	7.84	7.22	7.20	7.17	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.16	3.95	4.77	2.89	3.58	4.33	2.69	2.90	2.79	3.06
TiO ₂	1.10	1.11	1.12	1.02	1.02	1.03	0.39	0.39	0.32	0.32
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.08	0.08	0.12	0.12	0.13	0.13
Composition (mol. %)	Ex. 400	Ex. 401	Ex. 402	Ex. 403	Ex. 404	Ex. 405	Ex. 406	Ex. 407	Ex. 408	Ex. 409
SiO ₂	56.36	55.99	54.22	52.88	52.13	53.33	52.17	51.72	52.21	51.97
Al ₂ O ₃	9.82	9.38	9.10	9.87	9.34	9.24	9.39	9.44	9.40	9.81
MgO	20.41	21.54	23.03	23.71	26.11	24.63	26.24	26.53	26.25	26.13
Na ₂ O	5.04	4.80	4.69	5.00	3.74	4.26	3.74	3.73	3.74	3.72
K ₂ O	5.08	4.78	4.76	4.23	3.72	4.20	3.72	3.72	3.72	3.71
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	2.85	2.98	3.13	3.39	3.66	3.42	3.61	3.55	3.66	3.64
TiO ₂	0.32	0.40	0.95	0.79	1.18	0.79	1.02	1.17	0.94	0.94
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.13	0.13	0.13	0.13	0.12	0.13	0.12	0.12	0.08	0.08
Composition (mol. %)	Ex. 410	Ex. 411	Ex. 412	Ex. 413	Ex. 414	Ex. 415	Ex. 416	Ex. 417	Ex. 418	Ex. 419
SiO ₂	52.45	51.53	52.87	52.12	52.30	53.76	54.38	53.76	56.55	57.67
Al ₂ O ₃	8.98	9.27	9.51	9.38	9.41	11.84	11.81	10.52	11.92	11.72
MgO	26.38	27.22	25.33	26.21	26.30	18.78	18.73	21.19	16.30	15.07
Na ₂ O	3.76	3.69	3.78	3.73	3.74	6.11	6.09	5.28	6.15	7.03
K ₂ O	3.74	3.67	3.77	3.72	3.73	5.75	5.73	5.25	5.79	4.69
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

TABLE 12-continued

FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	3.68	3.61	3.70	3.83	3.49	2.81	2.80	3.07	2.83	3.70
TiO ₂	0.94	0.93	0.95	0.94	0.94	0.82	0.33	0.80	0.33	0.00
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.08	0.08	0.08	0.08	0.08	0.13	0.13	0.13	0.13	0.13
Composition (mol. %)	Ex. 420	Ex. 421	Ex. 422	Ex. 423	Ex. 424	Ex. 425	Ex. 426	Ex. 427	Ex. 428	Ex. 429
SiO ₂	54.71	50.34	48.51	52.12	52.19	52.21	39.99	55.56	55.41	54.57
Al ₂ O ₃	9.19	10.39	11.88	9.42	9.40	9.36	6.71	11.10	11.03	11.95
MgO	23.08	26.28	26.36	26.32	26.27	26.30	46.60	18.62	18.55	18.76
Na ₂ O	4.56	3.81	3.76	3.75	3.75	3.70	0.00	6.26	5.77	5.78
K ₂ O	4.82	3.52	3.30	2.73	1.73	3.75	0.00	5.22	5.66	5.67
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	1.00	2.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	0.00	3.78	3.78	3.79	3.77	2.26	4.01	2.78	3.11	2.80
TiO ₂	0.41	0.80	0.81	0.79	0.79	2.33	2.70	0.32	0.33	0.33
P ₂ O ₅	0.00	0.99	1.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.13	0.08	0.09	0.08	0.08	0.08	0.00	0.13	0.13	0.13
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Cs ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
B ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PbO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ThO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SeO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	3.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Composition (mol. %)	Ex. 430	Ex. 431	Ex. 432	Ex. 433	Ex. 434	Ex. 435	Ex. 436	Ex. 437	Ex. 438	Ex. 439
SiO ₂	53.60	57.90	53.71	47.52	49.04	50.53	52.15	52.18	52.21	53.73
Al ₂ O ₃	12.99	13.97	13.01	7.86	8.49	9.14	9.81	9.42	9.42	11.47
MgO	18.72	10.02	10.01	31.96	29.42	26.79	24.04	26.25	26.33	19.50
Na ₂ O	5.82	6.96	5.85	3.93	4.20	4.58	4.97	3.77	3.68	5.87
K ₂ O	5.64	6.98	5.66	3.91	4.18	4.52	4.88	2.76	1.71	5.59
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ZrO ₂	2.77	3.00	2.77	0.00	0.74	1.50	2.25	3.64	3.65	2.90
TiO ₂	0.33	1.03	0.34	4.76	3.86	2.86	1.81	0.89	0.92	0.81
P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.13	0.14	0.00	0.08	0.08	0.08	0.08	0.09	0.09	0.13
MnO ₂	0.00	0.00	8.65	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.98	0.00
Composition (mol. %)	Ex. 440	Ex. 441	Ex. 442	Ex. 443	Ex. 444	Ex. 445	Ex. 446			
SiO ₂	53.74	53.73	53.65	53.84	53.64	50.90	50.80			
Al ₂ O ₃	11.17	10.81	12.12	11.84	10.79	9.80	9.80			
MgO	20.07	20.63	18.66	18.72	21.07	29.8	29.80			
Na ₂ O	5.64	5.51	6.12	6.14	5.29	2.70	2.70			
K ₂ O	5.50	5.40	5.69	5.71	5.19	2.70	2.70			
CaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
SrO	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
BaO	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
ZnO	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
FeO	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
Li ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
Y ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
ZrO ₂	2.94	2.98	2.81	3.62	3.08	2.10	2.10			
TiO ₂	0.81	0.81	0.82	0.00	0.81	0.00	0.00			

TABLE 12-continued

P ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SnO ₂	0.13	0.13	0.13	0.13	0.13	0.10	0.10
MnO ₂	0.00	0.00	0.00	0.00	0.00	0.00	0.00
La ₂ O ₃	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HfO ₂	0.00	0.00	0.00	0.00	0.00	1.90	0.00
Ta ₂ O ₅	0.00	0.00	0.00	0.00	0.00	0.00	1.00

[0249] Table 13 shows example precursor glass compositions DI and DJ (in terms of mol. %) and glass-ceramic GC 102 and GC 103 resulting from subjecting example precursor glass compositions DI and DJ, respectively, to a ceram schedule of 780° C. for 4 hours and 850° C. for 4 hours.

TABLE 13

Batched Comp. (mol. %)	DI	DJ
SiO ₂	41.1	52.7
Al ₂ O ₃	11.0	5.0
MgO	28.9	27.1
CaO	4.7	1.1
ZnO	4.6	4.9
Na ₂ O	3.0	2.6
K ₂ O	0.8	0.8
ZrO ₂	3.3	3.4
TiO ₂	2.6	2.4
Glass-ceramic	GC102	GC103
Crystalline Phase	Jeffbenite, ZrO ₂	Jeffbenite, ZrO ₂

[0250] Table 14 shows example precursor glass compositions DK-DP (in terms of mol. %), the observable color of the precursor glass, and the observable color and appearance of glass-ceramics GC104 to GC108 resulting from subjecting example precursor glass compositions DK-DP to a ceram schedule of 780° C. for 4 hours and 850° C. for 4 hours.

TABLE 14

	Precursor glass (mol. %)					
	DK	DL	DM	DN	DO	DP
SiO ₂	50.06	50.03	50.06	50.03	50.07	50.03
Al ₂ O ₃	9.63	9.63	9.63	9.63	9.63	9.63
MgO	29.11	29.10	29.11	29.10	29.11	29.10
Na ₂ O	3.09	3.09	3.09	3.09	3.09	3.09
K ₂ O	3.08	3.08	3.08	3.08	3.08	3.08
ZrO ₂	3.06	3.06	3.06	3.06	3.06	3.06
TiO ₂	1.93	1.93	1.93	1.93	1.93	1.93
Cr ₂ O ₃	0.04	—	—	—	—	—
CoO	—	0.08	—	—	—	—
NiO	—	—	0.04	—	—	—
CuO	—	—	—	0.08	—	—
V ₂ O ₅	—	—	—	—	0.02	—
MnO	—	—	—	—	—	0.09
Precursor glass color	Green	Pale blue	Amber	Green	Pale amber	Black
Glass-ceramic	GC104	GC105	GC106	GC107	GC108	GC109
Glass-ceramic color	Darker brownish green	Pale purple	Blue turquoise	Turquoise	Bluish grey	Dark brown
Glass-ceramic appearance	Transparent	Transparent	Transparent	Transparent	Transparent	Transparent

[0251] The mol. % of the oxide of glass-ceramics GC 22 and GC 23 having a thickness of 0.6 mm after being subjected to ion exchange in a bath of 95 wt % KNO₃ and 5 wt % NaNO₃ at 550 C for 0.5 hour as a function of depth are depicted in FIGS. 34 and 35, respectively.

[0252] Each of U.S. App. Nos. 63/309,667 filed Feb. 14, 2022, Ser. No. 17/887,012 filed Aug. 12, 2022, 63/444,715 filed Feb. 10, 2023, 63/449,498 filed Mar. 2, 2023, 63/449,394 filed Mar. 2, 2023, 63/326,308 filed Apr. 1, 2022, 63/420,952 filed Oct. 31, 2022, 63/428,773 filed Nov. 30, 2022, and 63/530,542 filed Aug. 3, 2023 is hereby incorporated by reference herein in its entirety.

[0253] It will be apparent to those skilled in the art that various modifications and variations may be made to the embodiments described herein without departing from the spirit and scope of the claimed subject matter. Thus, it is intended that the specification cover the modifications and variations of the various embodiments described herein provided such modification and variations come within the scope of the appended claims and their equivalents.

What is claimed is:

1. A method of making a glass-ceramic article, the method comprising:

heat treating a glass comprising, in terms of representative oxides, SiO₂, Al₂O₃, and MgO to nucleate a crystalline phase in the glass,

growing the crystalline phase in the glass to form glass-ceramic,

- wherein the crystalline phase comprises a tetragonal structure; and
- wherein the glass-ceramic exhibits an X-ray diffraction spectrum comprising:
- a first peak between 2-theta angles of 30° to 32°;
 - a second peak between 2-theta angles of 33° to 35°;
 - a third peak between 2-theta angles of 40° to 42°; and
 - a fourth peak and a fifth peak between 2-theta angles of 55° to 58°.
2. The method of claim 1, wherein the growing is in a surrounding environment at less than 10 GPa of pressure.
3. The method of claim 2, wherein the growing is at a temperature of less than 1400K.
4. The method of claim 3, wherein the heat treating comprises heating the glass to a temperature range of greater than or equal to 700° to less than or equal to 950° C.
5. The method of claim 4, wherein the heat treating comprises maintaining the glass in the temperature range for a time greater than 0.25 hours and less than 6 hours.
6. The method of claim 1, wherein the glass-ceramic exhibits an “a” lattice parameter greater than or equal to 6.5 Å and less than or equal to 6.7 Å.
7. The method of claim 6, wherein the glass-ceramic exhibits a “c” lattice parameter greater than or equal to 18.0 Å and less than or equal to 18.5 Å.
8. A method of making a glass-ceramic article, the method comprising:
- heat treating a glass comprising, in terms of representative oxides, SiO₂, Al₂O₃, and MgO to nucleate a crystalline phase in the glass,
 - growing the crystalline phase in the glass to form glass-ceramic,
 - wherein the crystalline phase comprises a tetragonal structure; and
 - wherein the crystalline phase has a composition according to the formula:
- $$(Mg,R^{2+})_{3+x}(Zr,R^{4+})_xAl_{2-2x}Si_3O_{12},$$
- where R²⁺ represents one or more divalent metal cations, R⁴⁺ represents one or more tetravalent metal cations, and x is greater than or equal to 0 to less than or equal to 1.
9. The glass-ceramic article of claim 8, wherein R²⁺ is one or more divalent metal cations selected from Ca²⁺, Mn²⁺,

Fe²⁺, Zn²⁺, and wherein R⁴⁺ is one or more tetravalent metal cations selected from Ti⁴⁺, Sn⁴⁺, Hf⁴⁺.

10. The glass-ceramic article of claim 8, wherein the glass comprises less than or equal to 3 mol. % Li₂O.

11. The glass-ceramic article of claim 8, wherein the glass-ceramic article has an average transmittance of at least 75% for a light in a wavelength range from 400 nm to 800 nm at through a pathlength of 0.6 mm.

12. The glass-ceramic article of claim 8, wherein grains of the crystalline phase are homogeneously distributed throughout the glass-ceramic.

13. The glass-ceramic article of claim 8, wherein grains of the crystalline phase are randomly oriented within the glass-ceramic.

14. The glass-ceramic article of claim 8, wherein grains of the crystalline phase overlap and interlock with one another within the glass-ceramic to a degree that fracture toughness of the glass-ceramic is 0.75 MPa·m^{1/2} or greater.

15. The glass-ceramic article of claim 8, wherein a thickness of the glass-ceramic article is greater than 200 μm and less than 5 mm, and wherein a length and a width of the glass-ceramic article are both greater than 5 mm.

16. A method of making a glass-ceramic article, the method comprising:

- heat treating a glass comprising, in terms of representative oxides, SiO₂, Al₂O₃, and MgO to nucleate a crystalline phase in the glass,

- growing the crystalline phase in the glass to form glass-ceramic,

- wherein the crystalline phase comprises crystals comprising a jeffbenite crystalline structure.

17. The glass-ceramic article of claim 16, wherein the crystals comprising the jeffbenite crystalline structure are the primary form of crystals in the crystalline phase.

18. The glass-ceramic article of claim 16, wherein the glass-ceramic comprises greater than 25 wt. % and less than 75 wt. % of the crystalline phase.

19. The glass-ceramic article of claim 16, wherein at least some of the crystals comprising the jeffbenite crystalline structure have a largest dimension greater than 20 nm and less than 100 nm.

20. The glass-ceramic article of claim 16, wherein the crystals having the jeffbenite crystalline structure are homogeneously distributed throughout the glass-ceramic.

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